

sudden, bilateral loss of muscle tone precipitated by emotions. A range of positive emotions can trigger cataplexy, including those associated with laughter, anticipation, or surprise. Less commonly, cataplexy can be triggered by negative emotions such as anger and embarrassment. Muscles affected include those of the neck, jaw, arms, legs, or whole body, resulting in head bobbing, jaw dropping, or complete falls. Individuals are awake and aware during cataplexy. Cataplexy should not be confused with “weakness” occurring in

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the context of athletic activities (physiological) or exclusively after unusual emotional triggers such as stress or anxiety (suggesting possible psychopathology).

In children and rarely in adults with acute NT1 symptom onset, cataplexy may manifest as continuous hypotonia rather than episodic bouts of weakness triggered by strong emotions. This continuous hypotonia may result in gait instability, ptosis, and slack jaw. Superimposed on this muscle weakness, some individuals may demonstrate phenomena such as tongue protrusion and grimacing. This static cataplexy is most common within 6 months of a rapid onset.

NT1 is caused by loss of hypothalamic neurons that produce the hypocretin (orexin) neuropeptides, and CSF hypocretin levels are typically less than one-third of control values (< 110 pg/mL in most laboratories). Individuals with cataplexy have been shown to have low CSF hypocretin levels in 85%–90% of cases. In contrast, most individuals with NT2 have normal or intermediate levels of CSF hypocretin. Thus, hypocretin deficiency is a sufficient diagnostic test for NT1 (Criterion B2). If CSF hypocretin is measured and not low, an NT2 diagnosis is based on clinical symptoms (Criterion A) and sleep study data outlined in Criterion B3.

A nocturnal polysomnogram followed by an MSLT is the conventional method for confirming the diagnosis of both NT1 (if hypocretin testing is unavailable or not feasible) and NT2 (Criterion B3). These tests must be performed after the individual has stopped all psychotropic medications (for a duration based on elimination half-life) and obtained adequate sleep time on a normal sleep-wake schedule (as documented with sleep diaries or, preferably, actigraphy), ideally for 2 weeks. Notably, the abrupt discontinuation of antidepressants, α -adrenergic agonist medications, or stimulants or use of these medications during testing can alter REM sleep physiology.

The MSLT result must be positive for a diagnosis of NT2, showing a mean sleep latency of ≤ 8 minutes plus at least two SOREMPs; specifically, REM sleep must occur in at least two of the five nap opportunities. Alternatively, a nocturnal sleep-onset REM period (nSOREMP; REM sleep-onset latency ≤ 15 minutes) during polysomnography is sufficient to confirm the diagnosis and meets Criterion B3. An nSOREMP is highly specific to NT1 (95%–97%) but only moderately sensitive (54%–57%). False positive findings of SOREMPs can occur with shift work, circadian rhythm sleep-wake disorders, severe obstructive sleep apnea, medication effects, and insufficient sleep disorder.

The nocturnal polysomnogram and MSLT are diagnostically limited, especially in NT2. While reliability of diagnostic MSLT testing is relatively high at 85%–95% for NT1, reliability for the NT2 diagnosis is poorer. Test-retest reliability may be $< 50\%$. This poor reliability may be due to day-to-day variability in NT2 physiology and technical aspects of the polysomnogram and MSLT testing, especially inadequate attention to prior sleep time/schedule and medication/drug use.

Normal or intermediate CSF hypocretin levels among individuals with cataplexy symptoms can decline to undetectable levels over time.

Associated Features

When sleepiness is severe, automatic behaviors may occur, with the individual continuing his or her activities in a semiautomatic, hazelike fashion without memory or consciousness. Approximately 20%–60% of individuals experience vivid hypnagogic hallucinations before or upon falling asleep or hypnopompic hallucinations just after awakening. These hallucinations are typically visual or auditory, and sometimes tactile. They are distinct from the less vivid, nonhallucinatory dream-like mentation at sleep onset that occurs in persons with normal sleep.

Approximately 20%–60% of affected individuals experience sleep paralysis upon falling asleep or awakening, leaving them awake but unable to move or speak. However, many normal sleepers also report occasional sleep paralysis, especially with stress or sleep

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deprivation. Individuals with narcolepsy can have a range of nocturnal sleep symptoms, including disrupted nighttime sleep (frequent, brief awakenings), vivid and realistic dreams, periodic limb movements of sleep, and REM sleep behavior disorder. Nocturnal eating may occur. Obesity is common. Individuals may appear sleepy or fall asleep in the waiting area or during clinical examination. During cataplexy, individuals may slump in a chair and have slurred speech or drooping eyelids. If the clinician is able to check reflexes during cataplexy (most attacks are < 10 seconds), reflexes are abolished during whole body cataplexy—an important finding distinguishing genuine cataplexy from functional neurological symptom disorder (conversion disorder).

Although IQ testing is generally normal in individuals with narcolepsy, impairments in working memory and executive functioning have been reported.

Prevalence

Narcolepsy-cataplexy (NT1) affects 0.02%–0.05% of the adult general population worldwide and has an incidence of 0.74 per 100,000 person-years in the United States. Some prevalence variation has been reported, including lower rates in Israel and higher rates in Japan than in Europe and the United States. The true prevalence of NT2 is unknown in part because of diagnostic variability. Narcolepsy affects both genders fairly equally, but this may vary among different populations.

Development and Course

Onset occurs most often in childhood and adolescence or young adulthood but rarely in old age. Peak age at onset is around 15–25 years. Onset can be abrupt or progressive, with cataplexy developing over years. It has been reported that children presenting with abrupt onset of NT1 symptoms have the highest disease severity but that disease severity in these cases tends to partially improve in the first few years after onset. Abrupt onset in young, prepubescent children can be associated with obesity and premature puberty. About 50% of individuals with narcolepsy diagnosed in adulthood recall symptom onset in childhood or adolescence, highlighting problems

of diagnostic delays for this condition. Once the disorder has manifested, the course is persistent and lifelong.

In 90% of cases, the first symptom to manifest is sleepiness or increased sleep, followed by cataplexy (within 1 year in 50% of cases, within 3 years in 85%). Sleepiness, hypnagogic hallucinations, vivid dreaming, and REM sleep behavior disorder (vocalizations or complex motor behavior during REM sleep) are early symptoms. Excessive sleep rapidly progresses to an inability to stay awake during the day, and to maintain good sleep at night, without a clear increase in total 24-hour sleep time. In the first months, cataplexy may be atypical, especially in children, manifesting with a generalized hypotonia rather than with episodic emotionally triggered weakness. In general, narcolepsy symptoms remain fairly stable but may fluctuate with life events such as pregnancy and stressors. Exacerbations of symptoms suggest lack of compliance with medications or development of a concurrent sleep disorder, notably sleep apnea, which has been identified in about a quarter of individuals with narcolepsy.

Young children and adolescents with narcolepsy often develop aggression or behavioral problems secondary to sleepiness and/or nighttime sleep disruption. Workload and social pressure increase through high school and college, reducing available sleep time at night. Pregnancy does not seem to modify symptoms consistently. After retirement, individuals typically have more opportunity for napping, reducing the need for stimulants. Maintaining a regular schedule benefits individuals at all ages.

Risk and Prognostic Factors

Temperamental. Individuals with narcolepsy commonly report that they need more sleep than other family members.

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Environmental. Group A streptococcal throat infection, influenza (notably pandemic H1N1 2009), or other winter infections, as well as vaccinations (specifically Pandemrix H1N1 vaccination), may trigger an autoimmune process in some individuals, producing narcolepsy a few months later. Head trauma and abrupt changes in sleep-wake patterns (e.g., job changes, stress) may be additional triggers.

Genetic and physiological. Monozygotic twins are 25%–32% concordant for narcolepsy. The prevalence of narcolepsy is 1%–2% in first-degree relatives (a 10- to 40-fold increase overall). Narcolepsy is strongly associated with HLA DQB1*06:02 (see “Diagnostic Markers”). DQB1*03:01 increases, while DQB1*05:01, DQB1*06:01, and DQB1*06:03 reduce risk in the presence of DQB1*06:02, but the effect is small. Polymorphisms within the T-cell receptor alpha gene and other immune-modulating genes also modulate risk slightly.

Culture-Related Diagnostic Issues

Narcolepsy has been described in many ethnoracial groups and cultural contexts. One study of 1,097 treatment-seeking individuals suggested that among African Americans, more cases may manifest without cataplexy or with atypical cataplexy (although CSF hypocretin is low), and with earlier onset compared with non-Latinx Whites. Diagnosis may be further complicated by the higher presence of obesity and obstructive sleep apnea in this population, which can be

related to differential exposure to social determinants of health, including food insecurity, food deserts, and limited access to safe and affordable places for physical activity. Individuals with narcolepsy often experience sleep paralysis, which may be attributed to supernatural forces (e.g., frightening spirit is sitting on the sleeper's chest) in some cultural contexts, contributing to the perceived dangerousness of the condition and to help-seeking decisions.

Diagnostic Markers

Nocturnal polysomnography followed by an MSLT is used to confirm the diagnosis of narcolepsy, especially when the disorder is first being diagnosed and before treatment has begun. In the presence of clear-cut cataplexy, the polysomnography and MSLT are confirmatory for NT1. In the absence of cataplexy and hypocretin deficiency (if measured), the MSLT is diagnostic of NT2. Drug or medication effects (e.g., REM-inhibiting antidepressants or sedating medications), stimulant withdrawal, prior sleep deprivation, shift work, or severe depression may result in an inaccurate MSLT result and must be ruled out prior to performance of the MSLT. In particular, chronically insufficient sleep is common and must be considered.

An nSOREMP is highly specific (approximately 1% positive in control subjects) but moderately sensitive (approximately 50%) for NT1. In contrast, an nSOREMP was only found in 10%–23% of NT2 persons with normal hypocretin levels, suggesting even lower sensitivity in this subtype. The MSLT result is considered positive for narcolepsy if it displays an average sleep latency of ≤ 8 minutes and SOREMPs in two or more naps on a four- or five-nap test. The MSLT result is positive in 90%–95% of individuals with NT1 versus 2%–4% of control subjects or individuals with other sleep disorders. As noted, poor test-retest reliability for NT2 precludes determination of comparable data for NT2. Additional polysomnographic findings among individuals with narcolepsy often include frequent arousals, decreased sleep efficiency, and increased stage 1 sleep. Periodic limb movements (found in about 40% of individuals with NT1) and sleep apnea are often noted.

Hypocretin deficiency is demonstrated by measuring CSF hypocretin-1 levels. The test is particularly useful in individuals with suspected pseudocataplexy and those without typical cataplexy, or in treatment-refractory cases. The diagnostic value of the test is not affected by medications, sleep deprivation, or the time of day or night when it is collected, but the findings are uninterpretable when the individual is severely ill with a concurrent

infection or head trauma or is comatose. CSF cytology, protein, and glucose are within normal range even when sampled within weeks of rapid onset of the disorder. When measured in individuals with typical cataplexy symptoms, CSF hypocretin-1 is often already very diminished or undetectable.

About 85%–95% of individuals with NT1 are positive for the HLA DQB1*06:02 haplotype. This gene influences immune system antigen presentation, supporting an underlying autoimmune pathophysiology of NT1. Outbreaks of NT1 after specific vaccinations and infections further support an autoimmune etiology. In contrast to NT1, there are no biomarkers of NT2. Only about 40%–50% of individuals with NT2 are positive for DQB1*06:02. As 12%–38% of the general population is DQB1*06:02 positive, testing for this allele is not very helpful for diagnosing NT2

but can be helpful for screening of NT1.

Functional Consequences of Narcolepsy

School performance, driving, work, or other activities that require sustained attention are impaired, and individuals with narcolepsy should avoid jobs that place themselves (e.g., working with machinery) or others (e.g., bus driver, pilot) in danger. Once the narcolepsy is controlled with therapy, individuals can usually drive, although rarely long distances alone. Untreated individuals are also at risk for social isolation and accidental injury to themselves or others. Social relations may suffer as these individuals strive to avert cataplexy by exerting control over emotions or stimuli that cause emotions.

Differential Diagnosis

Other hypersomnias. Hypersomnolence disorder (also known as idiopathic hypersomnia) and narcolepsy are similar with respect to the presence of chronic daytime sleepiness, age at onset (typically adolescence or early adulthood), and stable course over time, but can be distinguished based on distinctive clinical and laboratory features. Individuals with hypersomnolence disorder typically have longer and less disrupted nocturnal sleep, greater difficulty awakening, more persistent daytime sleepiness (as opposed to more discrete “sleep attacks” in narcolepsy), longer and less refreshing daytime sleep episodes, and little or no dreaming during daytime naps. By contrast, individuals with NT1 generally have cataplexy. Those with NT1 or NT2 may demonstrate recurrent intrusions of elements of REM sleep into the transition between sleep and wakefulness (e.g., sleep-related hallucinations and sleep paralysis). The MSLT typically demonstrates shorter sleep latencies (i.e., greater physiological sleepiness) as well as the presence of multiple SOREMPs in individuals with narcolepsy.

Sleep deprivation and insufficient nocturnal sleep. Sleep deprivation and insufficient nocturnal sleep are common in adolescents and shift workers. In adolescents, difficulties falling asleep at night are common, causing sleep deprivation. The MSLT result may be falsely positive if conducted while the individual is sleep deprived or while his or her sleep is phase delayed.

Sleep apnea syndromes. Obstructive sleep apnea is common in the general population and can be present in individuals with narcolepsy due to obesity. Because obstructive sleep apnea is more frequent than narcolepsy, cataplexy may be overlooked (or absent). Narcolepsy should be considered in individuals with persistent sleepiness despite treatment of their sleep apnea.

Insomnia disorder. Individuals with narcolepsy may focus on the presence of nocturnal sleep disruption and incorrectly attribute daytime sleepiness to insomnia disorder. Although individuals with narcolepsy, like those with insomnia disorder, may experience frequent awakenings during the night, individuals with narcolepsy typically have no difficulty initiating sleep or returning to sleep in contrast to those with insomnia disorder.

Moreover, insomnia disorder is not typically associated with the severity of daytime sleepiness observed in narcolepsy.

Major depressive disorder. Excessive daytime sleepiness is a common complaint of both

individuals with major depression and individuals with narcolepsy. The presence of cataplexy (which is not a feature of major depressive disorder) along with the severity of excessive daytime sleepiness indicates a diagnosis of NT1 rather than major depressive disorder. Moreover, in individuals with major depression, MSLT results are most often normal, and there is dissociation between subjective and objective sleepiness, as measured by the mean sleep latency during the MSLT. In a meta-analysis of individuals with psychiatric disorders evaluated for sleepiness, while 25% had a mean sleep latency of < 8 minutes on the MSLT, only rarely were two or more SOREMPs noted on the MSLT, highlighting the more specific REM sleep dysfunction of narcolepsy.

Functional neurological symptom disorder (conversion disorder; pseudocatataplexy). Individuals with functional neurological symptom disorder can present with weakness that may raise questions of cataplexy. However, in functional neurological symptom disorder, the weakness is often long-lasting, has unusual triggers, and can result in frequent falls. Individuals may report sleeping and dreaming during MSLT naps, yet the MSLT does not show the characteristic SOREMP. Home video recordings and video during sleep studies can be helpful to distinguish this condition from true cataplexy. The weakness is usually generalized in pseudocatataplexy, without partial attacks. Full-blown, long-lasting pseudocatataplexy may occur during consultation, allowing the examining physician enough time to verify reflexes, which remain intact.

Attention-deficit/hyperactivity disorder or other behavioral problems. In children and adolescents, sleepiness can cause behavioral problems, including aggressiveness and inattention, leading to a misdiagnosis of attention-deficit/hyperactivity disorder (ADHD).

Atonic seizures. Atonic seizures, a type of seizure that causes sudden loss of muscle strength, must be distinguished from cataplexy. Atonic seizures are not commonly triggered by emotions and tend to manifest as abrupt falls rather than the slower “melting” quality of cataplexy. Atonic seizures usually occur in individuals with additional seizure types and have distinct signatures on the electroencephalogram.

Syncope. Like syncope, cataplexy usually develops over several seconds, but individuals with cataplexy do not have presyncopal symptoms of dizziness, tunnel vision, and auditory changes.

Chorea and motor disorders. In young children, cataplexy can be misdiagnosed as chorea or pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS), especially in the context of a strep throat infection and high antistreptolysin O antibody levels. Some children may have an overlapping motor disorder close to onset of the cataplexy.

Schizophrenia. In the presence of florid and vivid hypnagogic hallucinations, individuals with narcolepsy may think these experiences are real—a feature that suggests the presence of a true hallucination characteristic of schizophrenia. However, clear differences have been described in the pattern of hallucinatory experiences in narcolepsy compared with schizophrenia. Individuals with narcolepsy tend to report sleep-related multisensory “holistic” hallucinations (visual, auditory, tactile) rather than the predominantly verbal-auditory sensory mode of individuals with schizophrenia. Moreover, high-dose stimulant treatment of individuals with narcolepsy may result in the development of persecutory delusions. If cataplexy is present with hallucinations or delusions, the first clinical supposition would be that these symptoms are secondary to narcolepsy before consideration of a co-occurring diagnosis of schizophrenia.

Comorbidity

Medical and psychiatric comorbidities are common among individuals with narcolepsy and include obesity, bruxism, enuresis, precocious puberty (among individuals with pediatric-onset narcolepsy), mood disorders, and ADHD. Rapid weight gain is common in young children with a sudden disease onset. Parasomnias (e.g., sleepwalking, REM sleep behavior disorder), obstructive sleep apnea, restless legs syndrome, and periodic limb movements are common in individuals who develop narcolepsy. Comorbid sleep apnea should be considered if there is a sudden aggravation of preexisting narcolepsy.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), differentiates two subtypes of narcolepsy: NT1 (narcolepsy with cataplexy or hypocretin deficiency) and NT2 (narcolepsy without cataplexy or hypocretin deficiency). NT1 secondary to another medical condition (G47.421) and NT2 secondary to another medical condition (G47.429) are reported in ICSD-3 as secondary narcolepsy subtypes.

Breathing-Related Sleep Disorders

The breathing-related sleep disorders category encompasses three relatively distinct disorders: obstructive sleep apnea hypopnea, central sleep apnea, and sleep-related hypoventilation.

Obstructive Sleep Apnea Hypopnea

Diagnostic Criteria

G47.33

A. Either (1) or (2):

1. Evidence by polysomnography of at least five obstructive apneas or hypopneas per hour of sleep and either of the following sleep symptoms:
 - a. Nocturnal breathing disturbances: snoring, snorting/gasping, or breathing pauses during sleep.
 - b. Daytime sleepiness, fatigue, or unrefreshing sleep despite sufficient opportunities to sleep that is not better explained by another mental disorder (including a sleep disorder) and is not attributable to another medical condition.
2. Evidence by polysomnography of 15 or more obstructive apneas and/or hypopneas per hour of sleep regardless of accompanying symptoms.

Specify current severity:

Mild: Apnea hypopnea index is less than 15.

Moderate: Apnea hypopnea index is 15–30.

Severe: Apnea hypopnea index is greater than 30.

Specifiers

Disease severity is measured by a count of the number of apneas plus hypopneas per hour of sleep (apnea hypopnea index) using polysomnography or other overnight monitoring.

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Apnea refers to the total absence of airflow, and *hypopnea* refers to a reduction in airflow. Overall severity is also informed by levels of nocturnal desaturation and sleep fragmentation (measured by brain cortical arousal frequency and sleep stages) and degree of associated symptoms and daytime impairment. However, the exact number and thresholds may vary according to the specific measurement techniques used, and these numbers may change over time. Regardless of the apnea hypopnea index (count) per se, the disorder is considered to be more severe when apneas and hypopneas are accompanied by significant oxygen hemoglobin desaturation (e.g., when more than 10% of the sleep time is spent at desaturation levels of < 90%) or when sleep is severely fragmented as shown by an elevated arousal index (arousals per hour of sleep > 30) or reduced time in deep sleep (e.g., percentage stage N3 [slow-wave sleep] < 5%).

Diagnostic Features

Obstructive sleep apnea hypopnea is the most common breathing-related sleep disorder. It is characterized by repeated episodes of upper (pharyngeal) airway obstruction (apneas and hypopneas) during sleep. Each apnea or hypopnea represents a reduction in breathing of at least 10 seconds in duration in adults or two missed breaths in children and is typically associated with drops in oxygen saturation of $\geq 3\%$ and/or an electroencephalographic arousal. Both sleep-related (nocturnal) and wake-time symptoms are common. The cardinal symptoms of obstructive sleep apnea hypopnea are snoring and daytime sleepiness.

Obstructive sleep apnea hypopnea in adults is diagnosed on the basis of findings from a polysomnogram (or sleep testing performed outside of the sleep center, referred to as *out of center sleep testing* [OCST]) and symptoms. The diagnosis is based on symptoms of 1) nocturnal breathing disturbances (i.e., snoring, snorting/gasping, breathing pauses during sleep), or 2) daytime sleepiness, fatigue, or unrefreshing sleep despite sufficient opportunities to sleep that are not better explained by another mental disorder and not attributable to another medical condition, along with 3) evidence by polysomnography (or OCST) of five or more obstructive apneas or hypopneas per hour of sleep (Criterion A1). Diagnosis can be made in the absence of these symptoms if there is evidence by polysomnography (or limited OCST) of 15 or more obstructive apneas and/or hypopneas per hour of sleep (Criterion A2).

Criteria for a diagnosis of obstructive sleep apnea hypopnea in children differ from those for a diagnosis in adults. An obstructive apnea hypopnea index of one or more events per hour or evidence of obstructive hypoventilation in association with snoring or polysomnographic evidence of airflow obstruction is used to define thresholds of abnormality in children.

Polysomnographic findings in children may differ from those in adults in that children may demonstrate labored breathing; partial obstructive hypoventilation (sustained reductions of tidal volume due to upper airway flow limitations) with cyclical oxygen desaturations; hypercapnia; and paradoxical breathing.

Most cases of obstructive sleep apnea remain undiagnosed. Therefore, specific attention to symptoms of disturbed sleep occurring in association with snoring or breathing pauses and physical findings that increase risk of obstructive sleep apnea hypopnea (e.g., central obesity, crowded pharyngeal airway, elevated blood pressure) is important to reduce the chance of failure to diagnose this treatable condition.

Associated Features

Because of the frequency of nocturnal awakenings that occur with obstructive sleep apnea hypopnea, individuals may report symptoms of insomnia. Other common, though nonspecific, symptoms of obstructive sleep apnea hypopnea are heartburn, nocturia, morning headaches, dry mouth, erectile dysfunction, and reduced libido. Individuals may complain of difficulty breathing while lying supine or sleeping. Hypertension may occur in more than 60% of individuals with obstructive sleep apnea hypopnea.

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Arterial blood gas measurements while the individual is awake are usually normal, but some individuals may demonstrate waking hypoxemia or hypercapnia. This pattern should alert the clinician to the possibility of coexisting lung disease or hypoventilation. Imaging procedures may reveal narrowing of the upper airway. Cardiac testing may show evidence of impaired ventricular function. Arrhythmias such as sinus pauses, frequent atrial and ventricular ectopic beats, or atrial fibrillation may be present during sleep. Individuals with severe nocturnal oxygen desaturation may also have elevated hemoglobin or hematocrit values.

Prevalence

Obstructive sleep apnea hypopnea is a very common disorder. Prevalence may be particularly high among men as compared with women, ranging from 2:1 to 4:1; older adults; and certain racial and ethnic groups. Prevalence varies cross-nationally, partly because of differences in assessment methods. Because the disorder is strongly associated with obesity, the rise in obesity rates has resulted in an increased prevalence of this disorder.

In the United States, 13% of men and 6% of women have polysomnographic evidence of 15 or more obstructive apneas or hypopneas per hour of sleep, and 14% of men and 5% of women have more than 5 obstructive apneas or hypopneas per hour of sleep, plus symptoms of daytime sleepiness. Gender differences decline in older age, possibly because of increased prevalence in females after menopause; postmenopausal females are 2.6–3.5 times more likely to have obstructive sleep apnea compared with premenopausal females.

In the general community, prevalence rates in the United States of undiagnosed obstructive sleep apnea hypopnea may be very high in elderly individuals. Obstructive sleep apnea also occurs in children, with an estimated prevalence of 1%–4%; there is no gender difference among prepubertal children. Children who are obese have higher rates.

Prevalence of obstructive sleep apnea appears to be higher among African Americans than among U.S. non-Latinx Whites. An increased prevalence among African Americans, American Indians, and Hispanics may be related to higher rates of obesity, which can be associated with differential exposure to social determinants of health, including food insecurity, food deserts, and limited access to safe and affordable places for physical activity.

Development and Course

The age distribution of obstructive sleep apnea hypopnea has several peaks. The first occurs in children ages 3–8 years when the nasopharynx may be compromised by a relatively large mass of tonsillar tissue compared with the size of the upper airway. With growth of the airway and regression of lymphoid tissue during later childhood, there is reduction in prevalence. However, with the increase of obesity in adolescents, a second peak in prevalence occurs in that age group. Finally, as obesity prevalence continues to increase in midlife and women enter menopause, rates of obstructive sleep apnea hypopnea further increase. The course in older age is unclear; prevalence of the disorder may plateau after age 65 years, but in some individuals, severity may worsen with aging. Polysomnographic results must be interpreted in light of other clinical data. Significant clinical symptoms of insomnia or hypersomnia should be investigated regardless of the individual's age.

Obstructive sleep apnea hypopnea usually has an insidious onset, gradual progression, and persistent course. Typically, the loud snoring has been present for many years, often since childhood, but an increase in its severity may lead the individual to seek evaluation. Weight gain may precipitate an increase in symptoms. Although obstructive sleep apnea hypopnea can occur at any age, it most commonly manifests among individuals ages 40–60 years. Over 4–5 years, the average apnea hypopnea index increases in adults and older individuals by approximately two apneas or hypopneas per hour. The apnea hypopnea index is increased and incident obstructive sleep apnea hypopnea is greater among

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individuals who are older, who are male, or who have a higher baseline body mass index (BMI) or increase their BMI over time. Spontaneous resolution of obstructive sleep apnea hypopnea has been reported with weight loss, particularly after bariatric surgery. In children, seasonal variation in obstructive sleep apnea hypopnea has been observed, as has improvement with overall growth.

In young children, the signs and symptoms of obstructive sleep apnea hypopnea may be more subtle than in adults, making diagnosis more difficult to establish. Polysomnography is useful in confirming diagnosis. Evidence of fragmentation of sleep on the polysomnogram may not be as apparent as in studies of older individuals, possibly because of the high homeostatic drive in young individuals. Symptoms such as snoring are usually parent-reported and thus have reduced sensitivity. Agitated arousals and unusual sleep postures, such as sleeping on the hands and knees, may occur. Nocturnal enuresis also may occur and should raise the suspicion of obstructive sleep apnea hypopnea if it recurs in a child who was previously dry at night. Children may also manifest excessive daytime sleepiness, although this is not as common or pronounced as in adults. Daytime mouth breathing, difficulty in swallowing, and poor speech articulation are

also common features in children. Children younger than 5 years more often present with nighttime symptoms, such as observed apneas or labored breathing, than with behavioral symptoms (i.e., the nighttime symptoms are more noticeable and more often bring the child to clinical attention). In children older than 5 years, daytime symptoms such as sleepiness and behavioral problems (e.g., impulsivity and hyperactivity), attention-deficit/hyperactivity disorder, learning difficulties, and morning headaches are more often the focus of concern. Children with obstructive sleep apnea hypopnea also may present with delayed growth, failure to thrive, and developmental delays. Although obesity is a less important risk factor in young children, it nevertheless contributes to the occurrence of obstructive sleep apnea.

Risk and Prognostic Factors

Genetic and physiological. The major risk factors for obstructive sleep apnea hypopnea are obesity and male sex. Others include maxillary-mandibular retrognathia or micrognathia, positive family history of sleep apnea, genetic syndromes that reduce upper airway patency (e.g., Down syndrome, Treacher Collins syndrome), adenotonsillar hypertrophy (especially in young children), menopause (in females), and various endocrine syndromes (e.g., acromegaly). Compared with premenopausal females, males are at increased risk for obstructive sleep apnea hypopnea, possibly reflecting the influences of sex hormones on ventilatory control and body fat distribution, as well as gender differences in airway structure. Medications for mental disorders and medical conditions that tend to induce somnolence may worsen the course of apnea symptoms if these medications are not managed carefully.

Obstructive sleep apnea hypopnea has a strong genetic basis, as evidenced by the significant familial aggregation of the apnea hypopnea index. The prevalence of obstructive sleep apnea hypopnea is approximately twice as high among the first-degree relatives of probands with obstructive sleep apnea hypopnea as compared with members of control families. One-third of the variance in the apnea hypopnea index is explained by shared familial factors. Although genetic markers with diagnostic or prognostic value are not yet available for use, eliciting a family history of obstructive sleep apnea hypopnea should increase the clinical suspicion for the disorder.

Culture-Related Diagnostic Issues

There is a potential for sleepiness and fatigue to be reported differently across cultures. In some groups, snoring may be considered a sign of normal health and thus may not trigger concerns, leading to underdiagnosis.

Sex- and Gender-Related Diagnostic Issues

Menopause, pregnancy, and polycystic ovarian syndrome increase the risk of obstructive sleep apnea in females. The transition from premenopause to postmenopause is associated with increased severity of obstructive sleep apnea. Women may more commonly report fatigue, lack of energy, or insomnia rather than sleepiness and may underreport snoring.

Diagnostic Markers

Polysomnography provides quantitative data on frequency of sleep-related respiratory disturbances and associated changes in oxygen saturation and sleep continuity. Validated sleep measures (e.g., multiple sleep latency test, maintenance of wakefulness test) may identify sleepiness.

Functional Consequences of Obstructive Sleep Apnea Hypopnea

More than 50% of individuals with moderate to severe obstructive sleep apnea hypopnea report symptoms of daytime sleepiness. A twofold increased risk of occupational accidents has been reported in association with symptoms of snoring and sleepiness. Motor vehicle crashes also have been reported to be as much as sevenfold higher among individuals with elevated apnea hypopnea index values. Clinicians should be cognizant of state government requirements for reporting this disorder, especially in relationship to commercial drivers. Reduced scores on measures of health-related quality of life are common in individuals with obstructive sleep apnea hypopnea. Although the greatest functional impact is observed in the “vitality” domain, severe obstructive sleep apnea negatively affects general health and physical and social functioning as well.

Differential Diagnosis

Primary snoring and other sleep disorders. Individuals with obstructive sleep apnea hypopnea must be differentiated from individuals with primary snoring (i.e., otherwise asymptomatic individuals who snore and do not have abnormalities on overnight polysomnography). Individuals with obstructive sleep apnea hypopnea may additionally report nocturnal gasping and choking, which can be confused with the presence of asthma or gastroesophageal reflux. The presence of sleepiness or other daytime symptoms not explained by other etiologies suggests the diagnosis of obstructive sleep apnea hypopnea, but this differentiation requires polysomnography. Definitive differential diagnosis between hypersomnolence disorder, central sleep apnea, sleep-related hypoventilation, and obstructive sleep apnea hypopnea also requires polysomnographic studies.

Obstructive sleep apnea hypopnea must be differentiated from other causes of sleepiness, such as narcolepsy, hypersomnolence disorder, insufficient sleep, and circadian rhythm sleep disorders. Obstructive sleep apnea hypopnea can be differentiated from narcolepsy by the absence of cataplexy, sleep-related hallucinations, and sleep paralysis and by the presence of loud snoring, gasping during sleep, or observed apneas in sleep. Daytime sleep episodes in narcolepsy are characteristically shorter, more refreshing, and more often associated with dreaming. Obstructive sleep apnea hypopnea shows characteristic apneas and hypopneas and oxygen desaturation during nocturnal polysomnographic studies. Narcolepsy results in multiple sleep-onset rapid eye movement (REM) periods during the MSLT. Narcolepsy, like obstructive sleep apnea hypopnea, may be associated with obesity, and some individuals have concurrent narcolepsy and obstructive sleep apnea hypopnea. A diagnosis of narcolepsy does not exclude the diagnosis of obstructive sleep apnea hypopnea, as the two conditions may co-occur.

Central sleep apnea. Central sleep apnea can be distinguished from obstructive sleep apnea by the presence of repetitive apneas or hypopneas due to reduction or absence of respiratory effort on

polysomnogram recording. Snoring may be present, although it may be less prominent than observed in obstructive sleep apnea hypopnea or absent altogether. Individuals with central sleep apnea often exhibit fragmented sleep and may also complain of daytime sleepiness. Central sleep apnea is seen most commonly in individuals with congestive heart failure (Cheyne-Stokes breathing) or neurological disease, or those using opioid medications.

Insomnia disorder. For individuals complaining of difficulty initiating or maintaining sleep or early-morning awakenings, insomnia disorder can be differentiated from obstructive sleep apnea hypopnea by the absence of snoring and the absence of the history, signs, and symptoms characteristic of the latter disorder. However, insomnia and obstructive sleep apnea hypopnea may coexist, and if so, both disorders may need to be addressed concurrently to improve sleep.

Panic attacks. Nocturnal panic attacks may include symptoms of gasping or choking during sleep that may be difficult to distinguish clinically from obstructive sleep apnea hypopnea. However, the lower frequency of episodes, intense autonomic arousal, and lack of excessive sleepiness differentiate nocturnal panic attacks from obstructive sleep apnea hypopnea. Polysomnography (or OCST) in individuals with nocturnal panic attacks does not reveal the typical pattern of apneas or oxygen desaturation characteristic of obstructive sleep apnea hypopnea. Individuals with obstructive sleep apnea hypopnea do not provide a history of daytime panic attacks.

Nocturnal asthma. Nocturnal asthma can often cause sudden awakening from sleep with symptoms of gasping or choking that are indistinguishable from dyspneic episodes resulting from obstructive sleep apnea. However, a history of asthma is generally present and polysomnography (or OCST) does not find evidence of apneas, hypopneas, or oxygen desaturation indicative of obstructive apnea. Nevertheless, nocturnal asthma and obstructive sleep apnea can coexist, and this can make it difficult to determine the relative contributions of each condition.

Attention-deficit/hyperactivity disorder. Attention-deficit/hyperactivity disorder in children may include symptoms of inattention, academic impairment, hyperactivity, and internalizing behaviors, all of which may also be symptoms of childhood obstructive sleep apnea hypopnea. The presence of other symptoms and signs of childhood obstructive sleep apnea hypopnea (e.g., labored breathing or snoring during sleep and adenotonsillar hypertrophy) would suggest the presence of obstructive sleep apnea hypopnea. Obstructive sleep apnea hypopnea and attention-deficit/hyperactivity disorder may commonly co-occur, and there may be causal links between them; therefore, risk factors such as enlarged tonsils, obesity, or a family history of sleep apnea may help alert the clinician to their co-occurrence.

Substance/medication-induced insomnia or hypersomnia. Substance use and substance withdrawal (including medications) can produce insomnia or hypersomnia. A careful history is usually sufficient to identify the relevant substance/medication, and follow-up shows improvement of the sleep disturbance after discontinuation of the substance/medication. In other cases, the use of a substance/medication (e.g., alcohol, barbiturates, benzodiazepines, opiates) has been shown to exacerbate obstructive sleep apnea hypopnea. An individual with symptoms and signs consistent with obstructive sleep apnea hypopnea should receive that diagnosis, even in the presence of concurrent substance use that is exacerbating the condition.

Comorbidity

Systemic hypertension, coronary artery disease, heart failure, stroke, diabetes, and increased mortality are consistently associated with obstructive sleep apnea hypopnea. Risk

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estimates vary from 30% to as much as 300% for moderate to severe obstructive sleep apnea hypopnea. Obstructive sleep apnea and cardiovascular disease are strongly related, and treatment of obstructive sleep apnea reduces morbidity and mortality of cardiovascular disease. Ethnic and racialized groups that have not received adequate health care may be at higher risk for undetected cardiovascular risk factors associated with obstructive sleep apnea. Evidence of pulmonary hypertension and right heart failure (e.g., cor pulmonale, ankle edema, hepatic congestion) is rare in obstructive sleep apnea hypopnea and when present indicates either very severe disease or associated hypoventilation or cardiopulmonary comorbidities. Obstructive sleep apnea hypopnea also may occur with increased frequency in association with a number of medical or neurological conditions (e.g., cerebrovascular disease, Parkinson's disease). Physical findings reflect the co-occurrence of these conditions.

As many as one-third of individuals referred for evaluation of obstructive sleep apnea hypopnea report symptoms of depression, with as many of 10% having depression scores consistent with moderate to severe depression. Severity of obstructive sleep apnea hypopnea, as measured by the apnea hypopnea index, has been found to be correlated with severity of symptoms of depression. This association may be stronger in men than in women.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), differentiates 11 subtypes of “sleep-related breathing disorders,” including central sleep apneas (CSAs) (e.g., primary CSA, CSA due to a medical/neurological condition, CSA due to a substance or medication), obstructive sleep apnea (adult and pediatric), and sleep-related hypoventilation disorders.

Central Sleep Apnea

Diagnostic Criteria

- A. Evidence by polysomnography of five or more central apneas per hour of sleep.
- B. The disorder is not better explained by another current sleep disorder.

Specify whether:

G47.31 Idiopathic central sleep apnea: Characterized by repeated episodes of apneas and hypopneas during sleep caused by variability in respiratory effort but without evidence of airway obstruction.

R06.3 Cheyne-Stokes breathing: A pattern of periodic crescendo-decrescendo variation in tidal volume that results in central apneas and hypopneas at a frequency of at least five events per hour, accompanied by frequent arousal.

G47.37 Central sleep apnea comorbid with opioid use: The pathogenesis of this subtype is attributed to the effects of opioids on the respiratory rhythm generators in the medulla as well as the differential effects on hypoxic versus hypercapnic respiratory drive.

Coding note (for G47.37 code only): When an opioid use disorder is present, first code the opioid use disorder: F11.10 mild opioid use disorder or F11.20 moderate or severe opioid use disorder; then code G47.37 central sleep apnea comorbid with opioid use. When an opioid use disorder is not present (e.g., after a one-time heavy use of the substance), code only G47.37 central sleep apnea comorbid with opioid use.

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Specify current severity:

Severity of central sleep apnea is graded according to the frequency of the breathing disturbances as well as the extent of associated oxygen desaturation and sleep fragmentation that occur as a consequence of repetitive respiratory disturbances.

Subtypes

There are several subtypes of central sleep apnea. Idiopathic central sleep apnea (alternatively termed *primary central sleep apnea*) and central sleep apnea with Cheyne-Stokes breathing are characterized by increased gain of the ventilatory control system, also referred to as *high loop gain*, which leads to instability in ventilation and PaCO₂ levels. This instability is termed *periodic breathing* and can be recognized by hyperventilation alternating with hypoventilation. Individuals with these disorders typically have pCO₂ levels while awake that are slightly hypocapnic or normocapnic. Central sleep apnea may also manifest during initiation of treatment of obstructive sleep apnea hypopnea (termed *treatment-emergent central sleep apnea*) or may occur in association with obstructive sleep apnea hypopnea syndrome. The occurrence of central sleep apnea in association with obstructive sleep apnea is also considered to be due to high loop gain. In contrast, the pathogenesis of central sleep apnea comorbid with opioid use has been attributed to the effects of opioids on the respiratory rhythm generators in the medulla as well as to its differential effects on hypoxic versus hypercapnic respiratory drive. These individuals may have elevated pCO₂ levels while awake. Individuals receiving chronic methadone maintenance therapy have been noted to have increased somnolence and depression, although the role of opioid-induced breathing disorders in causing these problems has not been studied. Similarly, central apnea due to a medical disorder without Cheyne-Stokes breathing is a result of a pathological process that affects brain-stem ventilatory control centers.

Specifiers

An increase in the central apnea index (i.e., number of central apneas per hour of sleep) reflects an increase in severity of central sleep apnea. Sleep continuity and quality may be markedly impaired with reductions in restorative stages of non-rapid eye movement (NREM) sleep (i.e.,

decreased slow-wave sleep [stage N3]). In individuals with severe Cheyne-Stokes breathing, the pattern can also be observed during resting wakefulness, a finding that is thought to be a prognostic marker for increased mortality.

Diagnostic Features

Central sleep apnea disorders are characterized by repeated episodes of apneas and hypopneas during sleep caused by variability in respiratory effort. These are disorders of ventilatory control in which respiratory events occur in a periodic or intermittent pattern. *Idiopathic central sleep apnea* is characterized by sleepiness, insomnia, and awakenings due to dyspnea in association with five or more central apneas per hour of sleep. Individuals with heart failure, stroke, or renal failure who have central sleep apnea typically have a breathing pattern called *Cheyne-Stokes breathing*, which is characterized by a pattern of periodic crescendo-decrescendo variation in tidal volume that results in central apneas and hypopneas occurring at a frequency of at least five events per hour. Events are often associated with arousal, but arousals are not required for the diagnosis. Central sleep apnea observed at high altitude occurs after ascent to high altitude, generally at least 2,500 meters above sea level. Central and obstructive sleep apneas may coexist; a diagnosis of central sleep apnea hypopnea requires that central events be > 50% of the total number of respiratory events.

Alterations in neuromuscular control of breathing can occur in association with medications or substances, which can cause or exacerbate impairments of respiratory rhythm and ventilation. Individuals taking medications with these effects may have a sleep-related breathing disorder that could contribute to sleep disturbances and symptoms such

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as sleepiness, confusion, and depression. Specifically, *chronic use of long-acting opioid medications* is often associated with impairment of respiratory control leading to central sleep apnea.

Associated Features

Individuals with central sleep apnea hypopneas may present with sleepiness or insomnia. They may have complaints of sleep fragmentation, including awakening with dyspnea. Some individuals are asymptomatic. Obstructive sleep apnea hypopnea can coexist with Cheyne-Stokes breathing, and thus snoring and abruptly terminated obstructive events may be observed during sleep.

Physical findings seen in individuals with a Cheyne-Stokes breathing pattern relate to its risk factors. Findings consistent with heart failure, such as jugular venous distension, S3 heart sound, lung crackles, and lower-extremity edema, may be present.

Prevalence

The prevalence of idiopathic central sleep apnea is unknown but thought to be rare. The prevalence of Cheyne-Stokes breathing is high in individuals with depressed cardiac ventricular ejection fraction. In individuals with an ejection fraction of < 45%, the prevalence has been reported to range from 15% to 44%. The gender ratio for prevalence in North America, Europe,

and Australia is even more highly skewed toward men than for obstructive sleep apnea hypopnea. Prevalence increases with age, and most individuals with the disorder are older than 60 years. Cheyne-Stokes breathing occurs in approximately 20% of individuals with acute stroke as assessed in Barcelona and Toronto. Central sleep apnea comorbid with opioid use occurs in approximately 24% of individuals taking opioids chronically for nonmalignant pain and similarly in individuals receiving methadone maintenance therapy as seen in several high-income countries. Higher opioid doses are associated with greater severity, especially at morphine-equivalent daily dosages > 200 mg. In children assessed in France and Canada, the prevalence ranges from 4% to 6%.

Development and Course

Polysomnography parameters for diagnosing central sleep apnea are different for children than for adults and comprise any of the following: 1) cessation of airflow and respiratory effort for more than 20 seconds, two breath cycles that are associated with an arousal from sleep, or > 3% oxygen desaturation; or 2) two breath cycles that are associated with bradycardia.

The onset of Cheyne-Stokes breathing appears tied to the development of heart failure. The Cheyne-Stokes breathing pattern is associated with oscillations in heart rate, blood pressure, and oxygen desaturation, and elevated sympathetic nervous system activity that can promote progression of heart failure. The clinical significance of Cheyne-Stokes breathing in the setting of stroke is not known, but Cheyne-Stokes breathing may be a transient finding that resolves with time after acute stroke. Central sleep apnea comorbid with opioid use has been documented with chronic use (i.e., several months).

Risk and Prognostic Factors

Genetic and physiological. Cheyne-Stokes breathing is frequently present in individuals with heart failure. The coexistence of atrial fibrillation further increases risk, as do older age and male sex. Cheyne-Stokes breathing is also seen in association with acute stroke and possibly renal failure. The underlying ventilatory instability in the setting of heart failure has been attributed to increased ventilatory chemosensitivity and hyperventilation due to pulmonary vascular congestion and circulatory delay. Central sleep apnea is seen in individuals taking long-acting opioids. In children, central sleep apnea can be found in

individuals with congenital abnormalities, particularly Arnold-Chiari malformation, or comorbid medical conditions such as gastroesophageal reflux. Rarely, central sleep apnea resulting from a congenital condition may not manifest until adulthood (e.g., Arnold-Chiari malformation and congenital central hypoventilation).

Diagnostic Markers

Polysomnography is used to characterize the breathing characteristics of each breathing-related sleep disorder subtype. Central sleep apneas are recorded when periods of breathing cessation for longer than 10 seconds occur. Cheyne-Stokes breathing is characterized by a pattern of periodic crescendo-decrescendo variation in tidal volume that results in central apneas and hypopneas

occurring at a frequency of at least five events per hour with the number of central apneas and hypopneas > 50% of the total number of apneas and hypopneas. The cycle length of Cheyne-Stokes breathing (or time from end of one central apnea to the end of the next apnea) is about 60 seconds.

Functional Consequences of Central Sleep Apnea

Idiopathic central sleep apnea has been reported to cause symptoms of disrupted sleep, including insomnia and sleepiness. Cheyne-Stokes breathing with comorbid heart failure has been associated with excessive sleepiness, fatigue, and insomnia, although many individuals may be asymptomatic. Coexistence of heart failure and Cheyne-Stokes breathing may be associated with increased cardiac arrhythmias and increased mortality or cardiac transplantation. Individuals with central sleep apnea comorbid with opioid use may present with symptoms of sleepiness or insomnia.

Differential Diagnosis

Idiopathic central sleep apnea must be distinguished from other breathing-related sleep disorders, other sleep disorders, and medical conditions and mental disorders that cause sleep fragmentation, sleepiness, and fatigue. This is achieved using polysomnography.

Other breathing-related sleep disorders and sleep disorders. Central sleep apnea can be distinguished from obstructive sleep apnea hypopnea by the presence of at least five central apneas per hour of sleep. These conditions may co-occur, but central sleep apnea is considered to predominate when central respiratory events are > 50% of the total number of respiratory events.

Cheyne-Stokes breathing can be distinguished from other mental disorders, including other sleep disorders, and other medical conditions that cause sleep fragmentation, sleepiness, and fatigue based on the presence of a predisposing condition (e.g., heart failure or stroke) and signs and polysomnographic evidence of the characteristic breathing pattern. Polysomnographic respiratory findings can help distinguish Cheyne-Stokes breathing from insomnia due to other medical conditions. For example, central sleep apnea due to high-altitude periodic breathing has a pattern that resembles Cheyne-Stokes breathing but has a shorter cycle time, occurs only at high altitude, and is not associated with heart failure.

Central sleep apnea comorbid with opioid use can be differentiated from other types of breathing-related sleep disorders based on the use of long-acting opioid medications in conjunction with polysomnographic evidence of central apneas and periodic or ataxic breathing. It can be distinguished from insomnia due to drug or substance use based on polysomnographic evidence of central sleep apnea.

Comorbidity

Central sleep apnea disorders are frequently present in users of long-acting opioids, such as methadone. Individuals taking these medications have a breathing-related sleep disorder that could contribute to sleep disturbances and symptoms such as sleepiness,

confusion, and depression. While the individual is asleep, breathing patterns such as central

apneas, periodic apneas, and ataxic breathing may be observed. Obstructive sleep apnea hypopnea may coexist with central sleep apnea, and features consistent with this condition can also be present (see “Obstructive Sleep Apnea Hypopnea” earlier in this chapter). Cheyne-Stokes breathing is more commonly observed in association with conditions that include heart failure, stroke, and renal failure and is seen more frequently in individuals with atrial fibrillation. Individuals with Cheyne-Stokes breathing are more likely to be older, to be male, and to have lower weight than individuals with obstructive sleep apnea hypopnea.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), includes eight subtypes of central sleep apnea (central sleep apnea with Cheyne-Stokes breathing, central apnea due to a medical disorder without Cheyne-Stokes breathing, central sleep apnea due to high-altitude periodic breathing, central sleep apnea due to a medication or substance, primary central sleep apnea, primary central sleep apnea of infancy, primary central sleep apnea of prematurity, and treatment-emergent central sleep apnea). As in DSM-5, most of these diagnoses require a frequency of 5 or more central events per hour of sleep. In addition, ICSD-3 criteria also require the presence of signs or symptoms (e.g., complaints of insomnia or daytime sleepiness). Central events must constitute at least 50% of the total number of apneas and hypopneas. Primary central sleep apnea of infancy and primary central sleep apnea of prematurity have their own distinct criteria sets that differ from adult forms of central sleep apnea.

Sleep-Related Hypoventilation

Diagnostic Criteria

- A. Polysomnography demonstrates episodes of decreased respiration associated with elevated CO₂ levels. (**Note:** In the absence of objective measurement of CO₂, persistent low levels of hemoglobin oxygen saturation unassociated with apneic/hypopneic events may indicate hypoventilation.)
- B. The disturbance is not better explained by another current sleep disorder.

Specify whether:

G47.34 Idiopathic hypoventilation: This subtype is not attributable to any readily identified condition.

G47.35 Congenital central alveolar hypoventilation: This subtype is a rare congenital disorder in which the individual typically presents in the perinatal period with shallow breathing, or cyanosis and apnea during sleep.

G47.36 Comorbid sleep-related hypoventilation: This subtype occurs as a consequence of a medical condition, such as a pulmonary disorder (e.g., interstitial lung disease, chronic obstructive pulmonary disease) or a neuromuscular or chest wall disorder (e.g., muscular dystrophies, postpolio syndrome, cervical spinal cord injury, kyphoscoliosis), or medications (e.g.,

benzodiazepines, opiates). It also occurs with obesity (obesity hypoventilation disorder), where it reflects a combination of increased work of breathing due to reduced chest wall compliance and ventilation-perfusion mismatch and variably reduced ventilatory drive. Such individuals usually are characterized by body mass index of greater than 30 and hypercapnia during wakefulness (with a $p\text{CO}_2$ of greater than 45), without other evidence of hypoventilation.

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Specify current severity:

Severity is graded according to the degree of hypoxemia and hypercarbia present during sleep and evidence of end organ impairment due to these abnormalities (e.g., right-sided heart failure). The presence of blood gas abnormalities during wakefulness is an indicator of greater severity.

Subtypes

Subtypes of sleep-related hypoventilation include the following:

- *Idiopathic hypoventilation*, also referred to as *idiopathic central alveolar hypoventilation*, is characterized by reduction of tidal volume and elevated CO_2 during sleep, in the absence of any identifiable comorbidity that would account for the hypoventilation.
- *Congenital central alveolar hypoventilation* is a rare disorder associated with mutation of the gene *PHOX2B*. It typically manifests at birth.
- *Comorbid sleep-related hypoventilation* is due to one of numerous potential comorbidities, including pulmonary disease (e.g., chronic obstructive pulmonary disease [COPD]), chest wall abnormalities (e.g., kyphoscoliosis), neuromuscular disease (e.g., amyotrophic lateral sclerosis), and obesity (referred to as obesity hypoventilation), as well as use of medications or substances, especially opioids.

Diagnostic Features

Sleep-related hypoventilation can occur independently or, more frequently, comorbid with medical or neurological disorders, medication use, or substance use disorder. Although symptoms are not mandatory to make this diagnosis, individuals often report excessive daytime sleepiness, frequent arousals and awakenings during sleep, morning headaches, and insomnia complaints.

Associated Features

Individuals with sleep-related hypoventilation can present with sleep-related complaints of insomnia or sleepiness. Episodes of orthopnea can occur in individuals with diaphragm weakness. Headaches upon awakening may be present. During sleep, episodes of shallow breathing may be observed, and obstructive sleep apnea hypopnea or central sleep apnea may coexist. Consequences of ventilatory insufficiency, including pulmonary hypertension, cor pulmonale (right heart failure), polycythemia, and neurocognitive dysfunction, can be present.

With progression of ventilatory insufficiency, blood gas abnormalities extend into wakefulness. Features of a medical condition causing sleep-related hypoventilation can also be present. Episodes of hypoventilation may be associated with frequent arousals or bradycardia. Individuals may complain of excessive sleepiness and insomnia or morning headaches or may present with findings of neurocognitive dysfunction or depression. Hypoventilation may not be present during wakefulness.

Prevalence

Idiopathic sleep-related hypoventilation in adults is very uncommon. The prevalence of congenital central alveolar hypoventilation is unknown, but the disorder is rare. Comorbid sleep-related hypoventilation (i.e., hypoventilation comorbid with other conditions, such as COPD, neuromuscular disorders, or obesity) is more common.

The prevalence of comorbid sleep-related hypoventilation due to obesity in the general population is estimated to be approximately 0.14%–0.6% based on national obesity rates and prevalence of obstructive sleep apnea across several countries. Increasing rates of obesity are associated with increasing prevalence of comorbid sleep-related hypoventilation

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due to obesity. In individuals referred to a sleep clinic who have a body mass index > 35 kg/m², prevalence may be as high as 42%.

Development and Course

Idiopathic sleep-related hypoventilation is thought to be a slowly progressive disorder of respiratory impairment. When sleep-related hypoventilation disorder occurs comorbidly with other disorders (e.g., COPD, neuromuscular disorders, obesity), disease severity reflects the severity of the underlying condition, and the disorder progresses as the condition worsens. Complications such as pulmonary hypertension, cor pulmonale, cardiac dysrhythmias, polycythemia, neurocognitive dysfunction, and worsening respiratory failure can develop with increasing severity of blood gas abnormalities.

Congenital central alveolar hypoventilation usually manifests at birth with shallow, erratic, or absent breathing. This disorder can also manifest during infancy, childhood, and adulthood because of variable penetrance of the *PHOX2B* mutation.

Risk and Prognostic Factors

Environmental. Ventilatory drive can be reduced in individuals who are using central nervous system depressants, including benzodiazepines, opiates, and alcohol.

Genetic and physiological. Idiopathic sleep-related hypoventilation is associated with reduced ventilatory drive due to a blunted chemoresponsiveness to CO₂ (reduced respiratory drive; i.e., “won’t breathe”), reflecting underlying neurological deficits in centers governing the control of ventilation. More commonly, sleep-related hypoventilation is comorbid with another medical condition, such as a pulmonary disorder, a neuromuscular or chest wall disorder, or hypothyroidism, or with use of medications (e.g., benzodiazepines, opiates). In these conditions,

the hypoventilation may be a consequence of increased work of breathing and/or impairment of respiratory muscle function (i.e., “can’t breathe”) or reduced respiratory drive.

Neuromuscular disorders influence breathing through impairment of respiratory motor innervation or respiratory muscle function. They include conditions such as amyotrophic lateral sclerosis, spinal cord injury, diaphragmatic paralysis, myasthenia gravis, Lambert-Eaton syndrome, toxic or metabolic myopathies, postpolio syndrome, and Charcot-Marie-Tooth syndrome.

Congenital central alveolar hypoventilation is a genetic disorder attributable to mutations of *PHOX2B*, a gene that is crucial for the development of the embryonic autonomic nervous system and neural crest derivatives. Children with congenital central alveolar hypoventilation show blunted ventilatory responses to hypercapnia, especially in non-rapid eye movement sleep.

Sex- and Gender-Related Diagnostic Issues

Gender distributions for sleep-related hypoventilation occurring in association with comorbid conditions reflect the gender distributions of the comorbid conditions. For example, COPD is more frequently present in men and with increasing age. Contrary to previous data, obesity hypoventilation is now thought to occur equally between genders, and in some studies there may be even a slightly greater prevalence in women.

Diagnostic Markers

Sleep-related hypoventilation is diagnosed using polysomnography, which demonstrates sleep-related hypoxemia and hypercapnia that is not better explained by another breathing-related sleep disorder. The documentation of 1) increased arterial $p\text{CO}_2$ levels to > 55 mmHg during sleep or 2) a ≥ 10 -mmHg increase in $p\text{CO}_2$ levels (to a level that also exceeds

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50 mmHg) during sleep in comparison to awake supine values, in each case exceeding 10 minutes’ duration, is the gold standard for diagnosis. However, obtaining arterial blood gas determinations during sleep is impractical, and non-invasive measures of $p\text{CO}_2$ have not been adequately validated during sleep and are not widely used during polysomnography in adults. Prolonged and sustained decreases in oxygen saturation (oxygen saturation of $< 90\%$ for more than 5 minutes with a nadir of at least 85%, or oxygen saturation of $< 90\%$ for at least 30% of sleep time) in the absence of evidence of upper airway obstruction are often used as an indication of sleep-related hypoventilation; however, this finding is not specific, as there are other potential causes of hypoxemia, such as that due to lung disease.

Children with congenital central alveolar hypoventilation are more likely to have disorders of the autonomic nervous system, Hirschsprung’s disease, neural crest tumors, and characteristic box-shaped face (i.e., the face is short relative to its width).

Functional Consequences of Sleep-Related Hypoventilation

The consequences of sleep-related hypoventilation are related to the effects of chronic exposure to hypercapnia and hypoxemia. These blood gas derangements cause vasoconstriction of the

pulmonary vasculature leading to pulmonary hypertension, which, if severe, can result in right-sided heart failure (cor pulmonale). Hypoxemia can lead to dysfunction of organs such as the brain, blood, and heart, leading to outcomes such as cognitive dysfunction, polycythemia, and cardiac arrhythmias. Hypercapnia can depress ventilatory drive, leading to progressive respiratory failure.

Differential Diagnosis

Other medical conditions affecting ventilation. In adults, the idiopathic variety of sleep-related hypoventilation is very uncommon and is determined by excluding the presence of lung diseases, skeletal malformations, neuromuscular disorders, and other medical and neurological disorders or medications that affect ventilation. Sleep-related hypoventilation must be distinguished from other causes of sleep-related hypoxemia, such as that due to lung disease.

Other breathing-related sleep disorders. Sleep-related hypoventilation can be distinguished from obstructive sleep apnea hypopnea and central sleep apnea based on clinical features and findings on polysomnography. Sleep-related hypoventilation typically shows more sustained periods of oxygen desaturation rather than the periodic episodes seen in obstructive sleep apnea hypopnea and central sleep apnea. Obstructive sleep apnea hypopnea and central sleep apnea also show a pattern of discrete episodes of repeated airflow reductions that can be absent in sleep-related hypoventilation. However, both obstructive and central apneas and hypopneas can occur in association with sleep-related hypoventilation. In obesity hypoventilation, most individuals will have comorbid obstructive sleep apnea.

Comorbidity

Sleep-related hypoventilation often occurs in association with a pulmonary disorder (e.g., interstitial lung disease, COPD), with a neuromuscular or chest wall disorder (e.g., muscular dystrophies, post-polio syndrome, cervical spinal cord injury, kyphoscoliosis), with obesity, or, most relevant to the clinician, with medication use (e.g., benzodiazepines, opiates). Congenital central alveolar hypoventilation often occurs in association with autonomic dysfunction and may occur in association with Hirschsprung's disease. COPD, a disorder of lower airway obstruction usually associated with cigarette smoking, can result in sleep-related hypoventilation and hypoxemia. The presence of coexisting obstructive sleep apnea

hypopnea is thought to exacerbate hypoxemia and hypercapnia during sleep and wakefulness. The relationship between congenital central alveolar hypoventilation and idiopathic sleep-related hypoventilation is unclear; in some individuals, idiopathic sleep-related hypoventilation may represent cases of late-onset congenital central alveolar hypoventilation.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), recognizes six subtypes of sleep hypoventilation disorders. Congenital central alveolar hypoventilation syndrome and idiopathic hypoventilation (idiopathic central alveolar hypoventilation in the ICSD-3) are identically classified in DSM-5 and ICSD-3. However, the ICSD-3 subtypes obesity

hypoventilation syndrome, sleep-related hypoventilation due to a medication or substance, and sleep-related hypoventilation due to a medical disorder are subsumed under comorbid sleep-related hypoventilation in DSM-5. The subtype late-onset central hypoventilation with hypothalamic dysfunction is not in DSM-5. The DSM-5 approach to classification reflects the frequent co-occurrence of disorders that lead to hypoventilation and hypoxemia. In contrast, the classification used in ICD-3 reflects evidence that there are distinct sleep-related pathogenetic processes leading to hypoventilation.

Circadian Rhythm Sleep-Wake Disorders

Diagnostic Criteria

- A. A persistent or recurrent pattern of sleep disruption that is primarily due to an alteration of the circadian system or to a misalignment between the endogenous circadian rhythm and the sleep-wake schedule required by an individual's physical environment or social or professional schedule.
- B. The sleep disruption leads to excessive sleepiness or insomnia, or both.
- C. The sleep disturbance causes clinically significant distress or impairment in social, occupational, and other important areas of functioning.

Specify whether:

G47.21 Delayed sleep phase type: A pattern of delayed sleep onset and awakening times, with an inability to fall asleep and awaken at a desired or conventionally acceptable earlier time.

Specify if:

Familial: A family history of delayed sleep phase is present.

Specify if:

Overlapping with non-24-hour sleep-wake type: Delayed sleep phase type may overlap with another circadian rhythm sleep-wake disorder, non-24-hour sleep-wake type.

G47.22 Advanced sleep phase type: A pattern of advanced sleep onset and awakening times, with an inability to remain awake or asleep until the desired or conventionally acceptable later sleep or wake times.

Specify if:

Familial: A family history of advanced sleep phase is present.

G47.23 Irregular sleep-wake type: A temporally disorganized sleep-wake pattern, such that the timing of sleep and wake periods is variable throughout the 24-hour period.

G47.24 Non-24-hour sleep-wake type: A pattern of sleep-wake cycles that is not synchronized to the 24-hour environment, with a consistent daily drift (usually to later and later times) of sleep onset and wake times.

G47.26 Shift work type: Insomnia during the major sleep period and/or excessive sleepiness (including inadvertent sleep) during the major awake period associated with a shift work schedule (i.e., requiring unconventional work hours).

G47.20 Unspecified type

Specify if:

Episodic: Symptoms last at least 1 month but less than 3 months.

Persistent: Symptoms last 3 months or longer.

Recurrent: Two or more episodes occur within the space of 1 year.

Delayed Sleep Phase Type

Diagnostic Features

The delayed sleep phase type is based primarily on a history of a delay in the timing of the major sleep period (usually more than 2 hours) in relation to the desired sleep and wake-up time, resulting in symptoms of insomnia and excessive sleepiness. When allowed to set their own schedule, individuals with delayed sleep phase type exhibit normal sleep quality and duration for age. Symptoms of sleep-onset insomnia, difficulty waking in the morning, and excessive sleepiness early in the day are prominent.

Associated Features

Common associated features of delayed sleep phase type include a history of mental disorders or a concurrent mental disorder. Extreme and prolonged difficulty awakening with morning confusion is also common. Insomnia disorder may develop as a result of maladaptive behaviors that impair sleep and increase arousal because of repeated attempts to fall asleep at an earlier time.

Prevalence

The prevalence of delayed sleep phase type is highest in adolescents and young adults, with rates estimated between 3.3% and 4.6% in Norway and Sweden. Studies of adult prevalence yield significantly lower rates, estimated to be 0.2%–1.7% in Norway and New Zealand. Although the prevalence of familial delayed sleep phase type has not been established, a family history of delayed sleep phase is often present in individuals with delayed sleep phase type.

Development and Course

Course is persistent, with intermittent exacerbations throughout adulthood in some individuals.

Although age at onset is variable, symptoms begin typically in adolescence and early adulthood and persist for several months to years before diagnosis is established. Severity may decrease with age. Relapse of symptoms is common.

Clinical expression may vary across the life span depending on social, school, and work obligations. Exacerbation is usually triggered by a change in work or school schedule that requires an early rise time. Individuals who can alter their work schedules to accommodate the delayed circadian sleep and wake timing can experience remission of symptoms.

Increased prevalence in adolescence may be a consequence of both physiological and behavioral factors. Hormonal changes may be involved specifically, as delayed sleep phase is associated with the onset of puberty. Thus, delayed sleep phase type in adolescents should be differentiated from the common delay in the timing of circadian rhythms in this age group. In the familial form, the course is persistent and may not improve significantly with age.

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Risk and Prognostic Factors

Genetic and physiological. Predisposing factors may include a longer than average circadian period, changes in light sensitivity, and impaired homeostatic sleep drive. Some individuals with delayed sleep phase type may be hypersensitive to evening light, which can serve as a delay signal to the circadian clock, or they may be hyposensitive to morning light such that its phase-advancing effects are reduced. Genetic factors may play a role in the pathogenesis of familial and sporadic forms of delayed sleep phase type. A study of unrelated families showing strong heritability of the disorder described a mutation in the clock gene, *CRY1*, occurring in about 0.6% of the population, which results in increased inhibition of transcription of the activator clock genes, *CLOCK* and *BMAL1*.

Diagnostic Markers

Confirmation of the diagnosis includes a complete history and use of a sleep diary or actigraph (i.e., a wrist-worn motion detector that monitors motor activity for prolonged periods; if measured for at least 7 days, motor activity can be used as a proxy for sleep-wake patterns). The period covered should include weekends, when social and occupational obligations are less strict, to ensure that the individual exhibits a consistently delayed sleep-wake pattern. The most commonly available laboratory-derived phase marker is salivary dim light melatonin onset (DLMO) time. However, not all individuals with diagnosed delayed sleep phase exhibit delayed DLMO. An investigation of rigorously diagnosed individuals found that only 57% exhibited physiological phase delays (as gauged by a DLMO time occurring subsequent to the desired bedtime), whereas the remaining 43% had DLMO times that occurred before the desired bedtimes. As noted above, behavior, rather than circadian physiological alteration, may play a more predominant role in the latter (earlier DLMO) group. Given this, phase markers may ultimately demonstrate more value for optimization of treatment timing and/or as a measure of treatment response.

Functional Consequences of Delayed Sleep Phase Type

Excessive early day sleepiness is prominent. Extreme and prolonged difficulty awakening with

morning confusion (i.e., sleep inertia) is also common. The severity of insomnia and excessive sleepiness symptoms varies substantially among individuals and largely depends on the occupational and social demands on the individual.

Differential Diagnosis

Normative variations in sleep. Delayed sleep phase type must be distinguished from “normal” sleep patterns in which a person has a late schedule that does not cause personal, social, or occupational distress (most commonly seen in adolescents and young adults).

Other sleep disorders. Insomnia disorder and other circadian rhythm sleep-wake disorders should be included in the differential. Excessive sleepiness may also be caused by other sleep disturbances, such as breathing-related sleep disorders, insomnias, restless legs syndrome, and medical, neurological, and mental disorders. Overnight polysomnography may help in evaluating for other comorbid sleep disorders, such as sleep apnea. The circadian nature of delayed sleep phase type, however, should differentiate it from other disorders with similar complaints.

Comorbidity

Delayed sleep phase type is associated with depressive disorders, personality disorders, somatic symptom disorder or illness anxiety disorder, obsessive-compulsive disorder, attention-deficit/hyperactivity disorder, and autism spectrum disorder. In addition, comorbid sleep disorders, such as insomnia disorder, restless legs syndrome, and sleep apnea, as

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well as depressive and bipolar disorders and anxiety disorders, can exacerbate symptoms of insomnia and excessive sleepiness. Delayed sleep phase type may overlap with another circadian rhythm sleep-wake disorder, non-24-hour sleep-wake type. Sighted individuals with non-24-hour sleep-wake type disorder commonly also have a history of delayed circadian sleep phase.

Advanced Sleep Phase Type

Specifiers

The presence of a family history of advanced sleep phase type may be indicated with the specifier “familial.” In the familial form, specific mutations demonstrate an autosomal dominant mode of inheritance, the course is persistent, and the severity of symptoms may increase with age. The prevalence of familial advanced sleep phase type has not been established.

Diagnostic Features

Advanced sleep phase type is characterized by sleep-wake times that are several hours earlier than desired or conventional times. Diagnosis is based primarily on a history of an advance in the timing of the major sleep period (usually more than 2 hours) in relation to the desired sleep and wake-up time, with symptoms of early-morning insomnia and excessive daytime sleepiness. When allowed to set their schedule, individuals with advanced sleep phase type will exhibit normal sleep quality and duration for age.

Associated Features

Individuals with advanced sleep phase type are “morning types,” having earlier sleep-wake times, with the timing of circadian biomarkers such as melatonin and core body temperature rhythms occurring 2–4 hours earlier than normal. When required to keep a conventional schedule requiring a delay of bedtime, these individuals will continue to have an early rise time, leading to persistent sleep deprivation and daytime sleepiness. Use of hypnotics or alcohol to combat sleep-maintenance insomnia and stimulants to reduce daytime sleepiness may lead to substance abuse in these individuals.

Prevalence

The estimated prevalence of advanced sleep phase type is approximately 1% in middle-age adults in the United States. Sleep-wake times and circadian phase advance in older individuals probably account for the higher prevalence in this population.

Development and Course

Onset is usually in late adulthood, although in the familial form, onset can be earlier (during childhood or early adulthood). The course is typically persistent, lasting more than 3 months, but the severity may increase depending on work and social schedules. The advanced sleep phase type is more common in older adults.

Clinical expression may vary across the life span depending on social, school, and work obligations. Individuals who can alter their work schedules to accommodate the advanced circadian sleep and wake timing can experience remission of symptoms. Increasing age tends to advance the sleep phase. However, it is unclear whether the common age-associated advanced sleep phase type is attributable solely to a change in circadian timing (as seen in the familial form) or also to age-related changes in the homeostatic regulation of sleep, resulting in earlier awakening. Severity, remission, and relapse of symptoms are dependent on adherence to behavioral and environmental treatments designed to control sleep and wake structure and light exposure.

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Risk and Prognostic Factors

Environmental. Decreased late afternoon/early evening exposure to light and/or increased exposure to early-morning light because of early-morning awakening can increase the risk of advanced sleep phase type by advancing circadian rhythms. By going to bed early, these individuals are not exposed to light in the phase delay region of the curve, resulting in perpetuation of advanced phase. In familial advanced sleep phase type, a shortening of the endogenous circadian period can result in an advanced sleep phase, although circadian period does not appear to systematically decrease with age.

Genetic and physiological. Advanced sleep phase type has demonstrated an autosomal dominant mode of inheritance, including a *PER2* gene mutation causing hypophosphorylation of the PER2 protein and a missense mutation in *CKI*.

Diagnostic Markers

A sleep diary and actigraphy are used as diagnostic markers, as described earlier for delayed sleep phase type.

Functional Consequences of Advanced Sleep Phase Type

Excessive sleepiness associated with advanced sleep phase can have a negative effect on cognitive performance, social interaction, and safety. Use of wake-promoting agents to combat sleepiness later in the day or sedatives to inhibit early-morning awakening may increase potential for substance abuse.

Differential Diagnosis

Normal variations in sleep. Behavioral factors such as irregular sleep schedules, voluntary early awakening, and exposure to light in the early morning should be considered, particularly in older adults.

Other disorders that cause early-morning awakening. Careful attention should be taken to rule out other sleep-wake disorders (e.g., insomnia disorder), other mental disorders (e.g., depressive disorders, bipolar disorders), and medical conditions that can cause early-morning awakening.

Comorbidity

Repetitive attempts to resume sleep and the development of maladaptive cognitions and sleep-related behaviors may result in the development of a comorbid insomnia disorder that requires clinical attention.

Irregular Sleep-Wake Type

Diagnostic Features

Irregular sleep-wake type is characterized by a lack of discernible sleep-wake circadian rhythm. The diagnosis of irregular sleep-wake type is based primarily on a history of symptoms of insomnia at night (during the usual sleep period) and excessive sleepiness (napping) during the day. There is no major sleep period, and sleep and wake periods across 24 hours are fragmented, with sleep fragmented into at least three periods during the 24-hour day. The longest sleep period tends to occur between 2:00 A.M. and 6:00 A.M. and is usually < 4 hours.

Associated Features

A history of isolation or reclusion may occur in association with the disorder and contribute to the symptoms via a lack of external stimuli to help entrain a normal pattern.

Individuals or their caregivers report frequent naps throughout the day. Irregular sleep-wake type is most commonly associated with neurodegenerative disorders, such as major neurocognitive disorder, and many neurodevelopmental disorders in children.

Prevalence

Prevalence of irregular sleep-wake type in the general population is unknown.

Development and Course

The course of irregular sleep-wake type is persistent. Age at onset is variable, but the disorder is more common in older adults.

Risk and Prognostic Factors

Environmental. Decreased exposure to environmental light and structured daytime activity can be associated with a low-amplitude circadian rhythm. Hospitalized individuals are especially prone to such weak external entraining stimuli, and even outside the hospital setting, individuals with major neurocognitive disorder are exposed to significantly less bright light.

Genetic and physiological. Neurodegenerative disorders, such as Alzheimer's disease, Parkinson's disease, and Huntington's disease, and neurodevelopmental disorders in children increase the risk for irregular sleep-wake type.

Diagnostic Markers

A detailed sleep history and a sleep diary (by a caregiver) or actigraphy help confirm the irregular sleep-wake pattern.

Functional Consequences of Irregular Sleep-Wake Type

Lack of a clearly discernible major sleep and wake period in irregular sleep-wake type results in insomnia or excessive sleepiness at irregular times of the day. Disruption of the caregiver's sleep also often occurs and is an important consideration.

Differential Diagnosis

Normative variations in sleep. Irregular sleep-wake type should be distinguished from a voluntary irregular sleep-wake schedule and poor sleep hygiene, which can result in insomnia and excessive sleepiness.

Other medical conditions and mental disorders. Other causes of insomnia and daytime sleepiness, including comorbid medical conditions and mental disorders or medication, should be considered.

Comorbidity

Irregular sleep-wake type is often comorbid with neurodegenerative and neurodevelopmental disorders, such as major neurocognitive disorder, intellectual developmental disorder (intellectual disability), and traumatic brain injury. It is also comorbid with other medical conditions and mental disorders in which there is social isolation and/or lack of light and structured activities.

Non-24-Hour Sleep-Wake Type

Diagnostic Features

The diagnosis of non-24-hour sleep-wake type is based primarily on a history of symptoms of insomnia or excessive sleepiness related to abnormal synchronization between the 24-hour light-dark cycle and the endogenous circadian rhythm. Individuals typically present with periods of insomnia, excessive sleepiness, or both, which alternate with short

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asymptomatic periods. Starting with the asymptomatic period, when the individual's sleep phase is aligned to the external environment, sleep latency will gradually increase and the individual will complain of sleep-onset insomnia. As the sleep phase continues to drift so that sleep time is now in the daytime, the individual will have trouble staying awake during the day and will complain of sleepiness. Because the circadian period is not aligned to the external 24-hour environment, symptoms will depend on when an individual tries to sleep in relation to the circadian rhythm of sleep propensity.

Associated Features

Non-24-hour sleep-wake type is most common among blind or visually impaired individuals who have decreased light perception. In sighted individuals, there is often a history of delayed sleep phase and of decreased exposure to light and structured social and physical activity. Sighted individuals with non-24-hour sleep-wake type also demonstrate increased sleep duration.

Prevalence

Prevalence of non-24-hour sleep-wake type in the general population is unclear, but the disorder appears rare in sighted individuals. The prevalence in blind individuals in the United States is estimated to be 50%.

Development and Course

Course of non-24-hour sleep-wake type is persistent, with intermittent remission and exacerbations as a result of changes in work and social schedules throughout the life span. Age at onset is variable, depending on the onset of visual impairment. In sighted individuals, because of the overlap with delayed sleep phase type, non-24-hour sleep-wake type may develop in adolescence or early adulthood. Remission and relapse of symptoms in blind and sighted individuals largely depend on adherence to treatments designed to control sleep and wake structure and light exposure.

Clinical expression may vary across the life span depending on social, school, and work obligations. In adolescents and adults, irregular sleep-wake schedules and exposure to light or lack of light at critical times of the day can exacerbate the effects of sleep loss and disrupt circadian entrainment. Consequently, symptoms of insomnia, daytime sleepiness, and school, occupational, and interpersonal functioning may worsen.

Risk and Prognostic Factors

Environmental. In sighted individuals, decreased exposure or sensitivity to light and social and

physical activity cues may contribute to a free-running circadian rhythm. With the high frequency of mental disorders involving social isolation and cases of non-24-hour sleep-wake type developing after a change in sleep habits (e.g., night shift work, job loss), behavioral factors in combination with physiological tendency may precipitate and perpetuate this disorder in sighted individuals. Hospitalized individuals with neurological and psychiatric disorders can become insensitive to social cues, predisposing them to the development of non-24-hour sleep-wake type.

Genetic and physiological. Blindness is a risk factor for non-24-hour sleep-wake type. Non-24-hour sleep-wake type has been associated with traumatic brain injury.

Diagnostic Markers

Diagnosis is confirmed by history and sleep diary or actigraphy for an extended period. Sequential measurement of phase markers (e.g., melatonin) can help determine circadian phase in both sighted and blind individuals.

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Functional Consequences of Non-24-Hour Sleep-Wake Type

Complaints of insomnia (sleep onset and sleep maintenance), excessive sleepiness, or both are prominent. The unpredictability of sleep and wake times (typically a daily delay drift) results in difficulty attending school or maintaining a steady job and may increase potential for social isolation.

Differential Diagnosis

Other circadian rhythm sleep-wake disorders. In sighted individuals, non-24-hour sleep-wake type should be differentiated from delayed sleep phase type, as individuals with delayed sleep phase type may display a similar progressive delay in sleep period for several days.

Depressive disorders. Depressive disorders may result in similar circadian dysregulation and symptoms.

Comorbidity

Blindness is often comorbid with non-24-hour sleep-wake type, as are depressive and bipolar disorders with social isolation.

Shift Work Type

Diagnostic Features

Diagnosis is primarily based on a history of the individual working outside of the normal 8:00 A.M. to 6:00 P.M. daytime window (particularly at night) on a regularly scheduled (i.e., non-overtime) basis. Symptoms of excessive sleepiness at work, and impaired sleep at home, on a persistent basis are prominent. Presence of both sets of symptoms are usually required for a diagnosis of shift work type. Typically, when the individual reverts to a day-work routine,

symptoms resolve.

Prevalence

The prevalence of shift work type is unclear, but the disorder is estimated to affect 5%–10% of the night worker population in the United States (16%–20% of the workforce). Prevalence rises with advancement into middle age and beyond.

Development and Course

Shift work type can appear in individuals of any age but is more prevalent in individuals older than 50 years and typically worsens with the passage of time if the disruptive work schedule persists. Although older adults may show similar rates of circadian phase adjustment to a change in routine as do younger adults, they appear to experience significantly more sleep disruption as a consequence of the circadian phase shift.

Risk and Prognostic Factors

Temperamental. Predisposing factors include a morning-type disposition and a need for long (i.e., more than 8 hours) sleep durations in order to feel well rested.

Environmental. Trying to balance strong competing social and domestic needs (e.g., in parents of young children) can lead to the development of the shift work type. Persons who are able to commit to a nocturnal lifestyle, with few competing day-oriented demands, appear at lower risk for shift work type.

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Genetic and physiological. Because shift workers are more likely than day workers to be obese, obstructive sleep apnea may be present and may exacerbate the symptoms.

Diagnostic Markers

A history and sleep diary or actigraphy may be useful in diagnosis, as discussed earlier for delayed sleep phase type.

Functional Consequences of Shift Work Type

Individuals with shift work type not only may perform poorly at work but also appear to be at risk for accidents both at work and on the drive home. Individuals with a history of bipolar disorder are particularly vulnerable to shift work type–related episodes of mania resulting from missed nights of sleep. Shift work type often results in interpersonal problems.

Differential Diagnosis

Normative variations in sleep with shift work. The diagnosis of shift work type, as opposed to the “normal” difficulties of shift work, depends to some extent on the severity of symptoms and/or level of distress experienced by the individual.

Other sleep disorders. The presence of shift work type symptoms even when the individual is able

to live on a day-oriented routine for several weeks at a time may suggest the presence of other sleep disorders, such as sleep apnea, insomnia, and narcolepsy, which should be ruled out.

Jet lag. Individuals who travel across many time zones on a very frequent basis may experience effects similar to those experienced by individuals with shift work type who work rotating shifts. The distinction should be clear, based on the travel history.

Comorbidity

Shift work type has been associated with increased alcohol use disorder, other substance use disorders, and depression. A variety of physical health disorders (e.g., gastrointestinal disorders, cardiovascular disease, diabetes, cancer) have been found to be associated with prolonged exposure to shift work.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition, circadian rhythm sleep-wake disorders closely parallel DSM-5 but also include jet lag type.

Parasomnias

Parasomnias are disorders characterized by abnormal behavioral, experiential, or physiological events occurring in association with sleep, specific sleep stages, or sleep-wake transitions. The most common parasomnias are non-rapid eye movement (NREM) sleep arousal disorders and rapid eye movement (REM) sleep behavior disorder. These conditions each have distinct pathophysiology, clinical characteristics, and prognostic and therapeutic considerations discussed in the following sections specific to each disorder.

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Non–Rapid Eye Movement Sleep Arousal Disorders

Diagnostic Criteria

- A. Recurrent episodes of incomplete awakening from sleep, usually occurring during the first third of the major sleep episode, accompanied by either one of the following:
 - 1. **Sleepwalking:** Repeated episodes of rising from bed during sleep and walking about. While sleepwalking, the individual has a blank, staring face; is relatively unresponsive to the efforts of others to communicate with him or her; and can be awakened only with great difficulty.
 - 2. **Sleep terrors:** Recurrent episodes of abrupt terror arousals from sleep,

usually beginning with a panicky scream. There is intense fear and signs of autonomic arousal, such as mydriasis, tachycardia, rapid breathing, and sweating, during each episode. There is relative unresponsiveness to efforts of others to comfort the individual during the episodes.

- B. No or little (e.g., only a single visual scene) dream imagery is recalled.
- C. Amnesia for the episodes is present.
- D. The episodes cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- E. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication).
- F. Coexisting mental disorders and medical conditions do not explain the episodes of sleepwalking or sleep terrors.

Specify whether:

F51.3 Sleepwalking type

Specify if:

With sleep-related eating

With sleep-related sexual behavior (sexsomnia)

F51.4 Sleep terror type

Diagnostic Features

The essential feature of non-rapid eye movement (NREM) sleep arousal disorders is the repeated occurrence of incomplete arousals, usually beginning during the first third of the major sleep episode (Criterion A), that typically are brief, lasting 1–10 minutes, but may be protracted, lasting up to 1 hour. The maximum duration of an event is unknown. The eyes are typically open during these events. Many individuals exhibit both subtypes of arousal (i.e., sleepwalking type and sleep terror type) on different occasions, which underscores the unitary underlying pathophysiology. The subtypes reflect varying degrees of simultaneous occurrence of wakefulness and NREM sleep, resulting in complex behaviors arising from sleep with varying degrees of conscious awareness, motor activity, and autonomic activation.

The essential feature of *sleepwalking* is repeated episodes of complex motor behavior initiated during sleep, including rising from bed and walking about (Criterion A1). Sleepwalking episodes begin during any stage of NREM sleep, most commonly during slow-wave sleep and therefore most often occurring during the first third of the night. During episodes, the individual has reduced alertness and responsiveness, a blank stare, and relative unresponsiveness to communication with others or efforts by others to awaken the individual. If awakened during the episode (or on awakening the following morning), the

individual has limited recall for the episode. After the episode, there may initially be a brief period of confusion or difficulty orienting, followed by full recovery of cognitive function and appropriate behavior.

The essential feature of *sleep terrors* is the repeated occurrence of precipitous awakenings from sleep, usually beginning with a panicky scream or cry (Criterion A2). Sleep terrors usually begin during the first third of the major sleep episode and last 1–10 minutes, but they may last considerably longer, particularly in children. The episodes are accompanied by impressive autonomic arousal and behavioral manifestations of intense fear. During an episode, the individual is difficult to awaken or comfort. If the individual awakens after the sleep terror, little or none of the dream, or only fragmentary, single images, are recalled. During a typical episode of sleep terrors, the individual abruptly sits up in bed screaming or crying, with a frightened expression and autonomic signs of intense anxiety (e.g., tachycardia, rapid breathing, sweating, dilation of the pupils). The individual may be inconsolable and is usually unresponsive to the efforts of others to awaken or comfort him or her. Sleep terrors are also called “night terrors” or “pavor nocturnus.”

For both subtypes of NREM sleep arousal disorders, the determination of “disorder” depends on a number of factors, which may vary on an individual basis and will depend on the frequency of events, potential for violence or injurious behaviors, embarrassment, or disruption/distress of other household members. Severity determination is best made based on the nature or consequence of the behaviors rather than simply on frequency.

Associated Features

Sleepwalking episodes can include a wide variety of behaviors. Episodes may begin with confusion: the individual may simply sit up in bed, look about, or pick at the blanket or sheet. This behavior then becomes progressively complex. The individual may actually leave the bed and walk into closets, out of the room, and even out of buildings. Individuals may use the bathroom, eat, talk, or engage in more complex behaviors. Running and frantic attempts to escape some apparent threat can also occur. Most behaviors during sleepwalking episodes are routine and of low complexity. However, cases of unlocking doors and even operating machinery (driving an automobile) have been reported. Sleepwalking can also include inappropriate behavior (e.g., commonly, urinating in a closet or wastebasket). Most episodes last for several minutes to a half hour but may be more protracted. Inasmuch as sleep is a state of relative analgesia, painful injuries sustained during sleepwalking may not be appreciated until awakening after the fact.

There are two “specialized” forms of sleepwalking: sleep-related eating behavior and sleep-related sexual behavior (sexsomnia or sleep sex). Individuals with *sleep-related eating* experience unwanted recurrent episodes of eating with varying degrees of amnesia, ranging from no awareness to full awareness without the ability to avoid or stop eating. During these episodes, inappropriate foods or even nonfood items (i.e., candy wrappers, small food boxes, or even small toys) may be ingested. Individuals with sleep-related eating disorder may find evidence of their eating only the next morning. In *sexsomnia*, varying degrees of sexual activity (e.g., masturbation, fondling, groping, sexual intercourse) occur as complex behaviors arising from sleep without conscious awareness. This condition is more common in males and may result in serious interpersonal relationship problems or medicolegal consequences.

During a typical episode of sleep terrors, there is often a sense of overwhelming dread, with a compulsion to escape. Although fragmentary vivid dream images may occur, a storylike dream sequence (as in nightmares) is not reported. Most commonly, the individual does not awaken

fully, but returns to sleep and has amnesia for the episode on awakening the next morning. Usually only one episode will occur on any one night. Occasionally, several episodes may occur at intervals throughout the night. These events rarely arise during daytime naps.

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Prevalence

Isolated or infrequent NREM sleep arousal behaviors are very common in the general population worldwide. From 10% to 30% of children have had at least one episode of sleepwalking, and the cross-national 12-month prevalence rate for sleepwalking in children is approximately 5%. The prevalence of sleepwalking episodes (not sleepwalking disorder) is estimated to be 12%–14.5% of children in Canada and 1.0%–7.0% among adults in the United Kingdom, with weekly to monthly episodes occurring in just 0.5%–0.7% of adults. Estimates for the lifetime prevalence of sleepwalking overall range from approximately 6.9% to 29.2% around the world, with a past-year prevalence of sleepwalking of 1.5%–3.6% in adults.

The prevalence of sleep terror disorder in the general population is unknown. The prevalence of sleep terror episodes (as opposed to sleep terror disorder, in which there is recurrence and distress or impairment) is approximately 34.4%–36.9% at 18 months of age and 19.7% at 30 months of age in Canadian toddlers, and 2.2% in Canadian and British adults.

Development and Course

NREM sleep arousal disorders occur most commonly in childhood and diminish in frequency with increasing age. Sleepwalking and sleep terrors are frequently outgrown following infancy and childhood and become less frequent by adolescence, with remission rates between 50% and 65%; for individuals ages 10–18 years, frequency is reported at 1.1% for sleepwalking and 0.6% for sleep terrors.

Violent or sexual activity during sleepwalking episodes is more likely to occur in adults. The onset of sleepwalking in adults with no history of sleepwalking as children should prompt a search for specific etiologies, such as obstructive sleep apnea, nocturnal seizures, or effect of medication.

Older children and adults may provide a more detailed recollection of fearful images associated with sleep terrors than do younger children, who are more likely to have complete amnesia or report only a vague sense of fear.

Risk and Prognostic Factors

Environmental. Sedative use, sleep deprivation, sleep-wake schedule disruptions, fatigue, and physical or emotional stress increase the likelihood of episodes. Fever and sleep deprivation can produce an increased frequency of NREM sleep arousal disorders.

Genetic and physiological. A family history of sleepwalking or sleep terrors may occur in up to 80% of individuals who sleepwalk. The risk for sleepwalking is further increased (to as much as 60% of offspring) when both parents have a history of the disorder. Familial aggregation of sleep terrors and sleepwalking has been described, as parental sleepwalking history predicts incident and persistent sleep terrors in their offspring. Individuals with sleep terrors frequently have a

positive family history of either sleep terrors or sleepwalking, with as high as a 10-fold increase in the prevalence of the disorder among first-degree biological relatives. Sleep terrors are much more common in monozygotic twins as compared with dizygotic twins. The exact mode of inheritance is unknown.

Sex- and Gender-Related Diagnostic Issues

Eating during sleepwalking episodes is more commonly seen in women. Sleepwalking occurs more often in girls during childhood but more often in men during adulthood.

Among children, sleep terrors are more common in boys than in girls. Among adults, they are equally common in men and women.

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Diagnostic Markers

NREM sleep arousal disorders arise from any stage of NREM sleep but most commonly from deep NREM sleep (slow-wave sleep). They are most likely to appear in the first third of the night and do not commonly occur during daytime naps. During the episode, the polysomnogram may be obscured with movement artifact. In the absence of such artifact, the electroencephalogram (EEG) may show a variety of patterns, including continuation of rhythmic delta activity into awakening, indicating partial or incomplete arousal; or alternatively, theta, alpha, or mixed frequency EEG activity may be seen during the episode, with frequent mixed slow/mixed frequency EEG arousals during slow-wave sleep being more common in individuals with NREM sleep arousal disorders than in control subjects. In distinction to an epileptic seizure, NREM sleep parasomnia disorders of arousal do not show features of spatiotemporal evolution of EEG rhythms during the episode.

Polysomnography in conjunction with audiovisual monitoring can be used to document episodes of sleepwalking. In the absence of actually capturing an event during a polysomnographic recording, there are no reliable polysomnographic features that can serve as a marker for sleepwalking. Sleep deprivation may increase the likelihood of capturing an event. As a group, individuals who sleepwalk show instability of deep NREM sleep, but the overlap in findings with persons who do not sleepwalk is great enough to preclude use of this indicator in establishing a diagnosis. Unlike arousals from REM sleep associated with nightmares, in which there is an increase in heart rate and respiration prior to the arousal, the NREM sleep arousals of sleep terrors begin precipitously from sleep, without anticipatory autonomic changes. The arousals are associated with impressive autonomic activity, with doubling or tripling of the heart rate. The pathophysiology is poorly understood, but there appears to be instability in the deeper stages of NREM sleep. Aside from capturing an event during a formal sleep study, there are no reliable polysomnographic indicators of the tendency to experience sleep terrors.

Functional Consequences of Non–Rapid Eye Movement Sleep Arousal Disorders

For the diagnosis of an NREM sleep arousal disorder to be made, the individual or household members must experience clinically significant distress or impairment, although such symptoms may occur occasionally in nonclinical populations and would be subthreshold for the diagnosis.

Embarrassment concerning the episodes can impair social relationships. Social isolation or occupational difficulties can result. Uncommonly, NREM sleep arousal disorders may result in serious injury to the individual or to someone trying to console the individual. Injuries to others are confined to those in close proximity; individuals are not “sought out.” For individuals with sleep-related eating behaviors, unknowingly preparing or eating food during the sleep period may create problems such as poor diabetes control, weight gain, injury (cuts and burns), or consequences of eating dangerous or toxic inedibles. NREM sleep arousal disorders may rarely result in violent or injurious behaviors with forensic implications.

Differential Diagnosis

Nightmare disorder. In contrast to individuals with NREM sleep arousal disorders, individuals with nightmare disorder typically awaken easily and completely, report vivid storylike dreams accompanying the episodes, and tend to have episodes later in the night. NREM sleep arousal disorders occur during NREM sleep, whereas nightmares usually occur during REM sleep. Parents of children with NREM sleep arousal disorders may misinterpret reports of fragmentary imagery as nightmares.

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Breathing-related sleep disorders. Breathing disorders during sleep can also produce confusional arousals with subsequent amnesia. However, breathing-related sleep disorders are also characterized by symptoms of snoring, breathing pauses, and daytime sleepiness. In some individuals, a breathing-related sleep disorder may precipitate episodes of sleepwalking.

REM sleep behavior disorder. REM sleep behavior disorder may be difficult to distinguish from NREM sleep arousal disorders. REM sleep behavior disorder is characterized by episodes of prominent, complex movements, often involving personal injury arising from sleep. In contrast to NREM sleep arousal disorders, REM sleep behavior disorder occurs during REM sleep. Individuals with REM sleep behavior disorder may be awakened easily during an episode and report more detailed and vivid dream content than do individuals with NREM sleep arousal disorders. These individuals and/or their bed partners often report that they “act out dreams.”

Parasomnia overlap syndrome. Parasomnia overlap syndrome consists of clinical and polysomnographic features of both sleepwalking and REM sleep behavior disorder.

Sleep-related seizures. Some types of seizures can produce episodes of very unusual behaviors that occur predominantly or exclusively during sleep. Nocturnal seizures may closely mimic NREM sleep arousal disorders but tend to be more stereotypic in nature, occur multiple times nightly, and be more likely to occur from daytime naps. In addition, seizures may arise from wakefulness, which does not occur with NREM sleep arousal disorders. The presence of sleep-related seizures does not preclude the presence of NREM sleep arousal disorders. When recurrent, sleep-related seizures are considered to be a form of epilepsy.

Alcohol-induced blackouts. Alcohol-induced blackouts may be associated with extremely complex behaviors in the absence of other suggestions of intoxication. They do not involve the loss of consciousness but rather reflect an isolated disruption of memory for events during a drinking episode. By history, these behaviors may be indistinguishable from those seen in NREM sleep

arousal disorders.

Dissociative amnesia, with dissociative fugue. Dissociative fugue may be extremely difficult to distinguish from sleepwalking. Unlike all other parasomnias, nocturnal dissociative fugue arises from a period of wakefulness during sleep, rather than precipitously from sleep without intervening wakefulness. A history of recurrent childhood physical or sexual abuse is usually present (but may be difficult to obtain).

Malingering or other voluntary behavior. As with dissociative fugue, malingering or other voluntary behavior occurs during wakefulness.

Panic disorder. Panic attacks may also cause abrupt awakenings from deep NREM sleep accompanied by fearfulness, but these episodes produce rapid and complete awakening without the confusion, amnesia, or motor activity typical of NREM sleep arousal disorders.

Medication-induced complex behaviors. Behaviors similar to those in NREM sleep arousal disorders can be induced by use of, or withdrawal from, substances or medications (e.g., benzodiazepines, nonbenzodiazepine sedative-hypnotics, opiates, cocaine, nicotine, antipsychotics or other dopamine receptor–blocking agents, tricyclic antidepressants, chloral hydrate). Such behaviors may arise from the sleep period and may be extremely complex. The underlying pathophysiology appears to be a relatively isolated amnesia. In such cases, substance/medication-induced sleep disorder, parasomnia type, should be diagnosed (see “Substance/Medication-Induced Sleep Disorder” later in this chapter).

Night eating syndrome. In contrast to the sleep-related eating form of sleepwalking, which is characterized by recurrent episodes of eating during incomplete arousals from sleep, night eating syndrome is considered to be an abnormality in the circadian rhythm of meal timing, with a normal circadian timing of sleep onset in which the individual wakes up in the middle of the night and overeats.

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Comorbidity

Typically, sleepwalking in both children and adults is not associated with significant mental disorders. However, in adults, there is an association between sleepwalking and major depressive episodes and obsessive-compulsive disorder. Children or adults with sleep terrors may have elevated scores for depression and anxiety on personality inventories.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition, includes “confusional arousal, sleep terrors, and sleepwalking” as NREM sleep arousal disorders.

Nightmare Disorder

Diagnostic Criteria

F51.5

- A. Repeated occurrences of extended, extremely dysphoric, and well-remembered dreams that usually involve efforts to avoid threats to survival, security, or physical integrity and that generally occur during the second half of the major sleep episode.
- B. On awakening from the dysphoric dreams, the individual rapidly becomes oriented and alert.
- C. The sleep disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- D. The nightmare symptoms are not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication).
- E. Coexisting mental disorders and medical conditions do not adequately explain the predominant complaint of dysphoric dreams.

Specify if:

During sleep onset

Specify if:

With mental disorder, including substance use disorders

With medical condition

With another sleep disorder

Coding note: The code F51.5 applies to all three specifiers. Code also the relevant associated mental disorder, medical condition, or other sleep disorder immediately after the code for nightmare disorder in order to indicate the association.

Specify if:

Acute: Duration of period of nightmares is 1 month or less.

Subacute: Duration of period of nightmares is greater than 1 month but less than 6 months.

Persistent: Duration of period of nightmares is 6 months or greater.

Specify current severity:

Severity can be rated by the frequency with which the nightmares occur:

Mild: Less than one episode per week on average.

Moderate: One or more episodes per week but less than nightly.

Severe: Episodes nightly.

Recording Procedures

The specifiers “with mental disorder, including substance use disorders”; “with medical condition”; and “with another sleep disorder” are available to allow the clinician to note clinically relevant comorbidities. In such cases, record F51.5 nightmare disorder with [name of

comorbid condition(s) or disorder(s)] followed by the diagnostic code(s) for the comorbid conditions or disorders (e.g., F51.5 nightmare disorder with moderate alcohol use disorder and rapid eye movement sleep behavior disorder; F10.20 moderate alcohol use disorder; G47.52 REM sleep behavior disorder).

Diagnostic Features

Nightmares are typically lengthy, elaborate, story-like sequences of dream imagery that seem real and that incite anxiety, fear, or other dysphoric emotions. Nightmare content typically focuses on attempts to avoid or cope with imminent danger but may involve themes that evoke other negative emotions. Nightmares occurring after traumatic experiences may replicate the threatening situation (“replicative nightmares”), but most do not. On awakening, nightmares are well remembered and can be described in detail. They arise almost exclusively during REM sleep and can thus occur throughout sleep but are more likely in the second half of the major sleep episode when dreaming is longer and more intense. Factors that increase early-night REM intensity, such as sleep fragmentation or deprivation, jet lag, and medications that affect REM sleep, might facilitate nightmares earlier in the night, including at sleep onset.

Nightmares usually terminate with awakening and rapid return of full alertness. However, the dysphoric emotions may persist into wakefulness and contribute to difficulty returning to sleep and lasting daytime distress. Some nightmares, known as “bad dreams,” may not induce awakening and are recalled only later. If nightmares occur during sleep-onset REM periods (*hypnagogic*), the dysphoric emotion is frequently accompanied by an awakening and being unable to move voluntarily (*sleep paralysis*), which may also occur in isolation without a preceding dream or nightmare.

Associated Features

Mild autonomic arousal, including sweating, tachycardia, and tachypnea, may characterize nightmares. Body movements and vocalizations are not characteristic because of REM sleep–related loss of skeletal muscle tone. When talking or emoting occurs in nightmare disorder, the vocal or motor behaviors are typically brief events terminating the nightmare. Distinct from such motor or vocal activity, true dream enactment behavior may occur when there is a loss of normal REM atonia (REM sleep behavior disorder).

Prevalence

Prevalence of nightmares during childhood is approximately 1%–5%. From 1.3% to 3.9% of parents report that their preschool children have nightmares “often” or “always.” Prevalence increases to 5.2% in children ages 5–15 years. Family history of nightmares, parasomnia symptoms, and daytime consequences of temper outbursts/mood disturbance and poor academic performance are associated with frequent nightmares during childhood and adolescence, with comorbid insomnia seen in approximately 20% of children with frequent nightmares. Among adults, prevalence of nightmares at least monthly is 6%. Among adults in several countries, prevalence of weekly nightmares is 2%–6%, whereas prevalence of frequent nightmares is 1%–5%. Estimates often combine idiopathic and posttraumatic nightmares indiscriminately.

Development and Course

Nightmares often begin between ages 3 and 6 years but reach a peak prevalence and severity in late adolescence or early adulthood. Nightmares most likely appear in children

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exposed to acute or chronic psychosocial stressors and thus may not resolve spontaneously. In a minority, frequent nightmares persist into adulthood, becoming virtually a lifelong disturbance. Although specific nightmare content may reflect the individual's age, the essential features of the disorder are the same across age groups.

Risk and Prognostic Factors

Frequent nightmares in middle-age adults in the general community population have been shown in two studies in Hong Kong and Finland to be associated with low income, mood disturbance, insomnia or sleep-disordered breathing, and use of antidepressants or frequent heavy alcohol use.

Environmental. Sleep deprivation or fragmentation, and irregular sleep-wake schedules that alter the timing, intensity, or quantity of REM sleep can put individuals at risk for nightmares. Individuals who experience nightmares report more frequent past adverse events, but not necessarily trauma.

Genetic and physiological. Twin studies have identified genetic effects on the disposition to nightmares and their co-occurrence with other nocturnal behaviors (e.g., sleepwalking).

Course modifiers. Adaptive parental bedside behaviors, such as soothing the child following nightmares, may protect against developing chronic nightmares.

Culture-Related Diagnostic Issues

The significance attributed to nightmares may vary by culture, and sensitivity to such beliefs may facilitate disclosure. In several cultural contexts, nightmares may be viewed as important indicators of the individual's spiritual status or the condition of those who have died (e.g., among Indonesian civil war survivors, American Indian veterans, and Cambodian refugees). Frequent nightmares among Cambodian refugees are strongly associated with the presence of posttraumatic stress disorder (PTSD); assessment of the temporal sequence and severity of nightmares relative to other symptoms is needed to determine whether a separate diagnosis of nightmare disorder is warranted. Among Hmong immigrants to the United States, frequent nightmares are more common than among non-Latinx Whites in the same region and are associated with traumatic experiences and other sleep disorders, such as sleep paralysis and restless sleep.

Sex- and Gender-Related Diagnostic Issues

Adult women report having nightmares more frequently than adult men, but this gender difference was not found in children and the elderly. Nightmare content differs by gender, with women tending to report themes of sexual harassment or of loved ones disappearing/dying, and men tending to report themes of physical aggression or war/terror.

Diagnostic Markers

Polysomnographic studies demonstrate abrupt awakenings from REM sleep, usually during the second half of the night, prior to report of a nightmare. Heart, respiratory, and eye movement rates may quicken or increase in variability before awakening. Nightmares following traumatic events may also arise during light non-REM (NREM) sleep, particularly stage 2 sleep (now called N2 sleep). The typical sleep of individuals with nightmares is mildly impaired (e.g., reduced efficiency, less slow-wave sleep, more awakenings), with more frequent periodic leg movements in sleep and relative sympathetic nervous system activation after REM sleep deprivation.

Association With Suicidal Thoughts or Behavior

Individuals with frequent nightmares are at substantially greater risk for suicidal thoughts or behavior, even when gender and mental illness are taken into account.

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Functional Consequences of Nightmare Disorder

Nightmares cause more significant subjective distress than demonstrable social or occupational impairment. However, if awakenings are frequent or result in sleep avoidance, individuals may experience excessive daytime sleepiness, poor concentration, depression, anxiety, or irritability. Frequent childhood nightmares (e.g., several per week) may cause significant distress to parents and children.

Differential Diagnosis

Sleep terror disorder. Both nightmare disorder and sleep terror disorder include awakenings or partial awakenings with fearfulness and autonomic activation, but the two disorders can be readily differentiated. Nightmares typically occur later in the night, during REM sleep, and produce vivid, story-like, and clearly recalled dreams; mild autonomic arousal; and complete awakenings. Sleep terrors typically arise in the first third of the night during deep NREM sleep (especially during stage 3 sleep, now called N3 sleep) and produce either no dream recall or images without an elaborate story-like quality. Sleep terrors are thought to be caused by partial awakenings intermixed with persisting sleep, with clinical manifestations of confusion, disorientation, and only partial responsiveness, and often with substantial autonomic arousal. There is usually amnesia for the event in the morning.

REM sleep behavior disorder. The presence of complex vocal and motor activity during frightening dreams should prompt further evaluation for REM sleep behavior disorder, which occurs more typically among late middle- and older-age men but may also affect women. Although nightmares are typically characteristic of REM sleep behavior disorder, unlike nightmare disorder, REM sleep behavior disorder is associated with dream enactment that may cause nocturnal injuries. If the nightmares precede REM sleep behavior disorder and warrant independent clinical attention, an additional diagnosis of nightmare disorder may be given.

Bereavement. Dysphoric dreams may occur during bereavement but typically involve loss and sadness and are followed by self-reflection and insight, rather than distress, on awakening.

PTSD or acute stress disorder. Nightmares in which the content or affect of the dream is related to a traumatic event may be a component of PTSD or acute stress disorder. An additional diagnosis of nightmare disorder may be warranted if the severity or frequency of the nightmares necessitates independent clinical attention.

Narcolepsy. Nightmares are a frequent complaint in narcolepsy, but the presence of excessive sleepiness, with or without cataplexy, differentiates this condition from nightmare disorder.

Sleep-related seizures. Nocturnal seizures usually involve stereotyped motor activity. Associated nightmares, if recalled, are often also repetitively stereotyped in nature or reflect epileptogenic features such as the content of diurnal auras, phosphenes (visual sensations in the absence of light input), or ictal imagery.

Breathing-related sleep disorders. Breathing-related sleep disorders can lead to awakenings with autonomic arousal, but these are not usually accompanied by recall of nightmares.

Panic disorder. Panic attacks arising during sleep can produce abrupt awakenings with autonomic arousal and fearfulness, but nightmares are typically not reported and symptoms are similar to panic attacks arising during wakefulness.

Sleep-related dissociative disorders. Individuals may recall actual physical or emotional trauma as a “dream” during electroencephalography-documented awakenings.

Substance or medication use. Numerous substances/medications can precipitate nightmares, including dopaminergic drugs; β -adrenergic antagonists and other antihypertensives; amphetamine-type substances, cocaine, and other stimulants; antidepressants; smoking cessation aids; and melatonin. Withdrawal of REM sleep-suppressant medications

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(e.g., antidepressants) and alcohol can produce REM sleep rebound accompanied by nightmares. If nightmares are sufficiently severe to warrant independent clinical attention, a diagnosis of substance/medication-induced sleep disorder should be considered.

Comorbidity

Nightmares may be comorbid with several medical conditions, including coronary heart disease, cancer, parkinsonism, and pain, and can accompany medical treatments, such as hemodialysis, or withdrawal from medications or substances of abuse. Nightmares frequently co-occur with other mental disorders, including PTSD, acute stress disorder, insomnia disorder, REM sleep behavior disorder, and psychotic, mood, anxiety, adjustment, and personality disorders, as well as with grief during bereavement. A concurrent nightmare disorder diagnosis should only be considered when independent clinical attention is warranted. These conditions should be listed with the appropriate comorbid category specifier (e.g., “with REM sleep behavior disorder”); see also “Recording Procedures.”

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition, presents similar diagnostic criteria for nightmare disorder.

Rapid Eye Movement Sleep Behavior Disorder

Diagnostic Criteria

G47.52

- A. Repeated episodes of arousal during sleep associated with vocalization and/or complex motor behaviors.
- B. These behaviors arise during rapid eye movement (REM) sleep and therefore usually occur more than 90 minutes after sleep onset, are more frequent during the later portions of the sleep period, and uncommonly occur during daytime naps.
- C. Upon awakening from these episodes, the individual is completely awake, alert, and not confused or disoriented.
- D. Either of the following:
 - 1. REM sleep without atonia on polysomnographic recording.
 - 2. A history suggestive of REM sleep behavior disorder and an established synucleinopathy diagnosis (e.g., Parkinson's disease, multiple system atrophy).
- E. The behaviors cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (which may include injury to self or the bed partner).
- F. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition.
- G. Coexisting mental disorders and medical conditions do not explain the episodes.

Diagnostic Features

The essential feature of rapid eye movement (REM) sleep behavior disorder is repeated episodes of vocalizations and/or complex motor behaviors arising from REM sleep (Criterion A). These behaviors often reflect motor responses to the content of action-filled or violent dreams of being attacked or trying to escape from a threatening situation, which may be termed *dream enacting behaviors*. The vocalizations are often loud, emotion-filled, and profane. These behaviors may be very bothersome to the individual and the bed partner and

may result in significant injury (e.g., falling, jumping, or flying out of bed; running, punching, thrusting, hitting, or kicking). However, individuals with REM sleep behavior disorder may also present with relatively subtle vocal or motor behaviors during REM sleep, which are typically not the primary presenting sleep complaint but manifest during history taking or polysomnography in sleep, neurological, and psychiatric clinical visits. Upon awakening, the individual is usually immediately awake, alert, and oriented (Criterion C) and is often able to recall dream mentation, which closely correlates with the observed behavior. The eyes typically

remain closed during these events. The presence of REM sleep without atonia during a polysomnogram is typically required for the diagnosis of REM sleep behavior disorder. Alternatively, if polysomnography has not been performed, a provisional diagnosis of probable REM sleep behavior disorder may be given if there is an established synucleinopathy diagnosis (e.g., Parkinson's disease, multiple system atrophy) and the history is suggestive of REM sleep behavior disorder (Criterion D). The diagnosis of REM sleep behavior disorder requires clinically significant distress or impairment (Criterion E); this determination will depend on a number of factors, including the frequency of events, the potential for violence or injurious behaviors, embarrassment, and distress in other household members.

Severity determination is best made based on the nature or consequence of the behavior rather than simply on frequency. Although the behaviors are typically prominent and violent, lesser behaviors may also occur.

Prevalence

The prevalence of REM sleep behavior disorder was approximately 1% in a middle- to older-age general population sample in Switzerland and approximately 2% in an elderly general population sample in South Korea. One prevalence study found an equal prevalence between men and women in individuals younger than 50 years, while another study reported a prevalence of just over 1% with no difference between men and women in a population with a mean age of 59 years. Prevalence in individuals with psychiatric disorders may be greater, possibly related to medications prescribed for the psychiatric disorder.

Development and Course

The onset of REM sleep behavior disorder may be gradual or rapid. Because of the very high association with the later appearance of an underlying neurodegenerative disorder, the neurological status of individuals with REM sleep behavior disorder should be closely monitored. In individuals with idiopathic REM sleep behavior disorder, the risk of developing a defined neurodegenerative disease, most often a synucleinopathy (i.e., Parkinson's disease, major or mild neurocognitive disorder with Lewy bodies, or multiple system atrophy), is approximately 75% within 10–15 years following diagnosis, with an annualized risk of approximately 6%–7% per year.

Symptoms in young individuals, particularly young women, should raise the possibility of narcolepsy; substance/medication-induced sleep disorder, parasomnia type; a brainstem lesion; or an autoimmune encephalopathy.

Culture-Related Diagnostic Issues

Chinese individuals diagnosed with REM sleep behavior disorder by a neurology service in Taiwan had similar clinical and laboratory characteristics as non-Latinx White individuals in the United States; however, they differed in having a higher rate of nocturnal wandering out of the bedroom and a lower rate of sleep-related injuries, possibly as a result of earlier detection by the family.

Sex- and Gender-Related Diagnostic Issues

REM sleep behavior disorder is more common in men older than 50 years, but increasingly this disorder is being identified in women and in younger individuals. Women are younger than men in age at onset and age at diagnosis.

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Diagnostic Markers

Associated laboratory findings from polysomnography indicate increased tonic and/or phasic electromyographic activity during REM sleep, which is normally associated with muscle atonia. The increased muscle activity variably affects different muscle groups; more extensive electromyographic monitoring with arm electromyography (e.g., biceps brachii) should be considered because this measure is more specific for a REM sleep behavior disorder diagnosis. It is suggested that electromyographic monitoring also include the submental, bilateral flexor digitorum superficialis, and bilateral anterior tibialis muscle groups. Continuous video monitoring should accompany the polysomnography. Other polysomnographic findings may include very frequent periodic and aperiodic extremity electromyography activity during non-REM (NREM) sleep. *REM sleep without atonia* is present in virtually all cases of REM sleep behavior disorder but may also be an asymptomatic polysomnographic finding. It is not known whether isolated REM sleep without atonia is a precursor to REM sleep behavior disorder, although a pilot study suggested that isolated REM sleep without atonia may also be associated with neurodegenerative markers (i.e., hyposmia, orthostatic hypotension, color vision loss) and that 7%–14% of individuals with isolated REM sleep without atonia later develop clinical REM sleep behavior disorder. Thresholds for normative REM sleep without atonia levels have also been published that may serve to distinguish borderline cases and those whose neurological status should be further monitored.

Functional Consequences of Rapid Eye Movement Sleep Behavior Disorder

The most serious consequences of REM sleep behavior disorder are the short-term risks for injury to the individual or bed partner related to attacks of dream enactment, and the long-term risk of developing a defined neurodegenerative disease. According to surveys of individuals and their bed partners, approximately 55% of individuals with REM sleep behavior disorder may experience injury as a consequence of their attacks, with 12% of injuries being serious (including long bone or rib fractures or subdural hematomas) and requiring medical attention.

Differential Diagnosis

Other parasomnias. Confusional arousals, sleepwalking, and sleep terrors can easily be confused with REM sleep behavior disorder. In general, these disorders occur in individuals younger than 50 years. Unlike REM sleep behavior disorder, they arise from NREM sleep and therefore tend to occur in the early portion of the sleep period. Awakening from a confusional arousal is associated with confusion, disorientation, and incomplete recall of dream mentation accompanying the behavior. Polysomnographic monitoring in the disorders of arousal generally reveals normal REM sleep atonia unless there is a comorbid parasomnia.

Medication-induced sleep disorder, parasomnia type. Many widely prescribed medications, including tricyclic antidepressants, selective serotonin reuptake inhibitors, and serotonin-norepinephrine reuptake inhibitors, may result in polysomnographic evidence of REM sleep without atonia and in frank REM sleep behavior disorder, which is diagnosed as a medication-induced sleep disorder, parasomnia type. It is not known whether the medications per se result in REM sleep without atonia and/or REM sleep behavior disorder, or whether they unmask an underlying predisposition.

Asymptomatic REM sleep without atonia. Clinical dream-enacting behaviors coupled with the polysomnographic finding of REM sleep without atonia is necessary for the diagnosis of REM sleep behavior disorder. REM sleep without atonia without a clinical history of dream-enacting behaviors is simply an asymptomatic polysomnographic observation with an as yet unknown clinical significance.

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Nocturnal seizures. Nocturnal seizures may mimic REM sleep behavior disorder, but the behaviors characteristic of nocturnal seizures are generally stereotyped. Polysomnographic monitoring employing a full electroencephalographic seizure montage may differentiate the two. REM sleep without atonia is generally not present on polysomnographic monitoring in individuals with epilepsy.

Obstructive sleep apnea. Obstructive sleep apnea may result in vocalizations and motor behaviors that very closely resemble REM sleep behavior disorder, such as talking, shouting, gesturing, and punching, along with unpleasant dreams. Polysomnographic monitoring is necessary to differentiate between these two disorders. In REM sleep behavior disorder, the parasomnia symptoms occur during periods of REM sleep without atonia. In obstructive sleep apnea, the parasomnia symptoms only occur during arousals at the end of the obstructive sleep apneic events and resolve following effective treatment of the obstructive sleep apnea (continuous positive airway pressure). REM sleep without atonia is not typically observed in obstructive sleep apnea.

Other specified dissociative disorder (sleep-related psychogenic dissociative disorder). Unlike virtually all other parasomnias, which arise precipitously from NREM or REM sleep, psychogenic dissociative behaviors arise from a period of well-defined wakefulness during the sleep period. Unlike REM sleep behavior disorder, this condition is more prevalent in young women.

Malingering. Many cases of malingering in which the individual reports problematic sleep movements mimic the clinical features of REM sleep behavior disorder, and polysomnographic documentation is mandatory.

Comorbidity

REM sleep behavior disorder is present concurrently in approximately 30% of patients with narcolepsy. When it occurs in narcolepsy, the demographics reflect the younger age range of narcolepsy, with equal frequency in men and women. Based on findings from individuals presenting to sleep clinics, most individuals (>70%) with initially “idiopathic” REM sleep behavior disorder will eventually develop a neurodegenerative disease—most notably, one of the

synucleinopathies (Parkinson's disease, multiple system atrophy, or major or mild neurocognitive disorder with Lewy bodies). REM sleep behavior disorder often predates any other sign of these disorders by many years (often more than a decade).

Relationship to International Classification of Sleep Disorders

REM sleep behavior disorder is virtually identical to REM sleep behavior disorder in the *International Classification of Sleep Disorders*, 3rd Edition.

Restless Legs Syndrome

Diagnostic Criteria

G25.81

- A. An urge to move the legs, usually accompanied by or in response to uncomfortable and unpleasant sensations in the legs, characterized by all of the following:
 - 1. The urge to move the legs begins or worsens during periods of rest or inactivity.
 - 2. The urge to move the legs is partially or totally relieved by movement.
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- 3. The urge to move the legs is worse in the evening or at night than during the day, or occurs only in the evening or at night.
- B. The symptoms in Criterion A occur at least three times per week and have persisted for at least 3 months.
- C. The symptoms in Criterion A are accompanied by significant distress or impairment in social, occupational, educational, academic, behavioral, or other important areas of functioning.
- D. The symptoms in Criterion A are not attributable to another mental disorder or medical condition (e.g., arthritis, leg edema, peripheral ischemia, leg cramps) and are not better explained by a behavioral condition (e.g., positional discomfort, habitual foot tapping).
- E. The symptoms are not attributable to the physiological effects of a drug of abuse or medication (e.g., akathisia).

Diagnostic Features

Restless legs syndrome (RLS) is a sensorimotor, neurological sleep disorder characterized by a desire to move the legs or arms, usually associated with uncomfortable sensations typically described as creeping, crawling, tingling, burning, or itching (Criterion A). Frequent movements

of the legs occur in an effort to relieve the uncomfortable sensations. Although symptoms can occur during the daytime, they commonly occur in the late afternoon or evening hours, and in some individuals, symptoms occur only in the evening or night. Symptoms are often most severe at night when the individual is at rest, such as sitting or lying in bed. Evening worsening occurs independently of any differences in activity. The diagnosis of RLS is based primarily on individual self-report and history. It is important to differentiate RLS from other conditions that cause leg discomfort, such as positional discomfort and leg cramps (Criterion D).

The symptoms of RLS can delay sleep onset and awaken the individual from sleep and are associated with significant sleep fragmentation. The relief obtained from moving the legs may no longer be apparent in severe cases. RLS is associated with daytime sleepiness and is frequently accompanied by significant clinical distress or functional impairment.

Associated Features

Periodic leg movements in sleep (PLMS) can serve as corroborating evidence for RLS, with up to 90% of individuals diagnosed with RLS demonstrating PLMS when recordings are taken over multiple nights. Periodic leg movements during wakefulness are also supportive of an RLS diagnosis. Reports of difficulty initiating and maintaining sleep and of excessive daytime sleepiness substantiate a diagnosis of RLS. Additional supportive features include a family history of RLS among first-degree relatives and a reduction in symptoms, at least initially, with dopaminergic treatment.

Prevalence

Prevalence rates of RLS vary widely when broad criteria are utilized. When frequency of symptoms is at least three times per week with moderate or severe distress, the prevalence rate in the United States and Europe has been estimated as 1.6%. RLS that is severe enough to significantly impair functioning or is associated with mental disorders, including depression and anxiety, occurs in approximately 2%–3% of the population, as assessed in Western Europe, the United States, and South Korea. RLS is about twice as common in women as men and increases in prevalence with age. Reports of RLS vary across geographic regions, with lower prevalence in several Asian populations (e.g., Japan, South Korea).

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Development and Course

The onset of RLS typically occurs in the second or third decade. Approximately 40% of individuals diagnosed with RLS during adulthood report having experienced symptoms before age 20 years, and 20% report having experienced symptoms before age 10 years. Prevalence rates of RLS increase steadily with age until about age 60 years, with symptoms remaining stable or decreasing slightly in older age groups. Compared with nonfamilial cases, familial RLS usually has a younger age at onset and a slower progressive course. The clinical course of RLS differs by age at onset. When onset occurs before age 45, there is often a slow progression of symptoms. In late-onset RLS, rapid progression is typical, and aggravating factors are common. The RLS phenotype appears similar across the life span.

Diagnosis of RLS in children can be difficult because of the centrality of self-report in

establishing the diagnosis. While Criterion A for adults assumes that the description of “urge to move” is by the individual, pediatric diagnosis also requires a description in the child’s own words rather than by a parent or caretaker. Typically children age 6 years or older are able to provide detailed, adequate descriptors of RLS. However, children rarely use or understand the word “urge,” reporting instead that their legs “have to” or “got to” move. Also, potentially related to prolonged periods of sitting during class, two-thirds of children and adolescents with RLS report daytime leg sensations. Thus, for diagnostic Criterion A3, it is important to compare equal duration of sitting or lying down in the day to sitting or lying down in the evening or night. Nocturnal worsening tends to persist even in pediatric RLS. As with RLS in adults, there is a significant negative impact on sleep, mood, cognition, and function. Impairment in children and adolescents is manifested more often in behavioral and educational domains.

Risk and Prognostic Factors

Genetic and physiological. Predisposing factors include female sex, advancing age, genetic risk variants, and family history of RLS. Precipitating factors such as iron deficiency are often time-limited, with most individuals resuming normal sleep patterns after the triggering event has disappeared. Genetic risk variants also play a role in RLS secondary to disorders such as uremia, suggesting that individuals with a genetic susceptibility develop RLS in the presence of additional risk factors.

Genome-wide association studies have found that RLS is significantly associated with multiple genetic variants in intronic or intergenic regions. The variant in *MEIS1* has the strongest association with RLS of these genes, with nearly double the risk of RLS in the 7% of the population with this polymorphism among European-ancestry samples studied.

Pathophysiological mechanisms in RLS also include disturbances in the central dopaminergic and opioidergic systems and disturbances in iron metabolism. Treatment efficacy of dopaminergic drugs, opioids, and iron provides further support that these systems play a role in the pathophysiology of RLS. RLS may predispose to depression, and the effective treatment of RLS may significantly reduce depressive symptoms. However, serotonergic antidepressants can induce or aggravate RLS in some individuals.

Culture-Related Diagnostic Issues

Among indigenous-descent Latin American adult populations in the United States, including Mexican Americans with low acculturation to U.S. society, the reported prevalence of RLS appears to be lower when compared with Mexican Americans with higher acculturation. Among participants who reported RLS in a large population-based survey, risk factors associated with RLS were different in Mexican Americans (higher among women and persons who smoke) compared with non-Latinx Whites (older age, defined as ≥ 48 years).

Sex- and Gender-Related Diagnostic Issues

Although RLS is more prevalent in women than in men, there are no diagnostic differences according to gender. The prevalence of RLS during pregnancy is two to three times greater

than in the general population. RLS associated with pregnancy peaks during the third trimester and improves or resolves in most cases soon after delivery. The sex difference in prevalence of RLS is explained in part by parity, with nulliparous females being at the same risk for RLS as age-matched males.

Diagnostic Markers

Polysomnography demonstrates significant abnormalities in RLS, including increased latency to sleep and higher arousal index. Periodic limb movements are the motor sign of RLS and are usually present on overnight polysomnography, as well as during waking immobilization tests and during quiet resting, both of which can provoke RLS symptoms.

Functional Consequences of Restless Legs Syndrome

Although the impact of milder symptoms is less well characterized, individuals with RLS complain of disruption in at least one activity of daily living, with up to 50% reporting a negative impact on mood and a lack of energy. A common consequence of RLS is sleep disturbance, including difficulty falling asleep and sleep fragmentation, with associated reduction in total sleep time. RLS is also associated with quality-of-life impairments. RLS can result in daytime sleepiness or fatigue and is frequently accompanied by significant distress or impairment in affective, social, occupational, educational, academic, behavioral, or cognitive functioning.

Differential Diagnosis

The most important conditions in the differential diagnosis of RLS are leg cramps, positional discomfort, arthralgias/arthritis, myalgias, positional ischemia (numbness), leg edema, peripheral neuropathy, radiculopathy, and habitual foot tapping. Muscle cramps, relief with a single postural shift, limitation to joints, soreness to palpation (myalgias), and other abnormalities on physical examination are not characteristic of RLS. Unlike RLS, nocturnal leg cramps do not typically manifest with the desire to move the limbs nor are there frequent limb movements. Less common conditions to be differentiated from RLS include neuroleptic-induced akathisia, myelopathy, symptomatic venous insufficiency, peripheral artery disease, eczema, other orthopedic problems, and anxiety-induced restlessness. Worsening at night and periodic limb movements are more common in RLS than in medication-induced akathisia or peripheral neuropathy.

While it is important that RLS symptoms not be solely accounted for by another medical or behavioral condition, it should also be appreciated that any of these similar conditions can occur in an individual with RLS. This necessitates a separate focus on each possible condition in the diagnostic process and when assessing impact. For cases in which the diagnosis of RLS is not certain, evaluation for the supportive features of RLS, particularly PLMS or a family history of RLS, may be helpful. Clinical features, such as response to a dopaminergic agent and positive family history for RLS, can help with the differential diagnosis.

Comorbidity

RLS is associated with higher rates of depression, generalized anxiety disorder, panic disorder, and posttraumatic stress disorder. The main medical condition comorbid with RLS is

cardiovascular disease. There may be an association with numerous other medical conditions, including hypertension, migraine, Parkinson's disease, multiple sclerosis, peripheral neuropathy, diabetes mellitus, fibromyalgia, osteoporosis, obesity, thyroid disease, and cancer, as well as other sleep disorders including narcolepsy and obstructive sleep apnea. RLS is common in those with iron deficiency, pregnancy, and chronic renal failure and can dramatically improve once these conditions resolve.

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Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition, presents similar diagnostic criteria for RLS but does not contain a criterion specifying frequency or duration of symptoms.

Substance/Medication-Induced Sleep Disorder

Diagnostic Criteria

- A. A prominent and severe disturbance in sleep.
- B. There is evidence from the history, physical examination, or laboratory findings of both (1) and (2):
 - 1. The symptoms in Criterion A developed during or soon after substance intoxication or withdrawal or after exposure to or withdrawal from a medication.
 - 2. The involved substance/medication is capable of producing the symptoms in Criterion A.
- C. The disturbance is not better explained by a sleep disorder that is not substance/medication-induced. Such evidence of an independent sleep disorder could include the following:

The symptoms precede the onset of the substance/medication use; the symptoms persist for a substantial period of time (e.g., about 1 month) after the cessation of acute withdrawal or severe intoxication; or there is other evidence suggesting the existence of an independent non-substance/medication-induced sleep disorder (e.g., a history of recurrent non-substance/medication-related episodes).
- D. The disturbance does not occur exclusively during the course of a delirium.
- E. The disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.

Note: This diagnosis should be made instead of a diagnosis of substance intoxication or substance withdrawal only when the symptoms in Criterion A predominate in the clinical picture and when they are sufficiently severe to warrant clinical attention.

Coding note: The ICD-10-CM codes for the [specific substance/medication]-induced sleep disorders are indicated in the table below. Note that the ICD-10-CM code depends on whether or not there is a comorbid substance use disorder present for the same class of substance. In any case, an additional separate diagnosis of a substance use disorder is not given. If a mild substance use disorder is comorbid with the substance-induced sleep disorder, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance-induced sleep disorder (e.g., “mild cocaine use disorder with cocaine-induced sleep disorder”). If a moderate or severe substance use disorder is comorbid with the substance-induced sleep disorder, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder (e.g., after a one-time heavy use of the substance), then the 4th position character is “9,” and the clinician should record only the substance-induced sleep disorder.

There are two exceptions to this coding convention as it applies to caffeine- and tobacco-induced sleep disorders. Because caffeine use disorder is not an official DSM-5 category,

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there is only a single ICD-10-CM code for caffeine-induced sleep disorder: F15.982. Moreover, because ICD-10-CM assumes that tobacco-induced sleep disorder can only occur in the context of moderate or severe tobacco use disorder, the ICD-10-CM code for tobacco-induced sleep disorder is F17.208.

	ICD-10-CM		
	With mild use disorder	With moderate or severe use disorder	Without use disorder
Alcohol	F10.182	F10.282	F10.982
Caffeine	NA	NA	F15.982
Cannabis	F12.188	F12.288	F12.988
Opioid	F11.182	F11.282	F11.982
Sedative, hypnotic, or anxiolytic	F13.182	F13.282	F13.982
Amphetamine-type substance (or other stimulant)	F15.182	F15.282	F15.982
Cocaine	F14.182	F14.282	F14.982
Tobacco	NA	F17.208	NA
Other (or unknown) substance	F19.182	F19.282	F19.982

Specify whether:

Insomnia type: Characterized by difficulty falling asleep or maintaining sleep, frequent nocturnal awakenings, or nonrestorative sleep.

Daytime sleepiness type: Characterized by predominant complaint of

excessive sleepiness/fatigue during waking hours or, less commonly, a long sleep period.

Parasomnia type: Characterized by abnormal behavioral events during sleep.

Mixed type: Characterized by a substance/medication-induced sleep problem characterized by multiple types of sleep symptoms, but no symptom clearly predominates.

Specify (see [Table 1](#) in the chapter “Substance-Related and Addictive Disorders,” which indicates whether “with onset during intoxication” and/or “with onset during withdrawal” applies to a given substance class; or *specify* “with onset after medication use”):

With onset during intoxication: If criteria are met for intoxication with the substance and the symptoms develop during the intoxication.

With onset during withdrawal: If criteria are met for withdrawal from the substance and the symptoms develop during, or shortly after, withdrawal.

With onset after medication use: If symptoms developed at initiation of medication, with a change in use of medication, or during withdrawal of medication.

Recording Procedures

The name of the substance/medication-induced sleep disorder begins with the specific substance (e.g., alcohol) that is presumed to be causing the sleep disturbance. The ICD-10-CM code that corresponds to the applicable drug class is selected from the table included in the criteria set. For substances that do not fit into any of the classes (e.g., fluoxetine), the ICD-10-CM code for the other (or unknown) substance class should be used and the name

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of the specific substance recorded (e.g., F19.982 fluoxetine-induced sleep disorder, insomnia type). In cases in which a substance is judged to be an etiological factor but the specific substance is unknown, the ICD-10-CM code for the other (or unknown) substance class is used and the fact that the substance is unknown is recorded (e.g., F19.982 unknown substance-induced sleep disorder, hypersomnia type).

To record the name of the disorder, the comorbid substance use disorder (if any) is listed first, followed by “with substance/medication-induced sleep disorder” (incorporating the name of the specific etiological substance/medication), followed by the specification of onset (i.e., with onset during intoxication, with onset during withdrawal, with onset after medication use), followed by the subtype designation (i.e., insomnia type, daytime sleepiness type, parasomnia type, mixed type). For example, in the case of insomnia occurring during withdrawal in a man with a severe lorazepam use disorder, the diagnosis is F13.282 severe lorazepam use disorder with lorazepam-induced sleep disorder, with onset during withdrawal, insomnia type. A separate diagnosis of the comorbid severe lorazepam use disorder is not given. If the substance-induced sleep disorder occurs without a comorbid substance use disorder (e.g., with medication use as

prescribed), no accompanying substance use disorder is noted (e.g., F19.982 bupropion-induced sleep disorder, with onset during medication use, insomnia type). When more than one substance is judged to play a significant role in the development of the sleep disturbance, each should be listed separately (e.g., F10.282 severe alcohol use disorder with alcohol-induced sleep disorder, with onset during intoxication, insomnia type; F14.282 severe cocaine use disorder with cocaine-induced sleep disorder, with onset during intoxication, insomnia type).

Specifiers

Depending on the substance involved, one of four types of sleep disturbances is indicated. Insomnia type and daytime sleepiness type are most common, whereas parasomnia type is seen less often. The mixed type is noted when more than one type of sleep disturbance–related symptom is present and none predominates.

Diagnostic Features

The essential feature of substance/medication-induced sleep disorder is a prominent sleep disturbance that is sufficiently severe to warrant independent clinical attention (Criterion A). The sleep disturbance may be characterized by insomnia, daytime sleepiness, a parasomnia, or some combination of these. The sleep disturbance is judged to be primarily associated with the pharmacological effects of a substance (i.e., a drug of abuse, a medication, toxin exposure) (Criterion B). The disturbance must not be better explained by another sleep disorder that is not substance/medication-induced (Criterion C). A substance/medication-induced sleep disorder is distinguished from insomnia disorder or a disorder associated with excessive daytime sleepiness by considering onset and course. For drugs of abuse, there must be evidence of intoxication or withdrawal from the history, physical examination, or laboratory findings. Substance/medication-induced sleep disorder arises only in association with intoxication or discontinuation/withdrawal states, whereas other sleep disorders may precede the onset of substance use or occur during times of sustained abstinence. As discontinuation/withdrawal states for some substances can be protracted, onset of the sleep disturbance can occur 4 weeks after cessation of substance use, and the disturbance may have features atypical of other sleep disorders (e.g., atypical age at onset or course). The diagnosis is not made if the sleep disturbance occurs only during a delirium (Criterion D). The symptoms must cause clinically significant distress or impairment in social, occupational, or other important areas of functioning (Criterion E). This diagnosis should be made instead of a diagnosis of substance intoxication or substance withdrawal only when the symptoms in Criterion A predominate in the clinical picture and when the symptoms warrant independent clinical attention.

Associated Features

During periods of substance/medication use, intoxication, or withdrawal, individuals frequently complain of dysphoric mood, including depression and anxiety, irritability, cognitive impairment, inability to concentrate, and fatigue.

Prominent and severe sleep disturbances can occur in association with intoxication with the following classes of substances: alcohol; caffeine; cannabis; opioids; sedatives, hypnotics, or

anxiolytics; stimulants (including cocaine); and other (or unknown) substances. Prominent and severe sleep disturbances can occur in association with withdrawal from the following classes of substances: alcohol; caffeine; cannabis; opioids; sedatives, hypnotics, or anxiolytics; stimulants (including cocaine); tobacco; and other (or unknown) substances. Medications that invoke sleep disturbances include adrenergic agonists and antagonists, dopamine agonists and antagonists, cholinergic agonists and antagonists, serotonergic agonists and antagonists, antihistamines, and corticosteroids.

Alcohol. Alcohol-induced sleep disorder typically occurs as insomnia type. During acute intoxication with doses > 1 g/kg, alcohol produces an immediate sedative effect depending on dose, accompanied by a reduction in sleep latency, increased non-rapid eye movement (NREM) sleep stages 2 and 3 (N2 and N3), and reduced rapid eye movement (REM) sleep. Following these initial effects, there may be increased wakefulness, restless sleep, and vivid and anxiety-laden dreams for the remaining sleep period. In parallel, N2 and N3 are reduced, and wakefulness and REM sleep are increased during the latter portion of the night. With habitual use, alcohol continues to show a short-lived sedative effect in the first half of the night, followed by sleep continuity disruption in the second half. During acute alcohol withdrawal, there is extremely disrupted sleep continuity, and an increased amount and intensity of REM sleep, associated frequently with vivid dreaming, which in extreme form constitutes part of alcohol withdrawal delirium. After acute withdrawal, chronic alcohol users may continue to complain of light, fragmented sleep for months to years associated with a persistent prolongation of sleep latency and deficit in slow-wave sleep. Alcohol also aggravates breathing-related sleep disorder, including obstructive sleep apnea and sleep-related hypoventilation.

Caffeine. Caffeine consumed in low to moderate doses during the morning hours typically produces no significant effect on nighttime sleep in normal sleepers or those with insomnia. Caffeine may produce insomnia in a dose- and timing-dependent manner, particularly when larger doses are consumed later in the day or during evening hours. Prolongation of sleep latency, reduction of slow-wave sleep, increased nocturnal awakening, and reduced sleep duration are reported. Some individuals, particularly high consumers, may present with daytime sleepiness and performance impairments related to withdrawal.

Cannabis. Acute administration of cannabis may shorten sleep latency, though arousing effects with increments in sleep latency also occur. Cannabis enhances slow-wave sleep and suppresses REM sleep after acute administration. In chronic users, tolerance to the sleep-inducing and slow-wave sleep-enhancing effects develops. Upon withdrawal, sleep difficulties and unpleasant dreams have been reported lasting for several weeks. Polysomnographic studies demonstrate reduced slow-wave sleep and increased REM sleep during this phase.

Opioids. Opioids may produce an increase in sleepiness and in subjective depth of sleep, and reduced REM and slow-wave sleep, during acute short-term use. With continued administration, tolerance to the sedative effects of opioids develops and there are complaints of insomnia. Polysomnographic studies demonstrate reduced sleep efficiency and total sleep time, with reduction of slow-wave sleep and possibly REM sleep. Consistent with their respiratory depressant effects, opioids exacerbate obstructive sleep apnea. Emergence of central sleep apnea is also observed, especially with chronic use of longer-acting opioids.

Sedative, hypnotic, or anxiolytic substances. Sedatives, hypnotics, and anxiolytics (e.g., barbiturates, benzodiazepine receptor agonists, meprobamate, glutethimide, methyprylon) have similar effects as opioids on sleep. During acute intoxication, sedative-hypnotic drugs produce the expected increase in sleepiness and decrease in wakefulness. Daytime sleepiness may occur, primarily with longer-acting agents. Chronic benzodiazepine use may be associated with development of tolerance, rebound insomnia, and potentially serious withdrawal effects. Newer benzodiazepine receptor agonists such as zolpidem and eszopiclone have been shown to maintain efficacy over periods of 6 months to 2 years, without evidence of dosage escalation or major withdrawal effects. Newer hypnotic agents such as ramelteon, low-dose doxepin, and suvorexant do not appear to have significant abuse potential, respiratory depression, or major withdrawal syndromes. Sedative, hypnotic, or anxiolytic drugs with short durations of action are most likely to produce complaints of rebound insomnia. Some sedative-hypnotic drugs may increase the frequency and severity of obstructive sleep apnea events, although neither benzodiazepines nor benzodiazepine receptor agonists have been found to definitively worsen obstructive sleep apnea. Hypoventilation may worsen in susceptible individuals. Parasomnias (sleepwalking and sleep-related eating) have been associated with use of benzodiazepine receptor agonists, especially when these medications are taken at higher doses and when they are combined with other sedative drugs.

Amphetamine-type substances, other stimulants, and MDMA. Sleep disorders induced by amphetamine-type substances and other stimulants are characterized by insomnia during intoxication and excessive sleepiness during withdrawal. During acute intoxication, stimulants reduce the total amount of sleep, increase sleep latency and sleep continuity disturbances, and decrease REM sleep. Slow-wave sleep tends to be reduced. During withdrawal from chronic stimulant use, there is both prolonged nocturnal sleep duration and excessive daytime sleepiness. Multiple sleep latency tests may show increased daytime sleepiness during the withdrawal phase. Drugs like 3,4-methylenedioxymethamphetamine (MDMA; “ecstasy”) and related substances lead to restless and disturbed sleep within 48 hours of intake; frequent use of these compounds is associated with persisting symptoms of anxiety, depression, and sleep disturbances, even during longer-term abstinence. There is also evidence that suggests an increased frequency of obstructive sleep apnea in young MDMA users, even after a period of abstention from the drug.

Tobacco. Chronic tobacco use is associated primarily with symptoms of insomnia, decreased slow-wave sleep with a reduction of sleep efficiency, and increased daytime sleepiness. Withdrawal from tobacco can lead to impaired sleep. Individuals who smoke heavily may experience regular nocturnal awakenings caused by tobacco craving.

Other or unknown substances/medications. Other substances/medications may produce sleep disturbances, particularly medications that affect the central or autonomic nervous systems (e.g., adrenergic agonists and antagonists, dopamine agonists and antagonists, cholinergic agonists and antagonists, serotonergic agonists and antagonists, antihistamines, corticosteroids).

Development and Course

Insomnia in children can be identified by either a parent or the child. Often the child has a clear sleep disturbance associated with initiation of a medication but may not report symptoms, although parents observe the sleep disturbances. The use of some recreational substances (e.g.,

cannabis, ecstasy) is prevalent in adolescence and early adulthood. Insomnia or any other sleep disturbance encountered in this age group should prompt careful consideration of whether the sleep disturbance is attributable to consumption of these substances. Help-seeking behavior for the sleep disturbance in these age groups is limited, and

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thus corroborative report may be elicited from a parent, caregiver, or teacher. Older individuals take more medications and are at increased risk for developing a substance/medication-induced sleep disorder. They may interpret sleep disturbance as part of normal aging and fail to report symptoms. Individuals with major neurocognitive disorder (e.g., dementia) are at risk for substance/medication-induced sleep disorders but may not report symptoms, making corroborative report from caregiver(s) particularly important.

Risk and Prognostic Factors

Risk and prognostic factors involved in substance or medication use are normative for certain age groups. They are relevant for, and likely applicable to, the type of sleep disturbance encountered (see the chapter “Substance-Related and Addictive Disorders” for descriptions of respective substance use disorders).

Temperamental. Substance use generally precipitates or accompanies insomnia in vulnerable individuals. Thus, presence of insomnia in response to stress or change in sleep environment or timing can represent a risk for developing substance/medication-induced sleep disorder. A similar risk may be present for individuals with other sleep disorders (e.g., individuals with hypersomnia who use stimulants).

Sex- and Gender-Related Diagnostic Issues

The same amount and duration of consumption of a given substance may lead to highly different sleep-related outcomes in males and females based on, for example, sex-specific differences in hepatic functioning.

Diagnostic Markers

Each of the substance/medication-induced sleep disorders produces electroencephalographic sleep patterns that are associated with, but cannot be considered diagnostic of, other disorders. The electroencephalographic sleep profile for each substance is related to the stage of use and whether it is in the context of intake/intoxication, chronic use, or withdrawal following discontinuation of the substance. All-night polysomnography can help define the severity of insomnia complaints, while the multiple sleep latency test provides information about the severity of daytime sleepiness. Monitoring of nocturnal respiration and periodic limb movements with polysomnography may verify a substance’s impact on nocturnal breathing and motor behavior. Sleep diaries for 2 weeks and actigraphy are considered helpful in confirming the presence of substance/medication-induced sleep disorder, especially in the case of suspected insomnia type. Drug screening can be of use when the individual is not aware or unwilling to relate information about substance intake.

Functional Consequences of Substance/Medication-Induced Sleep Disorder

While there are many functional consequences associated with sleep disorders, the only unique consequence for substance/medication-induced sleep disorder is increased risk for relapse. For example, the degree of sleep disturbance during alcohol withdrawal (e.g., REM sleep rebound) predicts risk of relapse of drinking. Monitoring of sleep quality and daytime sleepiness during and after withdrawal may provide clinically meaningful information on whether an individual is at increased risk for relapse.

Differential Diagnosis

Substance intoxication and substance withdrawal. Sleep disturbances are commonly encountered in the context of substance intoxication and substance withdrawal. A diagnosis of substance/medication-induced sleep disorder should be made instead of a

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diagnosis of substance intoxication or substance withdrawal only when the sleep disturbance is predominant in the clinical picture and is sufficiently severe to warrant independent clinical attention.

Delirium. If the substance/medication-induced sleep disturbance occurs exclusively during the course of a delirium, it is not diagnosed separately.

Other sleep disorders. A substance/medication-induced sleep disorder is distinguished from another sleep disorder if a substance/medication is judged to be etiologically related to the symptoms. A substance/medication-induced sleep disorder attributed to a prescribed medication for a mental disorder or medical condition must have its onset while the individual is receiving the medication or during discontinuation, if there is a discontinuation/withdrawal syndrome associated with the medication. Once treatment is discontinued, the sleep disturbance will usually remit within days to several weeks. If symptoms persist beyond 4 weeks, other causes for the sleep disturbance–related symptoms should be considered. Not infrequently, individuals with another sleep disorder use medications or drugs of abuse to self-medicate their symptoms (e.g., alcohol for management of insomnia). If the substance/medication is judged to play a significant role in the exacerbation of the sleep disturbance, an additional diagnosis of a substance/medication-induced sleep disorder may be warranted.

Sleep disorder associated with medical condition. Substance/medication-induced sleep disorder and sleep disorders with a medical condition (i.e., insomnia disorder, hypersomnolence disorder, and nightmare disorder) may produce similar symptoms of insomnia, daytime sleepiness, or nightmares, respectively. Many medical conditions that cause sleep disturbance are treated with medications that may also cause sleep disturbances. The chronology of symptoms is the most important factor in distinguishing between these two sources of sleep symptoms. Difficulties with sleep in an individual with a comorbid medical condition that clearly preceded the use of any medication for treatment of that medical condition would suggest a diagnosis of insomnia disorder, hypersomnolence disorder, or nightmare disorder with the specifier “with [specific medical condition]” applicable to the diagnosis. Conversely, sleep symptoms that appear only

after the initiation of a particular substance/medication suggest a substance/medication-induced sleep disorder. If the sleep disturbance is comorbid with another medical condition and is also exacerbated by substance use, both diagnoses are given (i.e., insomnia disorder, hypersomnolence disorder, or nightmare disorder, “with [specific medical condition]” respectively; and [specific substance/medication]-induced sleep disorder). When there is insufficient evidence to determine whether the sleep disturbance is attributable to a substance/medication or a medical condition, or independent (i.e., not attributable to either a substance/medication or a medical condition), a diagnosis of unspecified sleep-wake disorder is indicated.

Comorbidity

See the “Comorbidity” sections for other sleep disorders in this chapter, including insomnia disorder, hypersomnolence disorder, central sleep apnea, sleep-related hypoventilation, and circadian rhythm sleep-wake disorders, shift work type.

Relationship to International Classification of Sleep Disorders

The *International Classification of Sleep Disorders*, 3rd Edition (ICSD-3), lists sleep disorders “due to a medication or substance” under their respective phenotypes (e.g., hypersomnia, movement disorder, parasomnia). ICSD-3 does not identify a separate diagnosis for “insomnia due to a medication or substance” based on evidence that the reliability of distinguishing specific, single etiological factors for chronic insomnia is poor.

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Other Specified Insomnia Disorder

G47.09

This category applies to presentations in which symptoms characteristic of insomnia disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for insomnia disorder or any of the disorders in the sleep-wake disorders diagnostic class. The other specified insomnia disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for insomnia disorder or any specific sleep-wake disorder. This is done by recording “other specified insomnia disorder” followed by the specific reason (e.g., “short-term insomnia disorder”).

Examples of presentations that can be specified using the “other specified” designation include the following:

1. **Short-term insomnia disorder:** Duration is less than 3 months.
2. **Restricted to nonrestorative sleep:** Predominant complaint is nonrestorative sleep unaccompanied by other sleep symptoms such as difficulty falling asleep or remaining asleep.

Unspecified Insomnia Disorder

G47.00

This category applies to presentations in which symptoms characteristic of insomnia disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for insomnia disorder or any of the disorders in the sleep-wake disorders diagnostic class. The unspecified insomnia disorder category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for insomnia disorder or a specific sleep-wake disorder, and includes presentations in which there is insufficient information to make a more specific diagnosis.

Other Specified Hypersomnolence Disorder

G47.19

This category applies to presentations in which symptoms characteristic of hypersomnolence disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for hypersomnolence disorder or any of the disorders in the sleep-wake disorders diagnostic class. The other specified hypersomnolence disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for hypersomnolence disorder or any specific sleep-wake disorder. This is done by recording “other specified hypersomnolence disorder” followed by the specific reason (e.g., “brief-duration hypersomnolence,” as in Kleine-Levin syndrome).

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Unspecified Hypersomnolence Disorder

G47.10

This category applies to presentations in which symptoms characteristic of hypersomnolence disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for hypersomnolence disorder or any of the disorders in the

sleep-wake disorders diagnostic class. The unspecified hypersomnolence disorder category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for hypersomnolence disorder or a specific sleep-wake disorder, and includes presentations in which there is insufficient information to make a more specific diagnosis.

Other Specified Sleep-Wake Disorder

G47.8

This category applies to presentations in which symptoms characteristic of a sleep-wake disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the sleep-wake disorders diagnostic class and do not qualify for a diagnosis of other specified insomnia disorder or other specified hypersomnolence disorder. The other specified sleep-wake disorder category is used in situations in which the clinician chooses to communicate the specific reason that the presentation does not meet the criteria for any specific sleep-wake disorder. This is done by recording “other specified sleep-wake disorder” followed by the specific reason (e.g., “repeated arousals during rapid eye movement sleep without polysomnography or history of Parkinson’s disease or other synucleinopathy”).

Unspecified Sleep-Wake Disorder

G47.9

This category applies to presentations in which symptoms characteristic of a sleep-wake disorder that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning predominate but do not meet the full criteria for any of the disorders in the sleep-wake disorders diagnostic class and do not qualify for a diagnosis of unspecified insomnia disorder or unspecified hypersomnolence disorder. The unspecified sleep-wake disorder category is used in situations in which the clinician chooses *not* to specify the reason that the criteria are not met for a specific sleep-wake disorder, and includes presentations in which there is insufficient information to make a more specific diagnosis.

Sexual Dysfunctions

Sexual dysfunctions include delayed ejaculation, erectile disorder, female orgasmic disorder, female sexual interest/arousal disorder, genito-pelvic pain/penetration disorder, male hypoactive sexual desire disorder, premature (early) ejaculation, substance/medication-induced sexual dysfunction, other specified sexual dysfunction, and unspecified sexual dysfunction. Sexual dysfunctions are a heterogeneous group of disorders that are typically characterized by a clinically significant disturbance in a person's ability to respond sexually or to experience sexual pleasure. An individual may have several sexual dysfunctions at the same time. In such cases, all of the dysfunctions should be diagnosed.

Clinical judgment should be used to determine if the sexual difficulties are the result of inadequate sexual stimulation; in these cases, there may still be a need for care, but a diagnosis of a sexual dysfunction would not be made. These cases may include, but are not limited to, conditions in which lack of knowledge about effective stimulation prevents the experience of arousal or orgasm.

Subtypes are used to designate the onset of the difficulty. In many individuals with sexual dysfunctions, the time of onset may indicate different etiologies and interventions. *Lifelong* refers to a sexual problem that has been present from first sexual experiences, and *acquired* applies to sexual dysfunctions that develop after a period of relatively normal sexual function. *Generalized* refers to sexual difficulties that are not limited to certain types of stimulation, situations, or partners, and *situational* refers to sexual difficulties that only occur with certain types of stimulation, situations, or partners.

In addition to the lifelong/acquired and generalized/situational subtypes, a number of factors must be considered during the assessment of sexual dysfunction, given that they may be relevant to etiology or treatment and may contribute, to varying degrees, across individuals: 1) partner factors (e.g., partner's sexual problems; partner's health status); 2) relationship factors (e.g., poor communication; discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., poor body image; history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), or stressors (e.g., job loss, bereavement); 4) cultural or religious factors (e.g., inhibitions related to prohibitions against sexual activity or pleasure; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment.

Clinical judgment about the diagnosis of sexual dysfunction should take into consideration cultural factors that may influence expectations or engender prohibitions about the experience of sexual pleasure. Aging and relationship duration may be associated with a normative decrease in sexual response.

Sexual response has a requisite biological underpinning, yet is usually experienced in an intrapersonal, interpersonal, and cultural context. Thus, sexual function involves a complex interaction among biological, sociocultural, and psychological factors. In many clinical contexts, a precise understanding of the etiology of a sexual problem is unknown. Nonetheless, a sexual

dysfunction diagnosis requires ruling out problems that are better explained by a nonsexual mental disorder, by the effects of a substance (e.g., drug or

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medication), by a medical condition (e.g., due to pelvic nerve damage), or by severe relationship distress, partner violence, or other stressors.

The population of gender-diverse persons, including transgender, nonbinary, and agender, may not identify with or appear to fit into the existing sex- and gender-based diagnostic categories described in this chapter. Despite the names given to male hypoactive sexual desire disorder and female sexual interest arousal disorder, the diagnostic criteria describe symptoms and experiences that are not dependent on the individual's specific sex or gender. As such, either diagnosis can be applied to gender-diverse individuals based on clinical judgment. For diagnoses linked to reproductive anatomy (e.g., erectile dysfunction, premature [early] ejaculation, delayed ejaculation, and genito-pelvic pain/penetration disorder), diagnoses should be based on the individual's current anatomy and not on the individual's sex assigned at birth. Much more research is needed to understand experiences of sexual dysfunction among gender-diverse persons. In the meantime, as with all categories in DSM, clinicians should use their best judgment.

If the sexual dysfunction is mostly explainable by another nonsexual mental disorder (e.g., depressive or bipolar disorder, anxiety disorder, posttraumatic stress disorder, psychotic disorder), then only the other mental disorder diagnosis should be made. If the problem is thought to be better explained by the use/misuse or discontinuation of a drug or substance, it should be diagnosed accordingly as a substance/medication-induced sexual dysfunction. If the sexual dysfunction is attributable to another medical condition (e.g., peripheral neuropathy), the person would not receive a psychiatric diagnosis. If severe relationship distress, partner violence, or significant stressors better explain the sexual difficulties, then a sexual dysfunction diagnosis is not made, but an appropriate Z code for the relationship problem or stressor may be listed (e.g., Z63.0 Relationship distress with spouse or intimate partner); see the chapter "Other Conditions That May Be a Focus of Clinical Attention." In many cases, a precise etiological relationship between another condition (e.g., a medical condition) and a sexual dysfunction cannot be established. It is possible to have a diagnosis of a sexual dysfunction and a coexisting medical condition, nonsexual mental disorder, or use/misuse or discontinuation of a drug or substance; and it is possible to have one or more diagnoses of sexual dysfunction.

Delayed Ejaculation

Diagnostic Criteria

F52.32

- A. Either of the following symptoms must be experienced on almost all or all occasions (approximately 75%–100%) of partnered sexual activity (in identified situational contexts or, if generalized, in all contexts), and without the individual desiring delay:

1. Marked delay in ejaculation.
 2. Marked infrequency or absence of ejaculation.
- B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months.
- C. The symptoms in Criterion A cause clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

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Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

The essential feature of delayed ejaculation is a marked delay in or inability to achieve ejaculation or marked infrequency of ejaculation on all or almost all occasions of partnered sexual activity, despite the presence of adequate sexual stimulation and the desire to ejaculate (Criterion A). In order to qualify for a DSM-5 diagnosis of delayed ejaculation, the symptoms must have persisted for a minimum duration of approximately 6 months (Criterion B) and must cause clinically significant distress in the individual (Criterion C). The partnered sexual activity may include manual, oral, coital, or anal stimulation. In most cases, the diagnosis will be made by self-report, although for men in heterosexual partnered relationships, it is frequently the female partner's distress that motivates treatment seeking. It is common for men who present with delayed ejaculation to be able to ejaculate with self-stimulation, but not during partnered sexual activity.

The definition of "delay" does not have precise boundaries, as there is no consensus as to what constitutes a reasonable time to reach orgasm or what is unacceptably long for most men and their sexual partners. Although the definitions of delayed ejaculation apply equally well to

both heterosexual and homosexual orientation, the vast majority of the research focus has been based on the concept of intravaginal latency, and therefore male-female intercourse. The findings from those studies document that the majority of men's intravaginal ejaculatory latency time (IELT) range is approximately 4–10 minutes. There is also no clear diagnostic delineation between delayed ejaculation as a sexual dysfunction and delay that is a consequence of normal aging. Therefore, the diagnosis of delayed ejaculation is based on clinical judgment, taking into consideration the individual's psychosexual and medical history, age, relationship context, and sexual stimulation patterns and behaviors. The diagnosis of delayed ejaculation should not be given if the clinician judges that the individual's dissatisfaction is entirely attributable to unrealistic expectations.

Associated Features

The man and his partner may report prolonged thrusting to achieve orgasm to the point of exhaustion or genital discomfort and sometimes even injury to himself and/or his partner before finally ceasing. Some males may report avoiding sexual activity because of a repetitive pattern of difficulty ejaculating.

Delayed ejaculation is associated with highly frequent masturbation, use of masturbation techniques not easily duplicated by a partner, and marked disparities between sexual fantasies during masturbation and the reality of sex with a partner.

Males with delayed ejaculation typically report less coital activity, higher levels of relationship distress, sexual dissatisfaction, lower subjective arousal, anxiety about their sexual performance, and general health issues than sexually functional men.

In addition to considerations of applicable subtypes (i.e., whether the ejaculatory delay has been present since the individual became sexually active or began after a period of relatively normal sexual function, and whether the ejaculatory delay is generalized or occurs only with certain types of stimulation, situations, or partners), the following factors are important to consider in the assessment of delayed ejaculation: 1) partner factors (e.g., partner's sexual problems or health); 2) relationship factors (e.g., poor communication, discrepancies

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in desire for sexual activity); 3) individual vulnerability factors (e.g., hypoactive sexual desire), psychiatric comorbidity (e.g., depression, anxiety), or stressors such as job loss or stress; 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); 5) medical factors, particularly hypogonadism or neurological disorders (e.g., multiple sclerosis, diabetic neuropathy); and 6) use of substances or medications that might inhibit ejaculation (e.g., use of serotonergic drugs).

Prevalence

The prevalence of delayed ejaculation in the United States is estimated at 1%–5% but has ranged as high as 11% in international studies. However, variation in syndrome definitions across studies may have contributed to differences in the prevalence of the DSM-5 disorder.

Development and Course

Lifelong delayed ejaculation begins with early sexual experiences and continues throughout the course of an individual's life. Acquired delayed ejaculation begins after a period of normal sexual function. A number of biomedical, psychosocial, and cultural factors can contribute to the predisposition to or maintenance of lifelong or acquired delayed ejaculation (see the section "Risk and Prognostic Factors"), and either subtype can be generalized or situational in nature.

The prevalence of delayed ejaculation increases with age. As males age, they are more likely to have progressively more of the following changes in ejaculatory function, including, but not limited to, reduced ejaculatory volume, force, and sensation, and increased "refractory time." Refractory latency increases for males secondary to surgical, medical, and pharmaceutical complications, as well as aging.

Risk and Prognostic Factors

Ejaculatory latency is an end point consequence that is determined by a range of factors. A large number of psychosocial factors increase the probability of an individual experiencing delayed ejaculation, with depression and relationship dissatisfaction being predominant contributors.

Genetic and physiological. Numerous medical conditions may lead to delayed ejaculation, including procedures that disrupt sympathetic or somatic innervation to the genital region such as radical prostatectomy for cancer treatment. Neurological and endocrine disorders, including spinal cord injury, stroke, multiple sclerosis, pelvic-region surgery, severe diabetes, epilepsy, hormonal abnormalities, and sleep apnea, as well as alcohol abuse, bowel dysfunction, cannabis use, and environmental factors, may be associated with delayed ejaculation.

Additionally, medications that inhibit α -adrenergic innervation of the ejaculatory system (e.g., tamsulosin) are associated with delayed ejaculation, as well as antihypertensive agents, antidepressants (e.g., selective serotonin reuptake inhibitors), and antipsychotic drugs.

Age-related loss of the fast-conducting peripheral sensory nerves and age-related decreased sex steroid secretion may be associated with an increase in delayed ejaculation in males as they age. Reduced androgen levels with age may also be associated with delayed ejaculation.

Sex- and Gender-Related Diagnostic Issues

By definition, the diagnosis of delayed ejaculation is only given to males. Distressing difficulties with orgasm in women would be considered under female orgasmic disorder.

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Functional Consequences of Delayed Ejaculation

Delayed ejaculation is often associated with considerable psychological distress in one or both partners.

Difficulty with ejaculation may contribute to difficulties in conception and lead to significant fertility assessment, as the lack of ejaculation is not often spontaneously discussed by individuals unless there is direct inquiry from their physician.

Differential Diagnosis

Another medical condition or injury and/or its treatment. A major diagnostic challenge is

differentiating between a delayed ejaculation that is fully explained by another medical condition or injury (or its treatment) and a delayed ejaculation attributable to a variety of proportionally different biomedical-psychosocial and cultural factors that determine the symptom(s). A number of medical conditions or injuries, along with their treatments, may produce delays in ejaculation independent of psychosocial and cultural issues.

Delayed ejaculation must be differentiated from a number of urological conditions (especially other ejaculatory disorders), including retrograde ejaculation or *anejaculation*, which is typically the result of etiologies ranging from hormonal to neurological and/or anatomical abnormalities, including ejaculatory duct obstruction and other urological disorders.

Substance/medication use. A number of pharmacological agents, such as antidepressants, antipsychotics, α sympathetic drugs, alcohol, and opioid drugs, can cause ejaculatory problems. In such cases, the diagnosis is substance/medication-induced sexual dysfunction instead of delayed ejaculation.

Dysfunction with orgasm. It is important in the history to ascertain whether the complaint concerns delayed ejaculation or the sensation of orgasm, or both. Ejaculation occurs in the genitals, whereas the experience of orgasm is believed to be primarily subjective. Ejaculation and orgasm usually occur together but not always. For example, a male with a normal ejaculatory pattern may complain of decreased pleasure (i.e., anhedonic ejaculation). Such a complaint would not be coded as delayed ejaculation but could be coded as other specified sexual dysfunction or unspecified sexual dysfunction.

Comorbidity

There is some evidence to suggest that delayed ejaculation may be more common in severe forms of major depressive disorder.

Erectile Disorder

Diagnostic Criteria	F52.21
A. At least one of the three following symptoms must be experienced on almost all or all (approximately 75%–100%) occasions of sexual activity (in identified situational contexts or, if generalized, in all contexts): 1. Marked difficulty in obtaining an erection during sexual activity. 2. Marked difficulty in maintaining an erection until the completion of sexual activity. 3. Marked decrease in erectile rigidity. B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months. C. The symptoms in Criterion A cause clinically significant distress in the individual.	
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- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

The essential feature of erectile disorder is a marked difficulty in obtaining or maintaining an erection or a marked decrease in erectile rigidity in all or almost all occasions of sexual activity (Criterion A) that has persisted for at least 6 months (Criterion B) and that causes clinically significant distress in the individual (Criterion C). A careful sexual history is necessary to ascertain that the problem has been present for a significant duration of time (i.e., at least approximately 6 months) and occurs on the majority of sexual occasions (i.e., at least 75% of the time). Symptoms may occur only in specific situations involving certain types of stimulation or partners, or they may occur in a generalized manner in all types of situations, stimulation, or partners.

This chapter uses the terms *erectile disorder* and *erectile dysfunction*, which are not synonymous. *Erectile dysfunction* is a widely used descriptive term (including in ICD-10) that refers to difficulty getting and maintaining an erection. *Erectile disorder* is the more specific DSM-5 diagnostic category in which erectile dysfunction persists for at least 6 months and causes distress in the individual.

Associated Features

Many males with erectile disorder may have low self-esteem, low self-confidence, and a decreased sense of masculinity, and may experience depressed mood. Erectile dysfunction is also strongly associated with feelings of guilt, self-blame, sense of failure, anger, and concern about disappointing one's partner. Fear and/or avoidance of future sexual encounters may occur. Decreased sexual satisfaction and reduced sexual desire in the individual's partner are common.

In addition to considerations of applicable subtypes (i.e., whether the erectile dysfunction has

been present since the individual became sexually active or began after a period of relatively normal sexual function, and whether the erectile dysfunction is generalized or occurs only with certain types of stimulation, situations, or partners), the following factors are important to consider in the assessment of erectile disorder: 1) partner factors (e.g., partner's sexual problems or health); 2) relationship factors (e.g., poor communication, discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., hypoactive sexual desire), psychiatric comorbidity (e.g., depression, anxiety), or stressors such as job loss or stress; 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); 5) medical factors, particularly surgery (e.g., transurethral resection of the prostate), hypogonadism, or neurological conditions

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(e.g., multiple sclerosis, diabetic neuropathy); and 6) use of substances or medications that might inhibit ejaculation (e.g., use of serotonergic drugs).

Prevalence

The prevalence of lifelong versus acquired erectile disorder is unknown. There is a strong age-related increase in both prevalence and incidence of problems with erection, particularly after age 50 years. Internationally, the prevalence of erectile disorder in the general population is approximately 13%–21% of males ages 40–80 years. Rates appear to be lower than 10% in males younger than 40 years, about 20%–40% in males in their 60s, and 50%–75% in males older than 70 years. In a longitudinal study in Australia, 80% of males age 70 and older experienced erectile disorder. In a review of studies largely from Western countries, about 20% of males feared erectile problems on their first sexual experience, whereas approximately 8% experienced erectile problems that hindered penetration during their first sexual experience. Among U.S.-based respondents to an online survey, there was no statistically significant difference in the prevalence of erectile disorder by ethnoracial background. Nationally representative U.S. data show that the prevalence of erectile difficulties is similar in older males who have sex with males or with both males and females.

Development and Course

Erectile failure on first sexual attempt has been found to be related to having sex with a previously unknown partner, concomitant use of drugs or alcohol, not wanting to have sex, and peer pressure. There is minimal evidence regarding the persistence of such problems after the first attempt. It is assumed that most of these problems spontaneously remit without professional intervention, but some males may continue to have episodic problems. In contrast, acquired erectile disorder is often associated with biological factors such as diabetes and cardiovascular disease. Acquired erectile disorder is likely to be persistent in most men.

The natural history of lifelong erectile disorder is unknown. Clinical observation supports the association of lifelong erectile disorder with psychological factors that are self-limiting or responsive to psychological interventions, whereas, as noted above, acquired erectile disorder is more likely to be related to biological factors and to be persistent. The incidence of erectile disorder increases with age. A minority of males diagnosed as having moderate erectile failure

may experience spontaneous remission of symptoms without medical intervention. Distress associated with erectile disorder is lower in older males as compared with younger males.

Risk and Prognostic Factors

Course modifiers. Risk factors for acquired erectile dysfunction and, as a consequence, erectile disorder include age, smoking tobacco, lack of physical exercise, diabetes, and decreased desire.

Culture-Related Diagnostic Issues

Prevalence of erectile disorder varies across countries. It is unclear to what extent these variations represent differences in cultural expectations as opposed to genuine differences in the frequency of erectile failure. Differential endorsement may be related to cultural concerns about appearing weak or less masculine or to diverse cultural norms about changes in erectile function during healthy aging. Cultural expectations concerning marital relationships, sexual performance, fertility, and gender roles can influence anxieties that may contribute to erectile disorder. Based on responses to an online survey, erectile disorder may be associated with concern about genital size in the United States and the Middle East and with fears of male infertility more frequently in the Middle East.

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Sex- and Gender-Related Diagnostic Issues

By definition, the diagnosis of erectile dysfunction is only given to males. Distressing difficulties with sexual arousal in women would be considered under female sexual interest/arousal disorder.

Diagnostic Markers

Nocturnal penile tumescence testing and measured erectile turgidity during sleep can be employed to help differentiate organic from psychogenic erectile problems on the assumption that adequate erections during rapid eye movement sleep indicate a psychological etiology to the problem. A number of other diagnostic procedures may be employed depending on the clinician's assessment of their relevance given the individual's age, comorbid medical problems, and clinical presentation. Doppler ultrasonography and intravascular injection of vasoactive drugs, as well as invasive diagnostic procedures such as dynamic infusion cavernosography, can be used to assess vascular integrity. Pudendal nerve conduction studies, including somatosensory evoked potentials, can be employed when a peripheral neuropathy is suspected. Testing for low levels of serum bioavailable or free testosterone is appropriate especially when diabetes is present, for men who also experience hypoactive desire, and for those who do not respond to phosphodiesterase type 5 inhibitors. Thyroid function may also be assessed. Determination of fasting serum glucose is useful to screen for the presence of diabetes mellitus. The assessment of serum lipids is important, as erectile disorder in males 40 years and older is predictive of the future risk for coronary artery disease.

Association With Suicidal Thoughts or Behavior

Among males receiving treatment for erectile disorder with comorbid depression, elevated rates

of suicidal thoughts or behavior have been observed; while the affected males attributed the suicidal symptoms to their erectile disorder, the presence of depression was also a likely contributing factor. Elevated suicide rates among males with prostate cancer may in part be related to treatment-associated erectile dysfunction and consequent depressive symptoms.

Functional Consequences of Erectile Disorder

Erectile disorder can interfere with fertility and produce both individual and interpersonal distress. Fear and/or avoidance of sexual encounters may interfere with the ability to develop intimate relationships. Significant psychological distress may occur among males presenting with erectile disorder.

Differential Diagnosis

Nonsexual mental disorders. Major depressive disorder and erectile disorder are closely associated, and erectile disorder accompanying severe depressive disorder may occur. If the erectile difficulties are better explained by another mental disorder, such as major depression, then a diagnosis of erectile disorder would not be made.

Normal erectile function. The differential should include consideration of normal erectile function in males with excessive expectations.

Substance/medication use. An onset of erectile dysfunction that coincides with the beginning of substance/medication use and that dissipates with discontinuation of the substance/medication or dose reduction is suggestive of a substance/medication-induced sexual dysfunction, which should be diagnosed instead of erectile disorder.

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Another medical condition. The most difficult aspect of the differential diagnosis of erectile disorder is ruling out erectile problems that are fully explained by medical factors. Such cases would not receive a diagnosis of a mental disorder. The distinction between erectile disorder as a mental disorder and erectile dysfunction as the result of another medical condition is usually unclear, and many cases will have complex, interactive biological and psychiatric etiologies. If the individual is older than 40–50 years and/or has concomitant medical problems, the differential diagnosis should include medical etiologies, especially vascular disease. The presence of an organic disease known to cause erectile problems does not confirm a causal relationship. For example, a male with diabetes mellitus can develop erectile disorder in response to psychological stress. In general, erectile dysfunction due to organic factors is generalized and gradual in onset. An exception would be erectile problems after traumatic injury to the nervous innervation of the genital organs (e.g., spinal cord injury). Erectile problems that are situational and inconsistent and that have an acute onset after a stressful life event are most often attributable to psychological events. An age younger than 40 years is also suggestive of a psychological etiology to the difficulty.

Comorbidity

Erectile disorder can be comorbid with other sexual diagnoses, such as premature (early)

ejaculation and male hypoactive sexual desire disorder, as well as with anxiety and depressive disorders. The risk of depression is significantly higher in males with erectile disorder, with a markedly higher risk of depression in the first year after onset. In males diagnosed with posttraumatic stress disorder, erectile problems are common. Erectile disorder is common in males with lower urinary tract symptoms related to prostatic hypertrophy. Erectile disorder may be comorbid with dyslipidemia, cardiovascular disease, hypogonadism, multiple sclerosis, diabetes mellitus, and other diseases that interfere with the vascular, neurological, or endocrine function necessary for normal erectile function.

Female Orgasmic Disorder

Diagnostic Criteria

F52.31

- A. Presence of either of the following symptoms and experienced on almost all or all (approximately 75%–100%) occasions of sexual activity (in identified situational contexts or, if generalized, in all contexts):
 - 1. Marked delay in, marked infrequency of, or absence of orgasm.
 - 2. Markedly reduced intensity of orgasmic sensations.
- B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months.
- C. The symptoms in Criterion A cause clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress (e.g., partner violence) or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

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Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify if:

Never experienced an orgasm under any situation.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

Female orgasmic disorder is characterized by difficulty experiencing orgasm and/or markedly reduced intensity of orgasmic sensations (Criterion A). Women show wide variability in the type or intensity of stimulation that elicits orgasm. Similarly, subjective descriptions of orgasm are extremely varied, suggesting that it is experienced in very different ways, both across women and on different occasions by the same woman. For a diagnosis of female orgasmic disorder, symptoms must be experienced on almost all or all (approximately 75%–100%) occasions of sexual activity (in identified situational contexts or, if generalized, in all contexts) and have a minimum duration of approximately 6 months. The use of the minimum severity and duration criteria is intended to distinguish transient orgasm difficulties from more persistent orgasmic dysfunction. The inclusion of “approximately” in Criterion B allows for clinician judgment in cases in which symptom duration does not meet the recommended 6-month threshold.

For a woman to have a diagnosis of female orgasmic disorder, clinically significant distress must accompany the symptoms (Criterion C). In many cases of orgasm problems, the causes are multifactorial or cannot be determined. If female orgasmic disorder is deemed to be better explained by another mental disorder, the effects of a substance/medication, or a medical condition, then a diagnosis of female orgasmic disorder would not be made. Finally, if interpersonal or significant contextual factors, such as severe relationship distress, intimate partner violence, or other significant stressors, are present, then a diagnosis of female orgasmic disorder would not be made.

Many women require clitoral stimulation to reach orgasm, and a relatively small proportion of women report that they always experience orgasm during penile-vaginal intercourse. Thus, a woman’s experiencing orgasm through clitoral stimulation but not during intercourse does not meet criteria for a clinical diagnosis of female orgasmic disorder. It is also important to consider whether orgasm difficulties are the result of inadequate sexual stimulation; in these cases, there may still be a need for care, but a diagnosis of female orgasmic disorder would not be made.

Associated Features

Associations between specific patterns of personality traits or psychopathology and orgasmic dysfunction have generally not been supported. Compared with women without the disorder, some women with female orgasmic disorder may have greater difficulty communicating about sexual issues. Overall sexual satisfaction, however, is not strongly correlated with orgasmic experience. Many women report high levels of sexual satisfaction despite rarely or never experiencing orgasm. Orgasm difficulties in women often co-occur with problems related to sexual interest and arousal.

In addition to the subtypes “lifelong/acquired” and “generalized/situational,” the following five factors must be considered during assessment and diagnosis of female orgasmic disorder given that they may be relevant to etiology or treatment: 1) partner factors (e.g., partner’s sexual problems, partner’s health status); 2) relationship factors (e.g., poor

communication, discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., poor body image, history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), or stressors (e.g., job loss, bereavement); 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment. Each of these factors may contribute differently to the presenting symptoms of different women with this disorder.

Prevalence

Reported prevalence rates for orgasm problems in premenopausal women vary widely, from 8% to 72%, depending on multiple factors (e.g., age, cultural background and context, duration, severity of symptoms); however, these estimates do not take into account the presence of distress. Only a proportion of women experiencing orgasm difficulties also report associated distress. Variation in how symptoms are assessed (e.g., the duration of symptoms and the recall period) also influence prevalence rates. Internationally, approximately 10% of women do not experience orgasm throughout their lifetime.

Development and Course

By definition, lifelong female orgasmic disorder indicates that the orgasm difficulties have always been present, whereas the acquired subtype would be assigned if the woman's orgasm difficulties developed after a period of normal orgasmic functioning.

A woman's first experience of orgasm can occur any time from the prepubertal period to well into adulthood. Women show a more variable pattern in age at first orgasm than do men, and women's reports of having experienced orgasm increase with age. Many women learn to experience orgasm as they experience a wide variety of stimulation and acquire more knowledge about their bodies. Women's rates of orgasmic consistency (defined as "usually or always" experiencing orgasm) are higher during masturbation than during sexual activity with a partner.

Risk and Prognostic Factors

Temperamental. A wide range of psychological factors, such as anxiety and concerns about pregnancy, can potentially interfere with a woman's ability to experience orgasm.

Environmental. There is a strong association between relationship problems, physical health, and mental health and orgasm difficulties in women. Sociocultural factors (e.g., gender role expectations and religious norms) are also important influences on the experience of orgasm difficulties.

Genetic and physiological. Many physiological factors may influence a woman's experience of orgasm, including medical conditions and medications. Conditions such as multiple sclerosis, pelvic nerve damage from radical hysterectomy, and spinal cord injury can all influence orgasmic functioning in women. Selective serotonin reuptake inhibitors are known to delay or inhibit orgasm in women. Women with vulvovaginal atrophy (characterized by symptoms such as vaginal dryness, itching, and pain) are significantly more likely to report orgasm difficulties than are women without this condition. Menopausal status is not consistently associated with the

likelihood of orgasm difficulties. There may be a significant genetic contribution to variation in female orgasmic function. However, psychological, sociocultural, and physiological factors likely interact in complex ways to influence women's experience of orgasm and of orgasm difficulties.

Culture-Related Diagnostic Issues

The degree to which lack of orgasm in women is regarded as a problem that requires treatment may vary depending on cultural context. Cultural views that undervalue women's

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sexual satisfaction or that perceive marital sex as a duty for women rather than a pleasurable activity are associated with lower help-seeking. In addition, women differ in how important orgasm is to their sexual satisfaction. There may be marked sociocultural and generational differences in women's orgasmic ability. For example, reports of the prevalence of inability to reach orgasm vary over a twofold range across world regions.

Sex- and Gender-Related Diagnostic Issues

By definition, the diagnosis of female orgasmic disorder is given only to women. Distressing difficulties with orgasm in men would be considered under delayed ejaculation.

Diagnostic Markers

Although measurable physiological changes occur during female orgasm, including changes in hormones, pelvic floor musculature, and brain activation, there is significant variability in these indicators of orgasm across women. In clinical situations, the diagnosis of female orgasmic disorder is based on a woman's self-report.

Association With Suicidal Thoughts or Behavior

Dysfunctions of sexual arousal and satisfaction have been associated with suicidal thoughts among female veterans and military service members even after adjustment for probable posttraumatic stress disorder, probable depression, history of deployment, married status, age, service in the army, and race.

Functional Consequences of Female Orgasmic Disorder

The functional consequences of female orgasmic disorder are unclear. Although there is a strong association between relationship problems and orgasm difficulties in women, it is unclear whether relationship factors are risk factors for orgasm difficulties or are consequences of those difficulties.

Differential Diagnosis

Nonsexual mental disorders. If the orgasm difficulties are better explained by another mental disorder, such as major depression, then a diagnosis of female orgasmic disorder would not be made.

Substance/medication-induced sexual dysfunction. An onset of orgasmic dysfunction that coincides with the beginning of substance/medication use and that dissipates with discontinuation of the substance/medication or dose reduction is suggestive of a substance/medication-induced sexual dysfunction, which should be diagnosed instead of female orgasmic disorder.

Another medical condition. If the disorder is attributable to another medical condition (e.g., multiple sclerosis, spinal cord injury), then a diagnosis of female orgasmic disorder would not be made.

Interpersonal factors. If interpersonal or significant contextual factors, such as severe relationship distress, intimate partner violence, or other significant stressors, are associated with the orgasm difficulties, then a diagnosis of female orgasmic disorder would not be made.

Other sexual dysfunctions. Female orgasmic disorder may occur in association with other sexual dysfunctions (e.g., female sexual interest/arousal disorder). The presence of another sexual dysfunction does not rule out a diagnosis of female orgasmic disorder. Occasional orgasm difficulties that are short-term or infrequent and are not accompanied by clinically significant distress or impairment are not diagnosed as female orgasmic disorder. A diagnosis is also not appropriate if the problems are the result of inadequate sexual stimulation.

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Comorbidity

Women with female orgasmic disorder may have co-occurring sexual interest/arousal difficulties. Women with diagnoses of other nonsexual mental disorders, such as major depressive disorder, may experience lower sexual interest/arousal, and this may indirectly increase the likelihood of orgasm difficulties.

Female Sexual Interest/Arousal Disorder

Diagnostic Criteria	F52.22
A. Lack of, or significantly reduced, sexual interest/arousal, as manifested by at least three of the following: 1. Absent/reduced interest in sexual activity. 2. Absent/reduced sexual/erotic thoughts or fantasies. 3. No/reduced initiation of sexual activity, and typically unreceptive to a partner's attempts to initiate. 4. Absent/reduced sexual excitement/pleasure during sexual activity in almost all or all (approximately 75%–100%) sexual encounters (in identified situational contexts or, if generalized, in all contexts). 5. Absent/reduced sexual interest/arousal in response to any internal or external sexual/erotic cues (e.g., written, verbal, visual). 6. Absent/reduced genital or nongenital sensations during sexual activity in	

almost all or all (approximately 75%–100%) sexual encounters (in identified situational contexts or, if generalized, in all contexts).

- B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months.
- C. The symptoms in Criterion A cause clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress (e.g., partner violence) or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

In assessing female sexual interest/arousal disorder, interpersonal context must be taken into account. A “desire discrepancy,” in which a woman has lower desire for sexual activity than her partner, is not sufficient to diagnose female sexual interest/arousal disorder. For the criteria for the disorder to be met, there must be absence or reduced frequency or

intensity of at least three of six indicators (Criterion A) for a minimum duration of approximately 6 months (Criterion B). There may be different symptom profiles across women, as well as variability in how sexual interest and arousal are expressed. For example, in one woman, sexual interest/arousal disorder may be expressed as a lack of interest in sexual activity, an absence of erotic or sexual thoughts, and reluctance to initiate sexual activity and respond to a partner’s sexual invitations. In another woman, an inability to become sexually excited, an inability to respond to sexual stimuli with sexual desire, and a corresponding lack of signs of physical sexual arousal may be the primary features. Difficulties in desire and sexual arousal may also occur simultaneously, as women with loss of sexual desire may be nine times more likely to also have lost sexual excitement or arousal. Short-term changes in sexual interest or arousal are common

and may be adaptive responses to events in a woman's life and do not represent a sexual dysfunction. Diagnosis of female sexual interest/arousal disorder requires a minimum duration of symptoms of approximately 6 months as a reflection that the symptoms must be a persistent problem. The estimation of persistence may be determined by clinical judgment when a duration of 6 months cannot be ascertained precisely.

There may be absent or reduced frequency or intensity of interest in sexual activity (Criterion A1), which was previously the single criterion for *hypoactive sexual desire disorder*; this condition is now represented by female sexual interest/arousal disorder. The frequency or intensity of sexual and erotic thoughts or fantasies may be absent or reduced (Criterion A2). The expression of fantasies varies widely across women and may include memories of past sexual experiences. The normative decline in sexual thoughts with age should be taken into account when this criterion is being assessed. Absence or reduced frequency of initiating sexual activity and of receptivity to a partner's sexual invitations (Criterion A3) is a behaviorally focused criterion. A couple's beliefs and preferences for sexual initiation patterns are highly relevant to the assessment of this criterion. There may be absent or reduced sexual excitement or pleasure during sexual activity in almost all or all (approximately 75%–100%) sexual encounters (Criterion A4). Lack of pleasure is a common presenting clinical complaint in women with low desire. Among women who report low sexual desire, there are fewer sexual or erotic cues that elicit sexual interest or arousal (i.e., there is a lack of "responsive desire"). Evidence suggests that there may be at least two types of female sexual interest/arousal disorder: one based on low sensitivity to sexual cues and a second based on overactivation of sexual inhibition. Assessment of the adequacy of sexual stimuli will assist in determining if there is a difficulty with responsive sexual desire (Criterion A5). Frequency or intensity of genital or nongenital sensations during sexual activity may be reduced or absent (Criterion A6). This may include reduced vaginal lubrication/vasocongestion, but because physiological measures of genital sexual response do not differentiate women who report sexual arousal concerns from those who do not, the self-report of reduced or absent genital or nongenital sensations is sufficient.

For a diagnosis of female sexual interest/arousal disorder to be made, clinically significant distress must accompany the symptoms in Criterion A. Distress may be experienced as a result of the lack of sexual interest/arousal or as a result of significant interference in a woman's life and well-being. If a lifelong lack of sexual desire is better explained by one's self-identification as "asexual," then a diagnosis of female sexual interest/arousal disorder would not be made.

Associated Features

Female sexual interest/arousal disorder is frequently associated with problems in experiencing orgasm, pain experienced during sexual activity, infrequent sexual activity, and couple-level discrepancies in desire. Relationship difficulties, chronic stress, and mood disorders are also frequently associated features of female sexual interest/arousal disorder. Unrealistic expectations and norms regarding the "appropriate" level of sexual

interest or arousal, along with poor sexual techniques and lack of information about sexuality, may also be evident in women diagnosed with female sexual interest/arousal disorder. The latter,

as well as normative beliefs about gender roles, are important factors to consider.

In addition to the subtypes “lifelong/acquired” and “generalized/situational,” the following five factors must be considered during assessment and diagnosis of female sexual interest/arousal disorder given that they may be relevant to etiology and/or treatment: 1) partner factors (e.g., partner’s sexual problems, partner’s health status, partner-related distress); 2) relationship factors (e.g., poor communication, discrepancies in desire for sexual activity; relationship duration); 3) individual vulnerability factors (e.g., poor body image, history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), or stressors (e.g., job loss, bereavement); 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment. Note that each of these factors may contribute differently to the presenting symptoms of different women with this disorder.

Prevalence

Approximately 30% of women experience chronic low desire, with approximately half of these experiencing significant partner-related distress and a quarter experiencing personal distress. The prevalence of low sexual desire and of problems with sexual arousal (with and without associated distress) may vary markedly in relation to age, cultural context, duration of symptoms, and presence of distress. Regarding duration of symptoms, there are striking differences in prevalence estimates between short-term and persistent problems related to lack of sexual interest. When distress about sexual functioning is required, prevalence estimates are markedly lower. Though there is a strong relationship between low desire and age, the prevalence of sex-related distress associated with low desire decreases as women age.

Development and Course

By definition, lifelong female sexual interest/arousal disorder suggests that the lack of sexual interest or arousal has been present for the woman’s entire sexual life. For Criteria A3, A4, and A6, which assess functioning during sexual activity, a subtype of lifelong would mean presence of symptoms since the individual’s first sexual experiences. The acquired subtype would be assigned if the difficulties with sexual interest or arousal developed after a period of nonproblematic sexual functioning. Adaptive and normative changes in sexual functioning may result from partner-related, interpersonal, or personal events and may be transient in nature. However, persistence of symptoms for approximately 6 months or more would constitute a sexual dysfunction.

There are normative changes in sexual interest and arousal across the life span. Furthermore, women in relationships of longer duration are more likely to report engaging in sex despite no obvious feelings of sexual desire at the outset of a sexual encounter compared with women in shorter-duration relationships. Vaginal dryness in older women is related to age and menopausal status.

Risk and Prognostic Factors

Temperamental. Temperamental factors include negative cognitions and attitudes about sexuality and past history of mental disorders. Differences in propensity for sexual excitation and sexual

inhibition may also predict the likelihood of developing sexual problems.

Environmental. Environmental factors include relationship difficulties, partner sexual functioning, and developmental history, such as early relationships with caregivers and childhood stressors.

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Genetic and physiological. Some medical conditions (e.g., diabetes mellitus, thyroid dysfunction) can be risk factors for female sexual interest/arousal disorder. There appears to be a strong influence of genetic factors on vulnerability to sexual problems in women. Psychophysiological research using vaginal photoplethysmography has not found differences between women with and without perceived lack of genital arousal.

Culture-Related Diagnostic Issues

There is marked variability in reported prevalence rates of low desire across world regions, ranging from 26% to 43%. Low levels of sexual desire have been reported by some ethnoracial and migrant groups. Although lower levels of reported sexual desire and arousal may reflect less interest in sex, such group differences may be an artifact of the measures used to quantify desire and of cultural factors affecting response, such as the desirability of reporting sexual activity by nonmarried, menopausal, or widowed women. A judgment about whether low sexual desire reported by a woman from a certain ethnocultural background meets criteria for female sexual interest/arousal disorder must take into account the fact that different cultural groups may vary in norms and expectations for sexual behavior.

Sex- and Gender-Related Diagnostic Issues

By definition, the diagnosis of female sexual interest/arousal disorder is only given to women. Distressing difficulties with sexual desire in men would be considered under male hypoactive sexual desire disorder. There are no data showing that rates or expressions of female sexual interest/arousal disorder differ between heterosexual and lesbian women.

Association With Suicidal Thoughts or Behavior

Dysfunctions of sexual arousal and satisfaction have been associated with suicidal thoughts among female veterans and military service members even after adjustment for probable PTSD, probable depression, history of deployment, married status, age, service in the army, and race.

Functional Consequences of Female Sexual Interest/Arousal Disorder

Difficulties in sexual interest/arousal are often associated with decreased relationship satisfaction.

Differential Diagnosis

Nonsexual mental disorders. Nonsexual mental disorders, such as major depressive disorder, in which there is “markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day,” may explain the lack of sexual interest/arousal. If the lack of interest

or arousal is completely attributable to another mental disorder, then a diagnosis of female sexual interest/arousal disorder would not be made.

Substance/medication use. An onset of difficulties in desire or arousal that coincides with the beginning of substance/medication use and that dissipates with discontinuation of the substance/medication or dose reduction is suggestive of a substance/medication-induced sexual dysfunction, which should be diagnosed instead of female sexual interest/arousal disorder.

Another medical condition. If the sexual symptoms are considered to be almost exclusively associated with the effects of another medical condition (e.g., diabetes mellitus,

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endothelial disease, thyroid dysfunction, central nervous system disease), then a diagnosis of female sexual interest/arousal disorder would not be made.

Interpersonal factors. If interpersonal or significant contextual factors, such as severe relationship distress, intimate partner violence, or other significant stressors, explain the sexual interest/arousal symptoms, then a diagnosis of female sexual interest/arousal disorder would not be made.

Other sexual dysfunctions. The presence of another sexual dysfunction does not rule out a diagnosis of female sexual interest/arousal disorder. It is common for women to experience more than one sexual dysfunction. For example, the presence of chronic genital pain may lead to a lack of desire for the (painful) sexual activity. Lack of interest and arousal during sexual activity may impair orgasmic ability. For some women, all aspects of the sexual response may be unsatisfying and distressing.

Inadequate or absent sexual stimuli. When differential diagnoses are being considered, it is important to assess the adequacy of sexual stimuli within the woman's sexual experience. In cases where inadequate or absent sexual stimuli are contributing to the clinical picture, there may be evidence for clinical care, but a sexual dysfunction diagnosis would not be made. Similarly, transient and adaptive alterations in sexual functioning that are secondary to a significant life or personal event must be considered in the differential diagnosis.

Comorbidity

Comorbidity between sexual interest/arousal problems and other sexual difficulties is extremely common. Sexual distress and dissatisfaction with sex life are also highly correlated in women with low sexual desire. Distressing low desire is associated with depression, thyroid problems, anxiety, urinary incontinence, and other medical factors. Arthritis and inflammatory or irritable bowel disease are also associated with sexual arousal problems. Low desire appears to be comorbid with depression, sexual and physical abuse in adulthood, and use of alcohol.

Genito-Pelvic Pain/Penetration Disorder

Diagnostic Criteria

F52.6

- A. Persistent or recurrent difficulties with one (or more) of the following:
 - 1. Vaginal penetration during intercourse.
 - 2. Marked vulvovaginal or pelvic pain during vaginal intercourse or penetration attempts.
 - 3. Marked fear or anxiety about vulvovaginal or pelvic pain in anticipation of, during, or as a result of vaginal penetration.
 - 4. Marked tensing or tightening of the pelvic floor muscles during attempted vaginal penetration.
- B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months.
- C. The symptoms in Criterion A cause clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of a severe relationship distress (e.g., partner violence) or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

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Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

Genito-pelvic pain/penetration disorder refers to four commonly comorbid symptom dimensions: 1) difficulty having intercourse, 2) genito-pelvic pain, 3) fear of pain or vaginal penetration, and 4) tension of the pelvic floor muscles (Criterion A). Because major difficulty in any one of these symptom dimensions is often sufficient to cause clinically significant distress, a diagnosis can be made on the basis of marked difficulty in only one symptom dimension. However, all four symptom dimensions should be assessed even if a diagnosis can be made on the basis of only one symptom dimension.

Marked difficulty having vaginal intercourse/penetration (Criterion A1) can vary from a total inability to experience vaginal penetration in any situation (e.g., intercourse, gynecological examinations, tampon insertion) to the ability to easily experience penetration in one situation but not in another. Although the most common clinical situation is when a female is unable to

experience intercourse or penetration with a partner, difficulties in undergoing required gynecological examinations may also be present. *Marked vulvovaginal or pelvic pain during vaginal intercourse or penetration attempts* (Criterion A2) refers to pain occurring in different locations in the genito-pelvic area. Location of pain as well as intensity should be assessed. Typically, pain can be characterized as superficial (vulvovaginal or occurring during penetration) or deep (pelvic; i.e., not felt until deeper penetration). The intensity of the pain is often not linearly related to distress or interference with vaginal penetration or other sexual activities. Some genito-pelvic pain only occurs when provoked (i.e., by intercourse or mechanical stimulation); other genito-pelvic pain may be spontaneous as well as provoked. Genito-pelvic pain can also be usefully characterized qualitatively (e.g., “burning,” “cutting,” “shooting,” “throbbing”). The pain may persist for a period after intercourse is completed and may also occur during urination. Typically, the pain experienced during vaginal intercourse can be reproduced during a gynecological examination.

Marked fear or anxiety about vulvovaginal or pelvic pain either in anticipation of, or during, or as a result of vaginal penetration (Criterion A3) is commonly reported by females who have regularly experienced pain during vaginal penetration. This “normal” reaction may lead to avoidance of sexual/intimate situations. In other cases, this marked fear does not appear to be closely related to the experience of pain but nonetheless leads to avoidance of intercourse and vaginal penetration situations. Some have described this as similar to a phobic reaction except that the phobic object may be vaginal penetration or the fear of pain.

Marked tensing or tightening of the pelvic floor muscles during attempted vaginal penetration (Criterion A4) can vary from reflexive-like spasm of the pelvic floor in response to attempted vaginal entry to “normal/voluntary” muscle guarding in response to the anticipated or the repeated experience of pain or to fear or anxiety. In the case of “normal/guarding” reactions, penetration may be possible under circumstances of relaxation. The characterization and assessment of pelvic floor dysfunction is often best undertaken by a specialist gynecologist or by a pelvic floor physical therapist.

Symptoms of genito-pelvic pain/penetration disorder may be characterized by previous terms, including *dyspareunia* (pain during sexual intercourse) and *vaginismus* (defined by

involuntary contraction of muscles making penetration painful or impossible). Specific medical disorders, such as vulvodynia (chronic idiopathic vulvar pain lasting at least 3 months) and provoked vestibulodynia (contact-induced vulvodynia localized to the vulvar vestibule), may be a primary cause of genito-pelvic pain/penetration disorder and may be a focus in studies of the disorder. Females diagnosed with these other conditions report significant distress, and their symptoms are likely to meet criteria for genito-pelvic pain/penetration disorder.

Associated Features

Genito-pelvic pain/penetration disorder is frequently associated with other sexual dysfunctions, particularly reduced sexual desire and interest (female sexual interest/arousal disorder). Sometimes desire and interest are preserved in sexual situations that are not painful or do not require penetration. Even when individuals with genito-pelvic pain/penetration disorder report

sexual interest/motivation, there is often behavioral avoidance of sexual situations and opportunities. Avoidance of gynecological examinations despite medical recommendations is also frequent. The pattern of avoidance is similar to that seen in phobic disorders. It is common for females who have not succeeded in having vaginal penetration to come for treatment only when they wish to conceive. Many females with genito-pelvic pain/penetration disorder will experience associated relationship/marital problems; they also often report that the symptoms significantly diminish their feelings of femininity.

In addition to the subtype “lifelong/acquired,” five factors should be considered during assessment and diagnosis of genito-pelvic pain/penetration disorder because they may be relevant to etiology or treatment: 1) partner factors (e.g., partner’s sexual problems, partner’s health status); 2) relationship factors (e.g., partner responses to the pain, including solicitous, negative, and facilitative responses; discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., poor body image, history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), or stressors (e.g., job loss, bereavement); 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment. Each of these factors may contribute differently to the presenting symptoms of different females with this disorder.

Prevalence

The prevalence of genito-pelvic pain/penetration disorder is unknown. However, approximately 10%–28% of females of reproductive age in the United States report recurrent pain during intercourse. Difficulties having intercourse appear to be a frequent referral to sexual dysfunction clinics and to specialist clinicians. Internationally, prevalence of genito-pelvic pain upon intercourse ranges from 8% to 28% among females of reproductive age and varies by country.

Prevalence of genito-pelvic pain during sexual activities involving vaginal penetration among lesbian women relative to heterosexual women remains uncertain but may be similar or lower. Prevalence rates among other sexual minorities, including transgender women, are unknown.

Development and Course

The development and course of genito-pelvic pain/penetration disorder is unclear. Because women generally do not seek treatment until they experience problems in sexual functioning, it can, in general, be difficult to characterize genito-pelvic pain/penetration disorder as lifelong (primary) or acquired (secondary). Although women typically come to clinical attention after the initiation of sexual activity, there are often earlier clinical signs. For example, difficulty with or the avoidance of use of tampons is an important predictor of later

problems. Difficulties with vaginal penetration (inability or fear or pain) may not be obvious until intercourse is attempted during sexual activity. Even once intercourse is attempted, the frequency of attempts may not be significant or regular. In cases where it is difficult to establish whether symptomatology is lifelong or acquired, it is useful to determine the presence of any consistent period of successful pain-, fear-, and tension-free intercourse. If the experience of such

a period can be established, then genito-pelvic pain/penetration disorder can be characterized as acquired. Once symptomatology is present for a period of approximately 6 months, the probability of spontaneous and significant symptomatic remission appears to diminish.

Complaints related to genito-pelvic pain peak during early adulthood and in the peri- and postmenopausal period. There may also be an increase in genito-pelvic pain-related symptoms in the postpartum period.

Risk and Prognostic Factors

Temperamental. Females with antecedent mood and anxiety disorders are four times more likely to develop symptoms of genito-pelvic pain/penetration disorder compared with those without these antecedent disorders. Psychosocial factors (e.g., pain catastrophizing, pain self-efficacy, avoidance of pain, negative mood) and interpersonal factors (e.g., insecure attachment, negative partner responses to the pain, sexual motives that focus on avoiding negative relationship outcomes) may exacerbate and maintain symptoms.

Environmental. Females with genito-pelvic pain/penetration disorder are more likely to report a history of sexual and/or physical abuse, and fear of abuse than females without this disorder, although not all women with presenting symptoms have this history.

Genetic and physiological. Females experiencing superficial pain during vaginal penetration often report the onset of the pain after a history of vaginal infections. Even after the infections have resolved and there are no known residual physical findings, the pain persists. Pain during tampon insertion or the inability to insert tampons before any sexual contact has been attempted is an important risk factor for genito-pelvic pain/penetration disorder.

Additional biomedical risk factors include early puberty, inflammation, early use of oral contraceptives, vulvar pain receptor proliferation (i.e., increase in the number of receptors) and sensitization (i.e., touch may become perceived as pain), and lower touch and pain thresholds. Abnormalities of the pelvic floor muscles while at rest, including hypertonicity, poor muscle control, hypersensitivity, and altered contractility, may close the vaginal hiatus and interfere with penetration.

Culture-Related Diagnostic Issues

Cultural contexts can affect the experience and reporting of genito-pelvic pain related to intercourse. Affected females experience negative implications related to social narratives of womanhood, sexuality, and femininity, including pressures to prioritize men's sexual desire and penetrative sex, and depictions of sex as easy and natural. Cultural views that devalue female sexual experience may affect the way women interpret the experience of pain during sex, their help-seeking choices, and how they discuss their symptoms with their caregivers. For example, some females may not report genito-pelvic pain specifically but rather refer to being unhappy in their marriages.

In the United States, Hispanic females endorse significantly higher rates of genito-pelvic pain and are more likely to report pain with first intercourse (i.e., primary genito-pelvic pain/penetration disorder) compared with non-Hispanic women. In a Minneapolis, Minnesota survey, only about half of females with genito-pelvic pain sought treatment, and those who did frequently reported feeling stigmatized. Such experiences may be heightened for

sexual minorities and underserved ethnic and racialized groups, especially given evidence of inequities in pain treatment for females and African Americans.

Sex- and Gender-Related Diagnostic Issues

Gendered social constructions relating to womanhood and femininity are implicated in the experience of genito-pelvic pain/penetration disorder, including the prioritization of both penetrative sex and men's sexual desires above women's own needs and desires. The disorder is associated with feelings of shame and inadequacy as a female, contributing further to enhanced psychological distress.

By definition, the diagnosis of genito-pelvic pain/penetration disorder is only given to females. There is relatively new research concerning urological chronic pelvic pain syndrome in males, suggesting that males may experience some similar problems. The prevalence of male genito-pelvic pain is estimated to be 2.2%–9.7% worldwide. The research and clinical experience are not sufficiently developed yet to justify the application of this diagnosis to males. Other specified sexual dysfunction or unspecified sexual dysfunction may be diagnosed in males appearing to fit this pattern.

Diagnostic Markers

Validated physiological measures of Criterion A2 symptoms (*Marked vulvovaginal or pelvic pain during vaginal intercourse or penetration attempts*) can be assessed in real time (e.g., the cotton-swab test, vulvalgesiometer, tampon test). Although these measures are well validated for pain intensity during penetration attempts, none approximate the sexual context in which the pain is experienced, which can only be assessed via self-report. Criterion A4 symptoms (*Marked tensing or tightening of the pelvic floor muscles during attempted vaginal penetration*) can also be measured (e.g., via electromyographic amplitude, dynamometer, 4D ultrasound by a qualified physical therapist). There are no validated physiological measures of component symptoms for Criterion A1 or A3. Validated psychometric inventories may be used to formally assess the pain and anxiety components related to genito-pelvic pain/penetration disorder.

Functional Consequences of Genito-Pelvic Pain/Penetration Disorder

Functional difficulties in genito-pelvic pain/penetration disorder are often associated with interference in various aspects of the romantic relationship—including the initiation of such relationships—and sometimes with the ability to conceive via penile/vaginal intercourse.

Differential Diagnosis

Another medical condition. In many instances, females with genito-pelvic pain/penetration disorder will also be diagnosed with another medical condition (e.g., lichen sclerosus, endometriosis, pelvic inflammatory disease, genitourinary syndrome of menopause). In some cases, treating the medical condition may alleviate the genito-pelvic pain/penetration disorder. Much of the time, this is not the case. There are no reliable tools or diagnostic methods to allow clinicians to know whether the medical condition or genito-pelvic pain/penetration disorder is

primary. Often, the associated medical conditions are difficult to diagnose and treat. For example, the increased incidence of postmenopausal pain during intercourse may sometimes be attributable to vaginal dryness or irritation associated with declining estrogen levels. The relationship, however, between genital symptoms, estrogen, and pain is not well understood.

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Somatic symptom and related disorders. Some females with genito-pelvic pain/penetration disorder may also be diagnosable with somatic symptom disorder. Since both genito-pelvic pain/penetration disorder and the somatic symptom and related disorders are new diagnoses in DSM-5, it is not yet clear whether they can be reliably differentiated. Some females diagnosed with genito-pelvic pain/penetration disorder will also be diagnosed with a specific phobia.

Inadequate sexual stimuli. It is important that the clinician, in considering differential diagnoses, assess the adequacy of sexual stimuli within the female's sexual experience. Sexual situations in which there is inadequate foreplay or arousal may lead to difficulties in penetration, pain, or avoidance. Erectile dysfunction or premature (early) ejaculation in the male partner may result in difficulties with penetration. These conditions should be carefully assessed. In some situations, a diagnosis of genito-pelvic pain/penetration disorder may not be appropriate.

Comorbidity

Comorbidity between genito-pelvic pain/penetration disorder and other sexual difficulties appears to be common. Comorbidity with relationship distress is also common and typically related to the lack of sexual intimacy rather than (solely) the pain itself. This is not surprising, because the inability to have (pain-free) intercourse with a desired partner and the avoidance of sexual opportunities may be either a contributing factor to or the result of other sexual or relationship problems. Because pelvic floor symptoms are implicated in the diagnosis of genito-pelvic pain/penetration disorder, there is likely to be a higher prevalence of other disorders related to the pelvic floor or reproductive organs (e.g., interstitial cystitis, constipation, vaginal infection, endometriosis, irritable bowel syndrome). Females with genito-pelvic pain/penetration disorder frequently experience comorbid chronic pain conditions (e.g., fibromyalgia, chronic headaches), and the prevalence of these comorbidities increases with the severity of vulvar pain symptoms.

Lesbian women also report genito-pelvic pain and penetration difficulties during sexual activities; the frequency of genito-pelvic pain/penetration symptoms among nonheterosexual women has been shown to be less than or the same as among heterosexual women.

Male Hypoactive Sexual Desire Disorder

Diagnostic Criteria	F52.0
A. Persistently or recurrently deficient (or absent) sexual/erotic thoughts or fantasies and desire for sexual activity. The judgment of deficiency is made by	

the clinician, taking into account factors that affect sexual functioning, such as age and general and sociocultural contexts of the individual's life.

- B. The symptoms in Criterion A have persisted for a minimum duration of approximately 6 months.
- C. The symptoms in Criterion A cause clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

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Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify current severity:

Mild: Evidence of mild distress over the symptoms in Criterion A.

Moderate: Evidence of moderate distress over the symptoms in Criterion A.

Severe: Evidence of severe or extreme distress over the symptoms in Criterion A.

Diagnostic Features

When an assessment for male hypoactive sexual desire disorder is being made, interpersonal context must be taken into account. A “desire discrepancy,” in which a man has lower desire for sexual activity than his partner, is not sufficient to diagnose male hypoactive sexual desire disorder. Both low/absent desire for sex and deficient/absent sexual thoughts or fantasies (Criterion A) are required for a diagnosis of the disorder. There may be variation across men in how sexual desire is expressed.

The lack of desire for sex and deficient/absent erotic thoughts or fantasies must be persistent or recurrent and must occur for a minimum duration of approximately 6 months. The inclusion of this duration criterion is meant to safeguard against making a diagnosis in cases in which a man's low sexual desire may represent a reactive but temporary response to adverse life conditions. For example, the man's low sexual desire may be related to an acute stressor or loss of self-esteem (e.g., being fired from a job or experiencing financial difficulty such as business failure). If these stressors persist beyond 6 months along with low sexual desire, then clinician judgment determines the appropriateness of the male hypoactive sexual desire disorder diagnosis.

Associated Features

Male hypoactive sexual desire disorder is sometimes associated with erectile and/or ejaculatory concerns. For example, persistent difficulties obtaining an erection may lead a man to lose interest in sexual activity. Men with hypoactive sexual desire disorder often report that they no longer initiate sexual activity and that they are minimally receptive to a partner's attempt to initiate. Sexual activities (e.g., masturbation or partnered sexual activity) may sometimes occur even in the presence of low sexual desire. Relationship-specific preferences regarding patterns of sexual initiation must be taken into account when making a diagnosis of male hypoactive sexual desire disorder. Although men are more likely to initiate sexual activity, and thus low desire may be characterized by a pattern of noninitiation, many men may prefer to have their partner initiate sexual activity. In such situations, the man's lack of receptivity to a partner's initiation should be considered when evaluating low desire.

In addition to the subtypes "lifelong/acquired" and "generalized/situational," the following five factors must be considered during assessment and diagnosis of male hypoactive sexual desire disorder given that they may be relevant to etiology and/or treatment: 1) partner factors (e.g., partner's sexual problems, partner's health status); 2) relationship factors (e.g., poor communication, discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., poor body image, history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), or stressors (e.g., job loss, bereavement); 4) cultural/religious factors (e.g., inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment. Each of these factors may contribute differently to the presenting symptoms of different men with this disorder.

Prevalence

The prevalence of male hypoactive sexual desire disorder varies depending on country of origin and method of assessment. Estimates of prevalence in representative samples range

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from 3% to 17%. Sexual desire problems are less common in younger men (ages 16–24), with prevalence rates between 3% and 14%, compared with older men (ages 60–74 years), with prevalence rates between 16% and 28%. However, a persistent lack of interest in sex, lasting 6 months or more, affects a smaller proportion of men (6%). Moreover, less than 2% of men report clinically significant distress associated with low desire. Studies on help-seeking behavior indicate that only 10.5% of men with sexual problems in the previous year sought help.

Development and Course

By definition, lifelong male hypoactive sexual desire disorder indicates that low or no sexual desire has always been present, whereas the acquired subtype would be assigned if the man's low desire developed after a period of normal sexual desire. There is a requirement that low desire persist for approximately 6 months or more; thus, short-term changes in sexual desire should not be diagnosed as male hypoactive sexual desire disorder.

There is a normative age-related decline in sexual desire. The prevalence of low sexual desire in men increases with age, from approximately 5.2% prevalence at age 27 years to 18.5% at age

50 years. Like women, men identify a variety of triggers for their sexual desire, and they describe a wide range of reasons that they choose to engage in sexual activity. Although erotic visual cues may be more potent elicitors of desire in younger men, the potency of sexual cues may decrease with age and must be considered when evaluating men for hypoactive sexual desire disorder.

Risk and Prognostic Factors

Temperamental. Mood and anxiety symptoms appear to be strong predictors of low desire in men. Up to half of men with a past history of psychiatric symptoms may have moderate or severe loss of desire, compared with only 15% of those without such a history. A man's feelings about himself, his perception of his partner's sexual desire toward him, feelings of being emotionally connected, and contextual variables may all negatively (as well as positively) affect sexual desire.

Beliefs about sexuality (particularly restrictive sexual attitudes and conservative beliefs) are commonly associated with low sexual desire in men. Moreover, lack of erotic thoughts and concerns about erection during sexual activity are significant predictors of low sexual desire, as well as low confidence levels in erectile function.

Environmental. Alcohol use may increase the occurrence of low desire. Other environmental determinants of low sexual desire include problematic dyadic relationships, reduced attraction toward the partner, living in a long-term relationship, sexual boredom, and professional stress. At the larger societal level, cohort studies in a few high-income countries indicate a trend toward the decrease of sexual desire in men in recent decades.

Genetic and physiological. Endocrine disorders such as hyperprolactinemia and hypogonadism significantly affect sexual desire in men. Age is a significant risk factor for low desire in men. It is unclear whether or not men with low desire also have abnormally low levels of testosterone; however, among hypogonadal men, low desire is common. There also may be a critical threshold below which testosterone will affect sexual desire in men and above which there is little effect of testosterone on men's desire.

Culture-Related Diagnostic Issues

There is marked variability in prevalence rates of low desire across world regions, ranging from 12.5% in Northern European men to 28% in Southeast Asian men ages 40–80 years. Distress about lack of sexual desire was significantly associated with sociocultural contexts (e.g., occupational stress) in a web-based survey across three European countries (Portugal, Croatia, and Norway).

Sex- and Gender-Related Diagnostic Issues

In contrast to the classification of sexual dysfunctions in women, desire and arousal disorders have been retained as separate constructs in men. Despite some similarities in the experience of desire across men and women, and the fact that desire fluctuates over time and is dependent on contextual factors, men do report a significantly higher intensity and frequency of sexual desire

compared with women. However, preliminary data suggest that the overlap between sexual desire and sexual arousal (erectile function) is also very common in men, particularly when they present for help regarding sexual problems. Regarding sexual orientation, data suggest that low sexual desire is more commonly reported by gay men (19%) than by heterosexual men (9%).

Differential Diagnosis

Nonsexual mental disorders. Nonsexual mental disorders, such as major depressive disorder, which is characterized by “markedly diminished interest or pleasure in all, or almost all, activities,” may explain the lack of sexual desire. If the lack of desire is better explained by another mental disorder, then a diagnosis of male hypoactive sexual desire disorder would not be made.

Substance/medication use. An onset of male hypoactive sexual desire that coincides with the beginning of substance/medication use and that dissipates with discontinuation of the substance/medication or dose reduction is suggestive of a substance/medication-induced sexual dysfunction, which should be diagnosed instead of male hypoactive sexual desire disorder.

Another medical condition. If the low/absent desire and deficient/absent erotic thoughts or fantasies are better explained by the effects of another medical condition (e.g., hypogonadism, diabetes mellitus, thyroid dysfunction, central nervous system disease), then a diagnosis of male hypoactive sexual desire disorder would not be made.

Interpersonal factors. If interpersonal or significant contextual factors, such as severe relationship distress or other significant stressors, are associated with the loss of desire in the man, then a diagnosis of male hypoactive sexual desire disorder would not be made.

Other sexual dysfunctions. The presence of another sexual dysfunction does not rule out a diagnosis of male hypoactive sexual desire disorder; there is some evidence that up to one-half of men with low sexual desire also have erectile difficulties, and slightly fewer may also have early ejaculation difficulties. If the man’s low desire is explained by self-identification as an asexual, then a diagnosis of male hypoactive sexual desire disorder is not made.

Comorbidity

Male hypoactive sexual desire disorder is rarely the sole sexual diagnosis in men. Erectile dysfunction, delayed ejaculation, and premature (early) ejaculation are often comorbid diagnoses. Depression and other mental disorders, as well as endocrinological factors, are often comorbid with male hypoactive sexual desire disorder.

Premature (Early) Ejaculation

Diagnostic Criteria

F52.4

- A. A persistent or recurrent pattern of ejaculation occurring during partnered sexual activity within approximately 1 minute following vaginal penetration and before the individual wishes it.

Note: Although the diagnosis of premature (early) ejaculation may be applied to individuals engaged in nonvaginal sexual activities, specific duration criteria have not been established for these activities.

- B. The symptom in Criterion A must have been present for at least 6 months and must be experienced on almost all or all (approximately 75%–100%) occasions of sexual activity (in identified situational contexts or, if generalized, in all contexts).
- C. The symptom in Criterion A causes clinically significant distress in the individual.
- D. The sexual dysfunction is not better explained by a nonsexual mental disorder or as a consequence of severe relationship distress or other significant stressors and is not attributable to the effects of a substance/medication or another medical condition.

Specify whether:

Lifelong: The disturbance has been present since the individual became sexually active.

Acquired: The disturbance began after a period of relatively normal sexual function.

Specify whether:

Generalized: Not limited to certain types of stimulation, situations, or partners.

Situational: Only occurs with certain types of stimulation, situations, or partners.

Specify current severity:

Mild: Ejaculation occurring within approximately 30 seconds to 1 minute of vaginal penetration.

Moderate: Ejaculation occurring within approximately 15–30 seconds of vaginal penetration.

Severe: Ejaculation occurring prior to sexual activity, at the start of sexual activity, or within approximately 15 seconds of vaginal penetration.

Diagnostic Features

Premature (early) ejaculation is manifested by ejaculation that occurs prior to or shortly after vaginal penetration, operationalized by an individual's estimate of ejaculatory latency (i.e., elapsed time before ejaculation) after vaginal penetration. Although the diagnostic criteria specify penile-vaginal sex, it is reasonable to assume that similar estimates of ejaculatory latency apply to males having sex with males, as well as to other sexual behaviors. Estimated and measured intravaginal ejaculatory latencies are highly correlated as long as the ejaculatory latency is of short duration; therefore, self-reported estimates of ejaculatory latency are sufficient for diagnostic purposes. A 60-second intravaginal ejaculatory latency time was previously considered to be an appropriate cutoff for the diagnosis of lifelong premature (early) ejaculation in men; however, expert consensus now considers this latency time to be too brief and instead

recommends a 120-second threshold.

Associated Features

Many males with premature (early) ejaculation complain of a sense of lack of control over ejaculation and report apprehension about their anticipated inability to delay ejaculation on future sexual encounters.

The following factors may be relevant in the evaluation of any sexual dysfunction: 1) partner factors (e.g., partner's sexual problems, partner's health status); 2) relationship factors (e.g., poor communication, discrepancies in desire for sexual activity); 3) individual vulnerability factors (e.g., history of sexual or emotional abuse), psychiatric comorbidity (e.g., depression, anxiety), and stressors (e.g., job loss, bereavement); 4) cultural/religious factors (e.g., lack of privacy, inhibitions related to prohibitions against sexual activity; attitudes toward sexuality); and 5) medical factors relevant to prognosis, course, or treatment.

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Prevalence

Estimates of the prevalence of premature (early) ejaculation vary widely depending on the definition utilized. Internationally, a prevalence range of 8%–30% has been reported across all ages, with even lower and higher rates in other studies. Prevalence of premature (early) ejaculation may increase with age. For example, the prevalence among males ages 18–30 in Switzerland and Turkey is about 9%–11%, while the reported prevalence of concern among males ages 50–59 in the United States about how rapidly they ejaculate may be as high as 55%. When premature (early) ejaculation is defined as ejaculation occurring within approximately 1 minute of vaginal penetration, only 1%–3% of males would be diagnosed with the disorder.

Development and Course

By definition, lifelong premature (early) ejaculation starts during a male's initial sexual experiences and persists thereafter. Some males may experience premature (early) ejaculation during their initial sexual encounters but gain ejaculatory control over time. It is the persistence of ejaculatory problems for longer than 6 months that determines the diagnosis of premature (early) ejaculation. In contrast, some males develop the disorder after a period of having a normal ejaculatory latency, known as *acquired premature (early) ejaculation*. There is far less known about acquired premature (early) ejaculation than about lifelong premature (early) ejaculation. The acquired form likely has a later onset, usually appearing during or after the fourth decade of life. Lifelong is relatively stable throughout life.

Risk and Prognostic Factors

Temperamental. Premature (early) ejaculation may be more common in males with anxiety disorders, especially social anxiety disorder.

Genetic and physiological. There is a moderate genetic contribution to lifelong premature (early) ejaculation. Premature (early) ejaculation may be associated with dopamine transporter gene polymorphism or serotonin transporter gene polymorphism. Thyroid disease, prostatitis, and

drug withdrawal are associated with acquired premature (early) ejaculation. Positron emission tomography measures of regional cerebral blood flow during ejaculation have shown primary activation in the mesocephalic transition zone, including the ventral tegmental area.

Culture-Related Diagnostic Issues

Perception of what constitutes a normal ejaculatory latency differs cross-culturally and may be related to varying awareness of sexual dysfunction, concern about sexual failure, and perceptions about the importance of sex. Measured ejaculatory latencies may differ in some countries. Cultural or religious factors may contribute to these differences. For example, reports of premature (early) ejaculation were more common in arranged marriages, because of factors such as anxiety over family pressures and lack of premarital sexual experience.

Sex- and Gender-Related Diagnostic Issues

Premature (early) ejaculation is a sexual dysfunction in men. Men and their sexual partners may differ in their perception of what constitutes an acceptable ejaculatory latency. There may be increasing concerns in women about early ejaculation in their sexual partners, which may be a reflection of changing societal attitudes concerning women's sexual activity.

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Diagnostic Markers

Ejaculatory latency is usually monitored in research settings by the sexual partner utilizing a timing device (e.g., stopwatch), though this is not ideal in real-life sexual situations. In clinical settings, the man's estimate of the time between intravaginal penetration and ejaculation should be accepted in lieu of stopwatch measurements.

Association With Suicidal Thoughts or Behavior

Among males receiving treatment for premature (early) ejaculation with comorbid depression, elevated rates of suicidal thoughts or behavior have been observed; although the affected males attributed the suicidal symptoms to their premature (early) ejaculation, the presence of depression was also a likely contributing factor.

Functional Consequences of Premature (Early) Ejaculation

A pattern of premature (early) ejaculation may be associated with decreased self-esteem and self-confidence, a sense of lack of control, and adverse consequences for partner relationships. It may also cause personal distress and decreased sexual satisfaction in the sexual partner. Single males are more bothered than partnered males by premature (early) ejaculation because of its interference with seeking and maintaining new relationships. Ejaculation prior to penetration may be associated with difficulties in conception.

Differential Diagnosis

Substance/medication-induced sexual dysfunction. When problems with premature (early) ejaculation are attributable exclusively to substance use, intoxication, or withdrawal,

substance/medication-induced sexual dysfunction should be diagnosed.

Ejaculatory concerns that do not meet diagnostic criteria. It is necessary to identify males with normal ejaculatory latencies who desire longer ejaculatory latencies and males who have episodic premature (early) ejaculation (e.g., during the first sexual encounter with a new partner when a short ejaculatory latency may be common or normative). Neither of these situations would lead to a diagnosis of premature (early) ejaculation, even though these situations may be distressing to some males.

Comorbidity

Premature (early) ejaculation may be associated with erectile problems. In many cases, it may be difficult to determine which difficulty preceded the other. Lifelong premature (early) ejaculation may be associated with certain anxiety disorders. Acquired premature (early) ejaculation may be associated with prostatitis, thyroid disease, or drug withdrawal (e.g., during opioid withdrawal).

Substance/Medication-Induced Sexual Dysfunction

Diagnostic Criteria

- A. A clinically significant disturbance in sexual function is predominant in the clinical picture.
- B. There is evidence from the history, physical examination, or laboratory findings of both (1) and (2):

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- 1. The symptoms in Criterion A developed during or soon after substance intoxication or withdrawal or after exposure to or withdrawal from a medication.
 - 2. The involved substance/medication is capable of producing the symptoms in Criterion A.
- C. The disturbance is not better explained by a sexual dysfunction that is not substance/medication-induced. Such evidence of an independent sexual dysfunction could include the following:

The symptoms precede the onset of the substance/medication use; the symptoms persist for a substantial period of time (e.g., about 1 month) after the cessation of acute withdrawal or severe intoxication; or there is other evidence suggesting the existence of an independent non-substance/medication-induced sexual dysfunction (e.g., a history of recurrent non-substance/medication-related episodes).
- D. The disturbance does not occur exclusively during the course of a delirium.
- E. The disturbance causes clinically significant distress in the individual.

Note: This diagnosis should be made instead of a diagnosis of substance intoxication or substance withdrawal only when the symptoms in Criterion A predominate in the clinical picture and are sufficiently severe to warrant clinical attention.

Coding note: The ICD-10-CM codes for the [specific substance/medication]-induced sexual dysfunctions are indicated in the table below. Note that the ICD-10-CM code depends on whether or not there is a comorbid substance use disorder present for the same class of substance. In any case, an additional separate diagnosis of a substance use disorder is not given. If a mild substance use disorder is comorbid with the substance-induced sexual dysfunction, the 4th position character is “1,” and the clinician should record “mild [substance] use disorder” before the substance-induced sexual dysfunction (e.g., “mild cocaine use disorder with cocaine-induced sexual dysfunction”). If a moderate or severe substance use disorder is comorbid with the substance-induced sexual dysfunction, the 4th position character is “2,” and the clinician should record “moderate [substance] use disorder” or “severe [substance] use disorder,” depending on the severity of the comorbid substance use disorder. If there is no comorbid substance use disorder (e.g., after a one-time heavy use of the substance), then the 4th position character is “9,” and the clinician should record only the substance-induced sexual dysfunction.

	ICD-10-CM		
	With mild use disorder	With moderate or severe use disorder	Without use disorder
Alcohol	F10.181	F10.281	F10.981
Opioid	F11.181	F11.281	F11.981
Sedative, hypnotic, or anxiolytic	F13.181	F13.281	F13.981
Amphetamine-type substance (or other stimulant)	F15.181	F15.281	F15.981
Cocaine	F14.181	F14.281	F14.981
Other (or unknown) substance	F19.181	F19.281	F19.981

Specify (see [Table 1](#) in the chapter “Substance-Related and Addictive Disorders,” which indicates whether “with onset during intoxication” and/or “with onset during withdrawal” applies to a given substance class; or *specify* “with onset after medication use”):

With onset during intoxication: If criteria are met for intoxication with the substance and the symptoms develop during intoxication.

With onset during withdrawal: If criteria are met for withdrawal from the substance and the symptoms develop during, or shortly after, withdrawal.

With onset after medication use: If symptoms developed at initiation of medication, with a change in use of medication, or during withdrawal of medication.

Specify current severity:

Mild: Occurs on 25%–50% of occasions of sexual activity.

Moderate: Occurs on 50%–75% of occasions of sexual activity.

Severe: Occurs on 75% or more of occasions of sexual activity.

Recording Procedures

The name of the substance/medication-induced sexual dysfunction begins with the specific substance (e.g., alcohol) that is presumed to be causing the sexual dysfunction. The ICD-10-CM code that corresponds to the applicable drug class is selected from the table included in the criteria set. For substances that do not fit into any of the classes (e.g., fluoxetine), the ICD-10-CM code for the other (or unknown) substance class should be used and the name of the specific substance recorded (e.g., F19.981 fluoxetine-induced sexual dysfunction). In cases in which a substance is judged to be an etiological factor but the specific substance is unknown, the ICD-10-CM code for the other (or unknown) substance class is used and the fact that the substance is unknown is recorded (e.g., F19.981 unknown substance-induced sexual dysfunction).

When recording the name of the disorder, the comorbid substance use disorder (if any) is listed first, followed by the word “with,” followed by the name of the substance-induced sexual dysfunction, followed by the specification of onset (i.e., onset during intoxication, onset during withdrawal, with onset after medication use), followed by the severity specifier (e.g., mild, moderate, severe). For example, in the case of erectile dysfunction occurring during intoxication in a man with a severe alcohol use disorder, the diagnosis is F10.281 severe alcohol use disorder with alcohol-induced sexual dysfunction, with onset during intoxication, moderate. A separate diagnosis of the comorbid severe alcohol use disorder is not given. If the substance-induced sexual dysfunction occurs without a comorbid substance use disorder (e.g., after a one-time heavy use of the substance), no accompanying substance use disorder is noted (e.g., F15.981 amphetamine-induced sexual dysfunction, with onset during intoxication). When more than one substance is judged to play a significant role in the development of the sexual dysfunction, each should be listed separately (e.g., F14.181 mild cocaine use disorder with cocaine-induced sexual dysfunction, with onset during intoxication, moderate; F19.981 fluoxetine-induced sexual dysfunction, with onset after medication use, moderate).

Diagnostic Features

The essential features of substance/medication-induced sexual dysfunction are clinically significant disturbances in sexual function that are predominant in the clinical picture (Criterion A) that are judged to be due to the effects of a substance (e.g., a drug of abuse or medication). The sexual dysfunction must have developed during or soon after substance intoxication or withdrawal or after exposure to or withdrawal from a medication, and the substances or medications must be capable of producing the symptoms (Criterion B2). Substance/medication-induced sexual dysfunction due to a prescribed treatment for a

mental disorder or another medical condition must have its onset while the individual is